MATH 251: Algebra II

Corrected Course Notes

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Definitions Results

Examples Notes

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Note

As always, some important results are given out as homework. I am listing them below. Let $\mathbb{F}$ be a field and V an $\mathbb{F}$-vector space. V may be infinite-dimensional, $S \subseteq V$ may be infinite unless stated.

Assignment 1, Question 4. $S \subseteq V$ a subset. Let X be all subspaces $W \leq V$ such that $S \subseteq W$. Then

$$\mathrm{Span}(S) = \bigcap_{W \in X} W.$$

Question 5. Let $S \subseteq V$, $v \in V$. Then

$$v \in \mathrm{Span}(S) \iff \mathrm{Span}(S) = \mathrm{Span}(S \cup \{v\}).$$

Question 6. Let $S_1, S_2 \subseteq V$. Then

$$\mathrm{Span}(S_1) + \mathrm{Span}(S_2) = \mathrm{Span}(S_1 \cup S_2).$$

Question 8. Let $S_1, S_2 \subseteq V$ be disjoint, linearly independent. Then

$$S_1 \cup S_2 \text{ is linearly independent} \iff \mathrm{Span}(S_1) \cap \mathrm{Span}(S_2) = \{0\}.$$

Assignment 2, Question 4. A vector space is infinite-dimensional

$$\iff \exists \text{ infinite linearly independent subset.}$$

Question 5. Let V, W be finite-dimensional $\mathbb{F}$-vector spaces. Then

$$\dim(V \oplus W) = \dim(V) + \dim(W),$$

where $V \oplus W$ is the external direct sum.

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Question 6. Let $\dim V < \infty$, $W_1, W_2 \leq V$. Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Question 8. Let $\dim V < \infty$, $W \leq V$. Then

$$\dim(W) + \dim(V/W) = \dim(V).$$

Assignment 3, Question 1.

$$\mathrm{tr} : M_{n\times n}(\mathbb{F}) \longrightarrow \mathbb{F}$$

is a linear transformation and satisfies

$$\mathrm{tr}(AB) = \mathrm{tr}(BA), \qquad A, B \in M_{n\times n}(\mathbb{F}).$$

Question 2. If $T : V \to V$ is nilpotent and $\dim V = n < \infty$, then $T^n = 0$.

Question 3 (Extended). Let $U, W \leq V$. Let $U \oplus W$ denote the internal direct sum. Then

$$\begin{aligned} V = U \oplus W &\iff V = U + W \text{ and } U \cap W = \{0\} \\ &\iff \forall v \in V,\ \exists!\ u \in U,\ w \in W \text{ such that } v = u + w. \end{aligned}$$

If V is finite-dimensional, these conditions are also equivalent to

$$U \cap W = \{0\} \quad \text{and} \quad \dim V = \dim U + \dim W.$$

Equivalently, the union of a basis of U and a basis of W is a basis of V.

Question 4. If T is a projection, that is $T^2 = T$, then

$$V = \ker T \oplus \mathrm{im}\, T.$$

Question 6. Let $\dim(V) < \infty$. Then $\exists \ell \geq 1$ such that $\forall k \geq \ell$,

$$\mathrm{im}(T^k) = \mathrm{im}(T^{k+1}), \qquad \ker(T^k) = \ker(T^{k+1}).$$

Question 8. Let $\dim(V) < \infty$, $T \in \mathrm{Hom}(V, V)$, $W \leq V$ such that W is T-invariant and has dimension k. Then $\exists \mathcal{B}$, a basis of V, such that

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}, \qquad A\ k \times k, \quad B\ k \times (n-k), \quad C\ (n-k) \times (n-k).$$

Assignment 4, Question 2. Let $\mathcal{B} = \{v_1, \ldots, v_n\}$, $\mathcal{C} = \{w_1, \ldots, w_m\}$ be bases of V, W, respectively. Define $T_{ij} : V \to W$ as

$$T_{ij}(v_\ell) = \begin{cases} w_j, & i = \ell, \\ 0, & i \neq \ell. \end{cases}$$

Then $\{T_{ij}\}_{i,j}$ gives a basis of $\mathrm{Hom}(V, W)$.

Question 3. The determinant of an upper triangular matrix is the product of its diagonal entries.

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Question 4. Let $M \in M_{n\times n}(\mathbb{F})$ be of form

$$M = \begin{pmatrix} A_{k\times k} & B_{k\times(n-k)} \\ 0 & C_{(n-k)\times(n-k)} \end{pmatrix}.$$

Then

$$\det M = (\det A)(\det C).$$

Question 5.

$$\det A = \det(A^t).$$

Assignment 5, Question 1. Let $A \in M_{m\times n}(\mathbb{F})$ such that $\operatorname{rank}(A) = m$. Then $\exists B \in M_{n\times m}(\mathbb{F})$ such that

$$AB = I_m.$$

Question 2. Let $B \in M_{n\times m}(\mathbb{F})$ such that $\operatorname{rank}(B) = m$. Then $\exists A \in M_{m\times n}(\mathbb{F})$ such that

$$AB = I_m.$$

Question 3. Alternative proof of multiplicativity of matrices.

Let $A \in M_{n\times n}(\mathbb{F})$. If $\operatorname{rank}(A) < n$, then $\det(A) = 0$.

If $A = E_k \cdots E_2 E_1$, with each E_i an elementary matrix, then

$$\det(A) = \det(E_k) \cdots \det(E_1).$$

For any $A, B \in M_{n\times n}(\mathbb{F})$,

$$\det(AB) = \det(A)\det(B).$$

Assignment 6, Question 1. Let $\dim V < \infty$, $W < V$. Then

$$\exists f \in V^* \setminus \{0_{V^*}\} \text{ such that } f(W) = \{0\}.$$

Question 2. Let $\dim V < \infty$, $W \leq V$. Then

$$W^\perp := \{f \in V^* \mid f(w) = 0,\ \forall w \in W\} \leq V^*.$$

Question 3. Let $\dim V < \infty$, $U, W \leq V$. Then

$$U = W \iff U^\perp = W^\perp.$$

Question 4. Let $\dim V, \dim W < \infty$. Then for $T : V \to W$ a linear transformation,

$$T^* : W^* \longrightarrow V^*, \qquad g \longmapsto g \circ T$$

is linear.

Question 5. Same assumption as Question 4,

$$T \text{ is surjective } \implies T^* \text{ is injective.}$$

Question 6. Same assumption as Question 4,

$$T \text{ is injective } \implies T^* \text{ is surjective.}$$

Question 7. Same assumption as Question 4. Let $\mathcal{B}, \mathcal{C}$ be bases of V, W, respectively, and $\mathcal{B}^*, \mathcal{C}^*$ their dual bases. Then

$$[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \left([T]_{\mathcal{B}}^{\mathcal{C}}\right)^t.$$

Assignment 7, Question 2. Let $\lambda_1, \lambda_2, \ldots, \lambda_k$ be distinct eigenvalues of $A \in M_{n\times n}(\mathbb{F})$ with algebraic multiplicity $m_1, \ldots, m_k$. Suppose $\exists Q \in GL_n(\mathbb{F})$ such that

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$$B = QAQ^{-1} \text{ is upper triangular.}$$

Then diagonal entries of B are exactly $\lambda_1, \ldots, \lambda_k$, appeared $m_1, \ldots, m_k$ times, $P_A(t)$ splits, and

$$\operatorname{tr}(A) = \sum_{i=1}^{k} m_i \lambda_i, \qquad \det(A) = \prod_{i=1}^{k} \lambda_i^{m_i}.$$

Question 4.

$$T \text{ is diagonalizable } \iff T^{-1} \text{ is diagonalizable.}$$

Question 5.

$$A \text{ is diagonalizable } \iff A^t \text{ is diagonalizable.}$$

Question 6. Let $\dim V < \infty$, $S, T \in \operatorname{Hom}(V, V)$, and $\exists \mathcal{B}$, a basis of V, such that $[S]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{B}}^{\mathcal{B}}$ are diagonal. Then

$$S \circ T = T \circ S.$$

Assignment 8, Question 1. Let $\dim V < \infty$, $T : V \to V$ linear, $0 \neq v \in V$, and

$$W = \operatorname{Span}\{T^i v \mid i \in \mathbb{N}\}.$$

Let $k \geq 1$ be the smallest integer such that

$$T^k(v) \in \operatorname{Span}\{T^i v \mid 0 \leq i < k\}.$$

Then

$$\{T^i v \mid 0 \leq i \leq k-1\}$$

gives a basis of W, and

$$m_{T|W}(t) = P_{T|W}(t) = t^k - a_{k-1}t^{k-1} - \cdots - a_1 t - a_0,$$

where the coefficients are the scalars such that

$$T^k v = a_{k-1}T^{k-1}v + \cdots + a_1 T v + a_0 v.$$

Note

This gives another way of computing the minimal polynomial of T: take a basis $\mathcal{B}$ of V. Consider $\{T^i v\}$ for every $v \in \mathcal{B}$. Find $m_{T|W}(t)$ for all $W = \operatorname{Span}\{T^i v\}$. Take their least common multiple; this gives us $m_T(t)$.

Question 2. Let $T : V \to V$ be a linear transformation, $\dim V < \infty$, and $W \leq V$ a T-invariant subspace. Then

$$P_{T|W}(t) \mid P_T(t).$$

As a result, this provides a new proof to Cayley–Hamilton theorem.

Question 3. If T, S are diagonalizable and commuting, then they are simultaneously diagonalizable, that is $\exists \mathcal{B}$, a basis of V, such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} \quad \text{and} \quad [S]_{\mathcal{B}}^{\mathcal{B}}$$

are both diagonal.

Question 4. Let $\dim V < \infty$, $T : V \to V$ be a linear operator,

$$m_T(t) = t^k + a_{k-1}t^{k-1} + \cdots + a_1 t + a_0.$$

Then

$$T \text{ is an isomorphism } \iff a_0 \neq 0.$$

If T is an isomorphism, then

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$$m_{T^{-1}}(t) = t^k + a_0^{-1}a_1 t^{k-1} + \cdots + a_0^{-1}a_{k-1}t + a_0^{-1}.$$

Question 5. Let $n \geq 2$ be an integer. Then

$$T : M_{n\times n}(\mathbb{F}) \longrightarrow M_{n\times n}(\mathbb{F}), \qquad T(A) = A^t$$

is diagonalizable

$$\iff \operatorname{char}(\mathbb{F}) \neq 2.$$

> [The result is somewhat trivial. The proof requires some level of cleverness—by noticing that $t^2 - 1$ annihilates T, and thus $m_T(t) \mid t^2 - 1$.]

Question 7. Assume the relevant characteristic polynomial splits over $\mathbb{F}$, so the Jordan canonical forms exist over $\mathbb{F}$. Then matrices are similar/conjugate

$$\iff$$

their Jordan canonical forms are the same up to rearrangement of Jordan blocks.

Assignment 9, Question 1. For $B \in M_{n\times n}(\mathbb{F})$, B^* its conjugate transpose,

$$\langle A, B\rangle = \operatorname{tr}(B^* A)$$

defines an inner product on $M_{n\times n}(\mathbb{F})$, $\mathbb{F}=\mathbb{C}$ or $\mathbb{R}$.

Question 2. Parallelogram law:

$$2\|u\|^2+2\|v\|^2=\|u+v\|^2+\|u-v\|^2.$$

Polarization identity:

$$\langle u,v\rangle=\frac{1}{4}\left[\|u+v\|^2-\|u-v\|^2+i\|u+iv\|^2-i\|u-iv\|^2\right].$$

Question 4. Let V be an inner product space and let $W\leq V$ be finite-dimensional. Let $T:V\to W$ be the projection on W along $W^\perp$. Then $\forall v\in V$, $\forall w\in W$,

$$\|v-w\|\geq\|v-Tv\|,\qquad \|v-w\|=\|v-Tv\|\iff w=Tv.$$

Question 5. Riesz representation theorem for finite dimension case:

$$\forall f\in V^*,\ \exists \text{ unique } v \text{ such that } f(w)=\langle w,v\rangle.$$

In particular, when $\mathbb{F}=\mathbb{R}$, this map is an isomorphism

$$V\cong V^*.$$

Question 8. Let V be a finite-dimensional $\mathbb{F}$-vector space, $T:V\to V$. If

$$T=T^*\quad\text{(self-adjoint)},\qquad \langle v,Tv\rangle=0,\quad\forall v\in V,$$

then $T=0$. As a corollary, if

$$\|Tv\|=\|v\|,\quad\forall v\in V,$$

then $T^{-1}=T^*$, that is T is unitary.

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Motivation

Linear Algebra $\longrightarrow$ algebraic topology / Lie algebra are the only things we know how to work with/compute. We reduce everything to linear.

$$x+y=1\longrightarrow \underset{\text{Pythagorean triples}}{x^2+y^2=1}\longrightarrow \underset{\text{Fermat's Last Theorem}}{x^n+y^n=1}.$$

1 Vector Spaces

Let $\mathbb{F}$ be a field. For example, $\mathbb{R},\mathbb{C},\mathbb{Q}$,

$$\mathbb{F}_p:=\mathbb{Z}/p\mathbb{Z},\quad p\text{ prime},\quad \mathbb{Q}_p,\dots$$

Definition

A *vector space over* $\mathbb{F}$ ($\mathbb{F}$-vector space) is a nonempty set V with two operations

$$+ : V \times V \longrightarrow V, \quad (v, w) \longmapsto v + w,$$
$$\cdot : \mathbb{F} \times V \longrightarrow V, \quad (a, v) \longmapsto av,$$

such that $(V, +)$ is an abelian group, and

(1) $1v = v, \qquad 1 \in \mathbb{F},\ v \in V,$
(2) $(ab)v = a(bv), \qquad \forall a, b \in \mathbb{F},\ v \in V,$
(3) $(a + b)v = av + bv, \qquad \forall a, b \in \mathbb{F},\ v \in V,$
(4) $a(v + w) = av + aw, \qquad \forall a \in \mathbb{F},\ v, w \in V.$

Counterexample

Let $V = \mathbb{R}$, $\mathbb{F} = \mathbb{R}$, and define $a \cdot v := v$. Then all other axioms hold, yet

$$\underbrace{(1 + 1) \cdot 1}_{1} \neq \underbrace{1 \cdot 1 + 1 \cdot 1}_{2}.$$

We call the elements of V *vectors* and elements of $\mathbb{F}$ *scalars*.

Example

$$\mathbb{F}^n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in \mathbb{F}\},$$

with scalar multiplication and $+$ defined component-wise.

$$\mathbb{F}[x] = \{\text{polynomials over } \mathbb{F}\}, \qquad \mathbb{F}^{\mathbb{F}} = \{\text{functions } f : \mathbb{F} \to \mathbb{F}\}.$$

Claim

$\mathbb{F}^{\mathbb{F}}$ satisfies axiom ④ under natural addition and scalar multiplication.

Definition

Let V be a vector space over $\mathbb{F}$. A *subspace* of V is a nonempty subset $W \subset V$ such that

(1) $\forall u, w \in W, \quad u + w \in W,$
(2) $\forall w \in W,\ a \in \mathbb{F}, \quad aw \in W.$

Example

For any $a_{ij} \in \mathbb{F}$,

$$i \in \{1, \ldots, n\}, \qquad j \in \{1, \ldots, m\}, \qquad n, m \geq 1 \text{ integers},$$

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then

$$W = \left\{ (x_1, \dots, x_n) \in \mathbb{F}^n \;\middle|\; \begin{array}{r} a_{11}x_1 + \cdots + a_{1n}x_n = 0, \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = 0 \end{array} \right\}$$

is a subspace of $\mathbb{F}^n$.

Example

Let $n \in \mathbb{N}$. Then

$$\mathbb{F}[x]_n = \{f \in \mathbb{F}[x] \mid \deg f(x) \leq n\}, \qquad \deg(0) := -\infty,$$

is a subspace of $\mathbb{F}[x]$ with natural scalar multiplication and addition.

Example

$$V = \mathbb{R}^{\mathbb{R}}, \qquad W = C(\mathbb{R}), \qquad U = D_n(\mathbb{R}).$$

Here W consists of continuous functions and U of n-th differentiable functions. Both U and W are subspaces of V.

Proposition

Let U, W be subspaces of the vector space V. Then

$$U + W := \{u + w \mid u \in U,\ w \in W\}$$

and $U \cap W$ are subspaces of V.

Proof. Trivial from the definition. □

Proposition and Definition

If V, W are $\mathbb{F}$-vector spaces, then

$$V \oplus W = \{(v, w) \mid v \in V,\ w \in W\}.$$

Then $V \oplus W$, with natural component-wise operation, is an $\mathbb{F}$-vector space. We call it the *direct sum*.

Proof. Exercise. □

Remark.

1. $V \oplus W = V \times W$ as additive abelian groups.
2. $V_1 \oplus V_2 \oplus \cdots \oplus V_n = \bigoplus_i V_i$ can be defined inductively.
3. $\oplus, \times$ are defined differently for infinitely many vector spaces.

Proposition

$$(1)\ 0_{\mathbb{F}} \cdot v = 0_V, \qquad (2)\ (-1)_{\mathbb{F}} \cdot v = (-v)_V, \qquad (3)\ a \cdot 0_V = 0_V,$$

and

$$(4) \text{ if } W \text{ is a subspace, then } 0_V \in W.$$

Proof. For (1),

$$0_{\mathbb{F}} \cdot v = (0+0) \cdot v = 0 \cdot v + 0 \cdot v.$$

Add additive inverse:

$$0 = 0 \cdot v.$$

The others follow similarly. □

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1.1 Spanning Sets and Linear Independence

Let $S \subset V$ be a subset.

Definition

The *span* of S, denoted $\operatorname{Span}(S)$, is the subset of V of all finite linear combinations of elements:

$$\operatorname{Span}(S) = \{a_1 v_1 + \cdots + a_n v_n\}.$$

In particular,

$$\operatorname{Span}(\varnothing) = \{0\}.$$

Note

$$S \subset T \implies \operatorname{Span}(S) \subset \operatorname{Span}(T),$$

and

$$S \subset \operatorname{Span}(T) \implies \operatorname{Span}(S) \subset \operatorname{Span}(T).$$

Remark

$\operatorname{Span} S$ is the smallest subspace of V which contains S.

Example

$$S = \{(1,0,0), (0,1,0), (1,1,0)\}.$$

Then

$$\begin{aligned} \operatorname{Span} S &= \{a(1,0,0) + b(0,1,0) + c(1,1,0)\} \\ &= \{(a+c, b+c, 0)\} \\ &= \{(d,e,0)\} \\ &= \operatorname{Span}\{(1,0,0), (0,1,0)\}. \end{aligned}$$

Proposition

$\operatorname{Span} S$ is a subspace of V.

Proof. $S = \varnothing$, $\operatorname{Span} S = \{0\}$. Otherwise trivial. □

Definition

If $\operatorname{Span} S = V$, we call S a *spanning set*. If S spans V and no proper subset $T \subsetneq S$ spans V, S is called a *minimal spanning set of* V.

Definition

$$\mathrm{St} := \{e_1, e_2, \ldots, e_n\} \subset \mathbb{F}^n$$

is a minimal spanning set.

Definition

S is *linearly dependent* if

$$\exists \text{ distinct } v_1, \ldots, v_n \in S, \quad a_1, \ldots, a_n \in \mathbb{F}$$

such that $a_1, \ldots, a_n$ are not all zero and

$$a_1 v_1 + \cdots + a_n v_n = 0,$$

that is, this linear relation is nontrivial. If S is not linearly dependent, it is called *linearly independent*.

Note

This definition does not require S finite.

Remark

$\varnothing$ is independent, $\{0\}$ is dependent, and $S \neq \varnothing$ is linearly independent if and only if

$$a_1 v_1 + \cdots + a_n v_n = 0 \implies a_i = 0, \quad \forall i.$$

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Example

For $v \in V$,

$$S = \{v\} \text{ is linearly independent} \iff v \neq 0.$$

For $v, w \in V$,

$$S = \{v, w\} \text{ is linearly dependent} \iff v = \lambda w \text{ or } w = \lambda v \quad \text{for some } \lambda \in \mathbb{F}.$$

Example

$$\mathrm{St} = \{e_1, \ldots, e_n\}$$

is linearly independent in $\mathbb{R}^n$.

Example

$$S = \{x^n \mid n \in \mathbb{N}\}$$

is linearly independent in $\mathbb{F}[x]$. If

$$a_1 x^{i_1} + \cdots + a_n x^{i_n} = 0(x),$$

then $a_1 = \cdots = a_n = 0$; each coefficient must be 0.

Example

$$S = \{e^x, \sin x, \cos x\}$$

is linearly independent in $C(\mathbb{R})$. Suppose $a, b, c \in \mathbb{R}$ with

$$ae^x + b\sin x + c\cos x = 0$$

(the zero function). If $a \neq 0$, e^x is unbounded, and $\sin x, \cos x, 0$ bounded; hence $a = 0$. Thus

$$b\sin x + c\cos x = 0.$$

If $x = 0$, $c\cos 0 = 0$, so $c = 0$. Lastly plug in any x so that $\sin x \neq 0$, and $b = 0$.

For $D^{(n-1)}(\mathbb{R})$ functions, linear dependence implies that the Wronskian vanishes identically:

$$W(f_1, \ldots, f_n) = \det \begin{pmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{pmatrix} = 0(x).$$

The converse requires additional hypotheses, such as analyticity or a common linear ordinary differential equation with suitable assumptions.

Definition

We say $S \subset V$ is a *maximal linearly independent set* if it is linearly independent in V, and

$$\forall v \in V \setminus S, \qquad S \cup \{v\} \text{ is linearly dependent.}$$

Example

St $= \{e_1, \ldots, e_n\}$ is a maximal linearly independent set: for every $v = (a_1, \ldots, a_n) \in \mathbb{R}^n$,

$$a_1e_1 + \cdots + a_ne_n + (-1)v = 0.$$

Theorem

Let $S \subset V$. The following are equivalent:

1. S is a minimal spanning set.
2. S is a maximal linearly independent set.
3. Every $v \in V$ can be written as a unique finite linear combination of elements in S.

Proof.

$$(1) \implies (2) \implies (3) \implies (1).$$

We do a brief discussion.
(1) $\Rightarrow$ (2): if S is not independent, then it is not minimal. Maximal, as S spans V: $v \notin S$ can be written as a linear combination of S, so $S \cup \{v\}$ is linearly dependent.
(2) $\Rightarrow$ (3): maximality implies that a linear combination exists. If another exists, it must be the same by linear independence.

(3) $\Rightarrow$ (1): spanning is assumed. If not minimal, then uniqueness in (3) fails. $\square$

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1.2 Bases and Dimension

Bases

A subset $\mathcal{B} \subset V$ is a (Hamel) *basis* if it satisfies:

$$\text{minimal spanning} \iff \text{maximal linearly independent} \iff \text{unique linear combination.}$$

Example

$$\mathrm{St} = \{e_1, \ldots, e_n\} \subset \mathbb{F}^n$$

is a basis.

Example

$$\{x^n \mid n \in \mathbb{N}\} \subset \mathbb{F}[x]$$

is a basis.

Proposition

If $\mathcal{B} \subset V$ is a (linearly independent) spanning set, then it is a basis of V.

Proof. Suppose $\mathcal{B}$ is not a basis (not minimal spanning). Then $\exists C \subsetneq \mathcal{B}$ such that

$$\mathrm{Span}(C) = \mathrm{Span}(\mathcal{B}).$$

For every $v \in \mathcal{B} \setminus C$, $v \in \mathrm{Span}(C)$, so $\mathcal{B}$ is linearly dependent as we write out the linear relation. $\square$

Example

$$W = \{(a, b, c) \mid a + b + c = 0\} \leq \mathbb{F}^3.$$

Then

$$\begin{aligned} W &= \{(a, b, -a - b) \mid a, b \in \mathbb{F}\} \\ &= \{a(1, 0, -1) + b(0, 1, -1) \mid a, b \in \mathbb{F}\} \\ &= \mathrm{Span}\{(1, 0, -1), (0, 1, -1)\}. \end{aligned}$$

We also check this spanning set is linearly independent, thus a basis.

Theorem

Any vector space V has a basis. Any two bases for V have the same cardinality.

Definition

The *dimension* of a vector space is the cardinality of a basis.

Example

$$\dim \mathbb{F}^n = |\mathrm{St}| = n, \qquad \dim(\mathbb{F}[x]) = |\mathbb{N}| \ (= \aleph_0).$$

Remark

dim depends on the ambient field. For example,

$$\dim_{\mathbb{R}}(\mathbb{C}^n) = 2n, \qquad \dim_{\mathbb{C}}(\mathbb{C}^n) = n.$$

Cantor–Bernstein–Schroder Lemma

Given sets A, B, if

$$|A| \leq |B|, \qquad |B| \leq |A|,$$

then $|B| = |A|$.

Proof. Claim: for a monotone map

$$\varphi : \mathcal{P}(X) \longrightarrow \mathcal{P}(X),$$

that is

$$A \subset B \implies \varphi(A) \subset \varphi(B),$$

$\exists A \in \mathcal{P}(X)$ such that $\varphi(A) = A$.
Consider

$$\mathcal{S} := \{A \in \mathcal{P}(X) \mid A \subset \varphi(A)\}, \qquad A_0 := \bigcup_{A \in \mathcal{S}} A.$$

Then

$$\varphi(A_0) \supseteq \bigcup_{A \in \mathcal{S}} \varphi(A) \supseteq \bigcup_{A \in \mathcal{S}} A = A_0.$$

Moreover, by monotonicity,

$$\varphi(\varphi(A_0)) \supset \varphi(A_0) \implies \varphi(A_0) \in \mathcal{S} \implies \varphi(A_0) \subset A_0.$$

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Thus $\varphi(A_0) = A_0$.
Since $|A| \leq |B|$, $|B| \leq |A|$, there are injective maps

$$f : A \hookrightarrow B, \qquad g : B \hookrightarrow A.$$

Define

$$\begin{aligned} \varphi : \mathcal{P}(A) &\longrightarrow \mathcal{P}(A), \\ X &\longmapsto A \setminus (g(B \setminus f(X))). \end{aligned}$$

It is easy to see that this map is monotone. Thus $\exists A_0 \subset A$ such that

$$\varphi(A_0) = A \setminus (g(B \setminus f(A_0))) = A_0.$$

Then notice

$$g|_{B\setminus f(A_0)} : B \setminus f(A_0) \longrightarrow A \setminus A_0$$

is bijective. Define $F : A \to B$ as

$$F = \begin{cases} g^{-1} : A \setminus A_0 \longrightarrow B \setminus f(A_0), \\ f : A_0 \longrightarrow f(A_0). \end{cases}$$

Then it is trivial to see that F is bijective, so $|A| = |B|$. □

Lemma

Given $|\mathbb{N}| = \aleph_0$, $|A| = \infty$, we have

$$\underbrace{\aleph_0 \cdot |A|}_{|\mathbb{N}\times A|} = |A|.$$

Note

This is not a simple statement, and it requires some equivalence of the Axiom of Choice. We omit the proof here. (Set theory, Cardinal Arithmetic.)

Theorem

If finite $S \subset V$ spans V, then

$$\exists S' \subset S \text{ such that } S' \text{ is a basis of } V.$$

Proof. Induction on $|S|$. Trivial. □

Steinitz Exchange Lemma

If $S, L \subset V$, S, L finite, and S spans V, L is linearly independent, then

1. $|L| \leq |S|$;
2. $\exists S' \subset S$ such that

$$|S'| = |S| - |L| \qquad \text{and} \qquad S' \cup L \text{ spans } V.$$

Note

This can be proven in a completely different fashion. See Artin page 90.

Proof. By induction on $|L|$. When $|L| = 0$, $L = \varnothing$, $S' = S$.
Suppose it is true for m. Suppose $|L| = m + 1$, $|S| = n$. By hypothesis,

$$\exists \{v_1, \ldots, v_{n-m}\} \subset S$$

such that

$$\{v_1, \ldots, v_{n-m}, w_1, \ldots, w_m\}$$

spans V. If $n = m$, then

$$w_{m+1} \in \text{Span}\{w_1, \ldots, w_m\},$$

which contradicts the fact that L is linearly independent. Thus $m < n$, so

$$m + 1 = |L| \leq |S| = n.$$

This proves (1).
Since $\{v_1, \dots, v_{n-m}, w_1, \dots, w_m\}$ spans V, there are $a_1, \dots, a_{n-m}, b_1, \dots, b_m \in \mathbb{F}$

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such that

$$a_1 v_1 + \cdots + a_{n-m} v_{n-m} + b_1 w_1 + \cdots + b_m w_m = w_{m+1}.$$

Again, since $L = \{w_1, \dots, w_{m+1}\}$ is independent, at least one of $a_1, \dots, a_{n-m}$ is non-zero. Say it is a_i. Then it is trivial to see

$$\begin{aligned} &\operatorname{Span}\{v_1, \dots, v_{n-m}, w_1, \dots, w_m\} \\ &\quad = \operatorname{Span}\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}, w_1, \dots, w_m, w_{m+1}\}. \end{aligned}$$

Proved by induction. □

Definition

Let E be a set. A *partial order* $\preceq$ on E is a relation satisfying

$$\begin{aligned} &(1) \quad \forall e \in E, \quad e \preceq e, \\ &(2) \quad \forall e, f \in E, \quad e \preceq f,\ f \preceq e \implies e = f, \\ &(3) \quad \forall e, f, g \in E, \quad e \preceq f,\ f \preceq g \implies e \preceq g. \end{aligned}$$

An *upper bound* for $D \subset E$ is an element $e \in E$ where

$$d \preceq e, \qquad \forall d \in D.$$

A *maximal element* $e \in E$ satisfies that if

$$e \preceq f, \qquad f \in E,$$

then $f = e$.

Given a partially ordered set $(E, \preceq)$, a *chain* is a subset D on which the partial order becomes a total order, that is for every $d, e \in D$, we have

$$d \preceq e \quad \text{or} \quad e \preceq d.$$

(Note the upper bound is not necessarily in D.)

Zorn's Lemma

Let $(E, \preceq)$ be a partially ordered set. If every chain C has an upper bound, then E has a maximal element.

Proof of the main theorem. Claim: every vector space V has a basis.
Define

$$\mathcal{S} := \{\text{linearly independent subsets of } V\},$$

with partial order inclusion. Let $\mathcal{C} \subset \mathcal{S}$ be a chain. Consider

$$S := \bigcup_{X \in \mathcal{C}} X.$$

S is linearly independent. Suppose otherwise. Then there is a nontrivial linear relation in S,

$$a_1 v_1 + \cdots + a_n v_n = 0,$$

where

$$v_1 \in S_1 \in \mathcal{C}, \quad v_2 \in S_2 \in \mathcal{C}, \quad \ldots, \quad v_n \in S_n \in \mathcal{C}.$$

Since $\mathcal{C}$ is a chain, $\exists S_j \in \{S_1, \ldots, S_n\}$ such that

$$S_i \subset S_j, \qquad i = 1, \ldots, n.$$

Then from the nontrivial linear relation, S_j is linearly dependent, a contradiction since $\mathcal{C} \subset \mathcal{S}$ consists of only linearly independent subsets of V.

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Then clearly S is an upper bound of chain $\mathcal{C}$. Any chain in $\mathcal{S}$ has an upper bound, so by Zorn's lemma, $\mathcal{S}$ has a maximal element $\mathcal{B}$. We claim $\mathcal{B}$ is a basis for V.
We will take advantage of how we've defined a Hamel basis (minimal spanning set / maximal linearly independent set). [If we have defined it as an independent spanning set, we still need to show that it is maximally independent.]
$\mathcal{B} \in \mathcal{S}$, so $\mathcal{B}$ is independent. Let $v \in V \setminus \mathcal{B}$. Then $\mathcal{B} \cup \{v\}$ is linearly dependent; otherwise,

$$\mathcal{B} \cup \{v\} \in \mathcal{S}, \qquad \mathcal{B} \subsetneq \mathcal{B} \cup \{v\},$$

and $\mathcal{B}$ is not a maximal element. Thus $\mathcal{B}$ is maximal linearly independent, so $\mathcal{B}$ is a basis. □

Claim

If $\mathcal{B}_1, \mathcal{B}_2$ are bases of the vector space V and $|\mathcal{B}_1| < \infty$, then

$$|\mathcal{B}_1| = |\mathcal{B}_2|.$$

Proof. 1. $|\mathcal{B}_2| < \infty$. If $|\mathcal{B}_2| = \infty$, then any finite subset of $\mathcal{B}_2$ is linearly independent; this contradicts the first part of the Steinitz exchange lemma: $\mathcal{B}_1$ spans V, yet there is $S \subset \mathcal{B}_2$ with

$$|\mathcal{B}_1| < |S| < \infty, \qquad S \text{ independent.}$$

2. Now $|\mathcal{B}_2| < \infty$. Applying Steinitz to $\mathcal{B}_1$ (spanning) and $\mathcal{B}_2$ (independent), we have

$$|\mathcal{B}_2| \leq |\mathcal{B}_1|.$$

Then $\mathcal{B}_1$ independent and $\mathcal{B}_2$ spanning implies

$$|\mathcal{B}_1| \leq |\mathcal{B}_2|.$$

Therefore $|\mathcal{B}_1| = |\mathcal{B}_2|$. □

Claim

If $\mathcal{B}_\Lambda, \mathcal{B}_\Omega$ are bases of the vector space V, then

$$|\mathcal{B}_\Lambda| = |\mathcal{B}_\Omega|.$$

Proof. Since $\mathcal{B}_\Lambda$ spans $\mathcal{B}_\Omega \subset V$, for every $w_\omega \in \mathcal{B}_\Omega$, there is a finite subset $S_\omega \subset \mathcal{B}_\Lambda$ such that

$$w_\omega \in \operatorname{Span}(S_\omega).$$

Thus $\operatorname{Span}(\bigcup_{\omega\in\Omega} S_\omega)$ contains each $w_\omega \in \mathcal{B}_\Omega$, and

$$\begin{aligned}\operatorname{Span}(\mathcal{B}_\Lambda) = V = \operatorname{Span}(\mathcal{B}_\Omega) &\subseteq \operatorname{Span}\left(\operatorname{Span}\left(\bigcup_{\omega\in\Omega} S_\omega\right)\right)\\ &= \operatorname{Span}\left(\bigcup_{\omega\in\Omega} S_\omega\right).\end{aligned}$$

Moreover, $\mathcal{B}_\Lambda$ is a basis, so it is minimal spanning—no subset of $\mathcal{B}_\Lambda$ could span V. Since

$$S_\omega \subset \mathcal{B}_\Lambda, \qquad \forall \omega \in \Omega,$$

and both $\bigcup_{\omega\in\Omega} S_\omega$ and $\mathcal{B}_\Lambda$ span V,

$$\bigcup_{\omega\in\Omega} S_\omega = \mathcal{B}_\Lambda.$$

Therefore, by the lemma on page 6 and the fact that S_ω are finite subsets, we have

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$$\begin{aligned}|\mathcal{B}_\Lambda| = \left|\bigcup_{\omega\in\Omega} S_\omega\right| &\le \sum_{\omega\in\Omega} |S_\omega| \le \sum_{\omega\in\Omega} |\mathbb{N}|\\ &= |\mathcal{B}_\Omega|\,|\mathbb{N}| = |\mathcal{B}_\Omega|\aleph_0 = |\mathcal{B}_\Omega|.\end{aligned}$$

Symmetrically,

$$|\mathcal{B}_\Omega| \le |\mathcal{B}_\Lambda|.$$

Therefore, by Cantor–Schroeder–Bernstein,

$$|\mathcal{B}_\Omega| = |\mathcal{B}_\Lambda|.$$

□

Note

Here $\le, =$ both refer to cardinality: $\exists$ injective/bijective function (since we are looking at infinite cases).

Proposition

Let V be a finite-dimensional $\mathbb{F}$-vector space,

$$n = \dim_{\mathbb{F}} V.$$

Let $S \subseteq V$, $|S| = n$. The following are equivalent:

1. S spans V;
2. S is linearly independent;
3. S is a basis.

Proof. (3) $\Rightarrow$ (1) and (3) $\Rightarrow$ (2) by definition.
(1) $\Rightarrow$ (3):
[1] Let $B \subsetneq S$. Then

$$|B| < |S| = n,$$

and if B spanned V, the Steinitz exchange lemma applied to a basis of n vectors and the spanning set B would imply $n \leq |B|$, a contradiction. Thus B does not span V, so S is minimal spanning and hence is a basis.
[2] Suppose S is not a basis. Then it spans V, so S is linearly dependent. By the theorem on page 6, we can find a subset

$$S' \subsetneq S$$

by removing the redundant vectors; moreover, S' is a basis of V. Then

$$\dim V = |S'| < |S| = n,$$

a contradiction.
(2) $\Rightarrow$ (3):
[1] By Steinitz exchange lemma, since S is independent, for any basis B, $\exists B' \subset B$ such that $B' \cup S$ is a basis. Yet

$$|B'| = |B| - |S| = 0.$$

Thus

$$S \cup B' = S \cup \varnothing = S$$

is a basis.
[2] Suppose it is not a basis. Then it is independent and it does not span V. Thus

$$\exists v \in V \setminus \operatorname{Span}(S) \implies S \cup \{v\} \text{ is independent}$$

Then $\dim V > n$, a contradiction. □

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Proposition

If $\dim_{\mathbb{F}} V < \infty$, $W \subseteq V$ a subspace, then $\dim_{\mathbb{F}} W < \infty$, and

$$\dim W \leq \dim V.$$

Moreover,

$$\dim_{\mathbb{F}} W \leq \dim_{\mathbb{F}} V,$$

where

$$\dim_{\mathbb{F}} W = \dim_{\mathbb{F}} V \iff V = W.$$

Proof. Let S be a basis for V, L a basis for W.
[1] By applying the exchange lemma to any finite subset of L, we have

$$|L| \leq |S| < \infty$$

(otherwise a contradiction to the lemma). Then applying it to L itself and S, we have

$$\dim W = |L| \leq |S| = \dim V.$$

If $\dim W = \dim V$, then L is independent, so L is a basis of V. Hence

$$\operatorname{Span} L = W = V.$$

[2] Take an arbitrary linearly independent finite subset of W, say L. Then by adding vectors in W that are not in $\operatorname{Span}(L)$, we eventually get a basis of W. By Steinitz exchange lemma, this process

of adding vectors to L must come to an end. Thus L is finite. If $|L| = n$, then L becomes a basis of V. □

Remark

The previous two propositions fail in the infinite-dimensional case.

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Example

$$W = \{f(x) \in \mathbb{F}[x] \mid f(0) = 0\}$$

is a subspace of $\mathbb{F}[x]$, but its basis is

$$\{x^n : n \geq 1\},$$

which is cardinality $\aleph_0$, same as

$$\{x^n \mid n \in \mathbb{N}\}.$$

Proposition

For a finite-dimensional $\mathbb{F}$-vector space V, every independent subset L can be expanded to a basis of V.

Proof. By the Steinitz exchange lemma, for any basis $S \subset V$, $\exists S' \subset S$ such that

$$S' \cup L$$

is a basis of V. In practice, we add the vectors in $V \setminus \text{Span}(L)$ to L. When L has n vectors, by a previous proposition it becomes a basis of V. □

Example

$$W := \{(a, b, c) \in \mathbb{F}^3 \mid a + b + c = 0\}.$$

Recall we have found its basis

$$w_1 = (1, 0, -1), \qquad w_2 = (0, 1, -1).$$

For every $v \in \mathbb{F}^3 \setminus W$,

$$\{v, w_1, w_2\}$$

is a basis of $\mathbb{F}^3$.

Lagrange Polynomials

Let

$$\alpha_0, \alpha_1, \ldots, \alpha_n \in \mathbb{F}$$

be distinct. For $0 \leq i \leq n$, let

$$f_i(x) := \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - \alpha_j}{\alpha_i - \alpha_j} \in \mathbb{F}[x].$$

Thus

$$f_i(x) = \frac{(x-\alpha_0)(x-\alpha_1)\cdots(x-\alpha_{i-1})(x-\alpha_{i+1})\cdots(x-\alpha_n)}{(\alpha_i-\alpha_0)(\alpha_i-\alpha_1)\cdots(\alpha_i-\alpha_{i-1})(\alpha_i-\alpha_{i+1})\cdots(\alpha_i-\alpha_n)}.$$

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Note that

$$f_i(\alpha_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Claim

$\{f_0, f_1, \ldots, f_n\}$ are linearly independent.

Proof. If $\exists a_0, a_1, \ldots, a_n$ such that

$$a_0 f_0 + a_1 f_1 + \cdots + a_n f_n = 0(x),$$

then by evaluating both sides at α_i, $i = 0, 1, \ldots, n$, we get

$$a_0 = a_1 = \cdots = a_n = 0.$$

Thus $\{f_0, f_1, \ldots, f_n\}$ is linearly independent. Moreover, it is a basis of $\mathbb{F}[x]_n$ (degree $\leq n$ polynomials in $\mathbb{F}[x]$). Thus $\forall g(x) \in \mathbb{F}[x]_n$, we can write

$$g(x) = \sum_{i=0}^{n} a_i f_i(x)$$

uniquely. □

Proposition and Definition

Plug in $x = \alpha_i$, we get

$$g(\alpha_i) = a_i.$$

Thus

$$\boxed{g(x) = \sum_{i=0}^{n} g(\alpha_i) f_i(x)}.$$

This is called the *Lagrange Interpolation formula.*

Proposition

Conversely, since $\{f_i(x)\}_{i=0}^n$ is a basis for $\mathbb{F}[x]_n$,

$$g(x) = \sum_{i=0}^{n} b_i f_i(x)$$

is the unique polynomial with $\deg g \leq n$ such that

$$g(\alpha_i) = b_i, \qquad i = 0, 1, \ldots, n.$$

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Example

Find unique $g(x) \in \mathbb{R}[x]_2$ such that

$$g(1) = 8, \qquad g(2) = 5, \qquad g(3) = -4.$$

Then

$$\alpha_0 = 1, \qquad \alpha_1 = 2, \qquad \alpha_2 = 3.$$

We have

$$f_0(x) = \prod_{\substack{j=0 \\ j\neq 0}}^{2} \frac{x-\alpha_j}{\alpha_0-\alpha_j} = \frac{(x-2)(x-3)}{(-1)(-2)} = \frac{x^2-5x+6}{2},$$
$$f_1(x) = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -x^2+4x-3,$$
$$f_2(x) = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{x^2-3x+2}{2}.$$

Therefore

$$\begin{aligned} g(x) &= \sum_{i=0}^{2} b_i f_i(x) = 8f_0(x) + 5f_1(x) + (-4)f_2(x) \\ &= 4x^2 - 20x + 24 + (-5)x^2 + 20x - 15 - 2x^2 + 6x - 4 \\ &= -3x^2 + 6x + 5. \end{aligned}$$

Definition

Given an $\mathbb{F}$-vector space V of dimension n, with basis

$$\mathcal{B} = \{v_1, v_2, \ldots, v_n\},$$

then any $v \in V$ can be uniquely (a previous proposition) expressed

$$v = \sum_{i=1}^{n} a_i v_i.$$

The n-tuple

$$[v]_{\mathcal{B}} := \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

is called the $\mathcal{B}$-coordinate of v.

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1.3 Coordinates and Ordered Bases

Notation

We sometimes write n-tuples of $\mathbb{F}$ as row vectors, but when multiplying with matrices later, it is by convention multiplying with the column vector.

Remark

The coordinate depends on the order and choice of basis $\mathcal{B}$.

Example

$$g(x) = -3x^2 + 6x + 5 \in \mathbb{F}[x]_2.$$

Under

$$\mathrm{St} := \{1, x, x^2\},$$
$$[g]_{\mathrm{St}} = \begin{pmatrix} 5 \\ 6 \\ -3 \end{pmatrix}.$$

Under

$$\mathcal{B} = \{f_0, f_1, f_2\},$$
$$[g]_{\mathcal{B}} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 8 \\ 5 \\ -4 \end{pmatrix}.$$

Under

$$\mathrm{St}' = \{1, x^2, x\},$$
$$[g]_{\mathrm{St}'} = \begin{pmatrix} 5 \\ -3 \\ 6 \end{pmatrix}.$$

2 Linear Transformations

Let $\mathbb{F}$ be a field, V, W be $\mathbb{F}$-vector spaces.

Definition

A *linear transformation* from V to W is a map

$$T : V \to W$$

such that

(1) $T(v_1 + v_2) = Tv_1 + Tv_2, \qquad \forall v_1, v_2 \in V,$
(2) $T(\lambda v_1) = \lambda T v_1, \qquad \forall \lambda \in \mathbb{F}.$

Remark

By (1), T is a group homomorphism from $(V, +_V)$ to $(W, +_W)$. That leads to

$$(1)\ T(0_V) = 0_W, \qquad (2)\ T(-v) = -Tv.$$

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Proposition and Definition

$$\ker T := \{v \in V \mid Tv = 0_W\},$$
$$\operatorname{im} T = T(V) := \{Tv \mid v \in V\}.$$

Moreover,

$$T \text{ is injective} \iff \ker T = \{0\},$$
$$T \text{ is surjective} \iff \operatorname{im} T = W.$$

Proposition

Let $T : V \to W$ be a linear transformation. Then

$$\ker T \leq V, \qquad \operatorname{im} T \leq W$$

(subspaces).

Example

Zero map:

$$T : V \to W, \qquad v \longmapsto 0_W, \quad \forall v \in V.$$

Example

Identity map:

$$T : V \to V, \qquad v \longmapsto v, \quad \forall v \in V.$$

Example

Let

$$T : \mathbb{R}[x] \longrightarrow \mathbb{R}[x], \qquad T(f) := f'.$$

Verifying T is linear is trivial.

$$\ker T = \{\text{constant polynomial}\},$$
$$\operatorname{im} T = \mathbb{R}[x] \qquad \text{(why?)}.$$

Also,

$$\ker(T|_{\mathbb{R}[x]_n}) = \{\text{constant polynomial}\},$$
$$\operatorname{im}(T|_{\mathbb{R}[x]_n}) = \mathbb{R}[x]_{n-1} \qquad \text{(why?)}.$$

Example

Let

$$M_{m\times n}(\mathbb{F}) = \{m \times n \text{ matrices with entries in } \mathbb{F}\}.$$

Let $A \in M_{m\times n}(\mathbb{F})$, define

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$$T : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \qquad Tv = Av.$$

Then T is a linear transformation.

Conversely, Proposition Let

$$T : \mathbb{F}^n \longrightarrow \mathbb{F}^m$$

be a linear transformation. Then $\exists$ a unique matrix

$$A \in M_{m\times n}(\mathbb{F})$$

such that

$$\forall v \in \mathbb{F}^n, \qquad Tv = Av.$$

Proof.

$$Te_i = \begin{pmatrix} a_{1i} \\ \vdots \\ a_{mi} \end{pmatrix} \in \mathbb{F}^m.$$

Write

$$A := \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

Then $Tv = Av$ follows by definition, and uniqueness follows by multiplying A to the e_i's. □

Note

We can generalize this to arbitrary vector spaces V and W; then we have A being the matrix representation of T relative to the chosen bases. The term *change-of-basis matrix* is reserved for the matrix of the identity map between two coordinate systems on the same vector space.

Proposition

Let V, W be $\mathbb{F}$-vector spaces, $\dim_{\mathbb{F}} V < \infty$. Let

$$\mathcal{B} = \{v_1, \ldots, v_n\}$$

be a basis of V, and let

$$\{w_1, \ldots, w_n\} \subseteq W.$$

Then $\exists!$ linear transformation $T : V \to W$ such that

$$T(v_i) = w_i.$$

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Remark

This basically says that the linear transformation is always uniquely determined by its effect on a set of basis vectors.

Proof. Let $v \in V$. Since $\mathcal{B}$ is a basis,

$$v = \sum_{i=1}^{n} a_i v_i.$$

Define

$$Tv := \sum_{i=1}^{n} a_i w_i.$$

Then other verifications become trivial. □

Proposition

Let U, V, W be $\mathbb{F}$-vector spaces. Let

$$S : U \to V, \qquad T : V \to W$$

be linear transformations. Then $T \circ S$ is a linear transformation from $U \to W$.

Proof. Trivial. □

Proposition

Let

$$S : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \qquad T : \mathbb{F}^m \longrightarrow \mathbb{F}^k$$

be linear transformations. Moreover, let matrix of S be B, matrix of T be A. Then matrix of $T \circ S$ is AB.

Proof. Say

$$C = AB, \qquad C_{k\times n}, \quad A_{k\times m}, \quad B_{m\times n}.$$

Then

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$$\begin{aligned} i\text{-th column of } C &= i\text{-th column of } AB \\ &= A \begin{pmatrix} b_{1i} \\ \vdots \\ b_{mi} \end{pmatrix} \\ &= T \begin{pmatrix} b_{1i} \\ \vdots \\ b_{mi} \end{pmatrix} \\ &= T(S(e_i)) = (T \circ S)(e_i), \end{aligned}$$

where the column displayed is the i-th column of B, namely Be_i. Thus uniqueness follows from the second-last proposition. □

Definition and Proposition

If the set of positive integers n such that

$$\underbrace{1_{\mathbb{F}} + 1_{\mathbb{F}} + \cdots + 1_{\mathbb{F}}}_{n} = 0_{\mathbb{F}}$$

is nonempty, its least element is called the *characteristic* of $\mathbb{F}$, denoted $\operatorname{char}(\mathbb{F})$. This least element is prime. If no such positive integer exists, then $\operatorname{char}(\mathbb{F}) = 0$.

Proof. Let n be the least positive integer with $n1_{\mathbb{F}} = 0_{\mathbb{F}}$. If $n = pq$ with $1 < p, q < n$, then

$$(p1_{\mathbb{F}})(q1_{\mathbb{F}}) = n1_{\mathbb{F}} = 0_{\mathbb{F}}.$$

Since a field has no zero divisors, $p1_{\mathbb{F}} = 0_{\mathbb{F}}$ or $q1_{\mathbb{F}} = 0_{\mathbb{F}}$, contradicting the minimality of n. Therefore n is prime. □

Proposition

Let V, W be $\mathbb{F}$-vector spaces. Denote

$$\operatorname{Hom}(V, W) := \{T : V \to W \mid T \text{ is linear}\}.$$

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Let the addition and scalar multiplication be the natural ones with functions. Then $\operatorname{Hom}(V, W)$ becomes an $\mathbb{F}$-vector space. Moreover,

$$\operatorname{End}(V) := \operatorname{Hom}(V, V)$$

is a ring under addition and composition.

Proof. We do one of the axioms. The others are left as exercises.
Distributivity over the vectors: let $\lambda \in \mathbb{F}$, $T_1, T_2 \in \operatorname{Hom}(V, W)$. Then $\forall v \in V$,

$$\begin{aligned}(\lambda(T_1 + T_2))(v) &= \lambda((T_1 + T_2)(v)) \\ &= \lambda(T_1 v + T_2 v) \\ &= \lambda(T_1 v) + \lambda(T_2 v) \\ &= (\lambda T_1)v + (\lambda T_2)v.\end{aligned}$$

Hence distributivity holds.
Distributivity as a ring.
Remark. $\operatorname{End}(V)$ is noncommutative in general; for example, this holds when $\dim V \geq 2$.
We verify the left distributivity. Let

$$T_1, T_2, T_3 \in \operatorname{End}(V).$$

Let $v \in V$. Then

$$\begin{aligned}(T_1 \circ (T_2 + T_3))v &= T_1[(T_2 + T_3)v] \\ &= T_1[T_2 v + T_3 v] \\ &= T_1(T_2 v) + T_1(T_3 v)\end{aligned}$$

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$$= (T_1 \circ T_2)v + (T_1 \circ T_3)v.$$

The other axioms are left as exercises. □

Definition

Let V, W be $\mathbb{F}$-vector spaces. A bijective linear transformation is called an *isomorphism* between V and W. We say V and W are isomorphic if $\exists$ isomorphism

$$T : V \to W.$$

We denote

$$V \cong W.$$

Proposition

If $T : V \to W$ is an isomorphism, then

$$T^{-1} : W \to V$$

is an isomorphism.

Proof. Of course inverse of bijective functions are bijective. We show T^{-1} is linear. Let $w_1, w_2 \in W$, $\lambda \in \mathbb{F}$. Since bijective, $\exists v_1, v_2 \in V$ such that

$$Tv_1 = w_1, \qquad Tv_2 = w_2.$$

Then

$$\begin{aligned} T^{-1}(w_1 + \lambda w_2) &= v_1 + \lambda v_2 && \text{(linearity of } T\text{)} \\ &= T^{-1}(w_1) + \lambda T^{-1}(w_2). \end{aligned}$$

□

Proposition

Let V, W be isomorphic $\mathbb{F}$-vector spaces. Then

$$\dim(V) = \dim(W).$$

Proof. We will do the argument with ∞-

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dimensional vector spaces, finite-dimensional cases are *relatively* more trivial.

Suppose $\dim V = \infty$, $\dim W = \infty$. Then for any basis $\mathcal{B}$ of V, we claim:

$$T(\mathcal{B}) \text{ is a basis of } W.$$

Let $\mathcal{B} := \{v_\alpha\}_{\alpha \in A}$. $T(\mathcal{B})$ is linearly independent:

Suppose not. Then $\exists a_1, \ldots, a_n \in \mathbb{F}$, $\alpha_1, \alpha_2, \ldots, \alpha_n \in A$ such that

$$a_1 T(v_{\alpha_1}) + a_2 T(v_{\alpha_2}) + \cdots + a_n T(v_{\alpha_n}) = 0$$

and $a_1, a_2, \ldots, a_n$ are not all zero. Then

$$T(a_1 v_{\alpha_1} + \cdots + a_n v_{\alpha_n}) = 0 \quad \Longrightarrow \quad a_1 v_{\alpha_1} + \cdots + a_n v_{\alpha_n} = 0 \quad \text{(by injectivity)}.$$

This contradicts independence of $\mathcal{B}$, as it has a linearly dependent subset.

Similarly, $\forall w \in W$, $\exists v \in V$ such that $Tv = w$, $\exists \alpha_1, \dots, \alpha_m \in A$ such that

$$v = \sum_{i=1}^{m} a_i v_{\alpha_i} \quad \Longrightarrow \quad w = \sum_{i=1}^{m} a_i T(v_{\alpha_i}).$$

Thus $T(\mathcal{B})$ is a basis of W, so

$$\dim V = |\mathcal{B}| = |T(\mathcal{B})| = \dim W. \qquad \square$$

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Remark. As a consequence,

$$V \cong W \quad \Longrightarrow \quad \dim V < \infty \quad \Longleftrightarrow \quad \dim W < \infty.$$

×

Theorem 2.1. *Let V be a finite-dimensional vector space over $\mathbb{F}$, and let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis of V. Then*

$$T : V \longrightarrow \mathbb{F}^n, \qquad v \longmapsto [v]_{\mathcal{B}} = \textit{coordinates of } v \textit{ with respect to } \mathcal{B}$$

is an isomorphism. Thus $\mathbb{F}^n \cong V$.

Proof. Only injectivity is slightly nontrivial. We claim $T^{-1}(\{0\}) = \{0\}$. Suppose

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Since $\mathcal{B}$ is a basis, it follows that

$$a_1 = a_2 = \cdots = a_n = 0.$$

Thus $\sum_{i=1}^{n} a_i v_i = 0$. $\square$

Corollary 2.2. *Let V, W be $\mathbb{F}$-vector spaces, $\dim V, \dim W < \infty$. Then*

$$V \cong W \quad \Longleftrightarrow \quad \dim V = \dim W.$$

Proof. ($\Rightarrow$) Last proposition.
($\Leftarrow$) $V \cong \mathbb{F}^n \cong W$. $\square$

Theorem 2.3 (First Isomorphism Theorem). SOURCE PAGE 28

Let $T : V \to W$ be a linear transformation between $\mathbb{F}$-vector spaces. Then

$$V/\ker T \cong \operatorname{Im} T.$$

Proof. Define the natural linear map

$$S : V/\ker T \longrightarrow \operatorname{Im} T, \qquad v + \ker T \longmapsto Tv.$$

It is well-defined: Let v, w such that

$$v + \ker T = w + \ker T \quad \Longrightarrow \quad v - w \in \ker T.$$

Thus

$$T(v-w) = 0 \quad \Longrightarrow \quad \underbrace{Tv}_{S(v+\ker T)} = \underbrace{Tw}_{S(w+\ker T)}.$$

Linearity: $\forall v, w \in V$,

$$\begin{aligned} S[(v+\ker T) + (w+\ker T)] &= S[(v+w)+\ker T] \\ &= T(v+w) = Tv + Tw \\ &= S(v+\ker T) + S(w+\ker T). \end{aligned}$$

Scalar multiplication is similar.
Injectivity:

$$S(v+\ker T) = 0 \quad \Longleftrightarrow \quad Tv = 0 \quad \Longleftrightarrow \quad v + \ker T = \ker T.$$

Surjectivity: Tautology. □

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Theorem 2.4 (The dimension theorem). *Let $T : V \to W$ be a linear transformation, where $\dim_{\mathbb{F}} V < \infty$. Then $\dim(\operatorname{Im} T) < \infty$ and*

$$\dim V = \dim \operatorname{Im} T + \dim \ker T.$$

Proof. Recall that we proved in Homework 2, Question 8:

$$\forall W \leq V, \qquad \dim V = \dim W + \dim V/W.$$

Then

$$\dim V = \dim \ker T + \dim(V/\ker T) = \dim \ker T + \dim(\operatorname{Im} T).$$

□

Corollary 2.5. *Let $T : V \to W$ be a linear transformation between finite-dimensional $\mathbb{F}$-vector spaces with $\dim V = \dim W$. Then the following are equivalent:*

① *T is injective;*
② *T is surjective;*
③ *T is an isomorphism.*

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Proof. Apply the dimension formula to

$$\begin{cases} \ker T, & \text{if injective;} \\ \operatorname{Im} T, & \text{if surjective,} \end{cases}$$

and conclude the dimension of the other. □

2.1 Matrix Representations

Theorem 2.6. *Let V and W be finite-dimensional vector spaces. Let $\mathcal{B}$ be an ordered basis of V, and $\mathcal{C}$ be an ordered basis of W. Let*

$$n = \dim V, \qquad m = \dim W.$$

① *$\exists!$ $m \times n$ matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ such that for every $v \in V$,*

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[v]_{\mathcal{B}}.$$

② *The map*

$$\mathrm{Hom}(V, W) \longrightarrow M_{m\times n}(\mathbb{F}), \qquad T \longmapsto [T]_{\mathcal{B}}^{\mathcal{C}}$$

is an isomorphism.

③ *If U is an $\mathbb{F}$-vector space with basis $\mathcal{D}$, and $R : W \to U$ is a linear transformation, then*

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$$[R \circ T]_{\mathcal{B}}^{\mathcal{D}} = [R]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}.$$

Proof. ① Write

$$\mathcal{B} = \{v_1, \ldots, v_n\}, \qquad \mathcal{C} = \{w_1, \ldots, w_m\}.$$

Remark. This is exactly how to compute the matrix of a linear map.

Let $A = (a_{ij}) \in M_{m\times n}(\mathbb{F})$ be defined by

$$[T(v_j)]_{\mathcal{C}} = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix} \in \mathbb{F}^m, \qquad j = 1, 2, \ldots, n.$$

Then $\forall v \in V$, if

$$[v]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n,$$

we have

$$\begin{aligned} A[v]_{\mathcal{B}} = A \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} &= \sum_{i=1}^{n} [T(v_i)]_{\mathcal{C}} \alpha_i \\ &= \sum_{i=1}^{n} [\alpha_i T(v_i)]_{\mathcal{C}} && \text{(the coordinate vector represents } v\text{)} \\ &= \left[T\left(\sum_{i=1}^{n} \alpha_i v_i \right) \right]_{\mathcal{C}} \end{aligned}$$

$$= [T(v)]_{\mathcal{C}}.$$

By multiplying T with standard basis vectors in $\mathbb{F}^n$,

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we can see the choice of A is unique:

$$i\text{th column of } [T]_{\mathcal{B}}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = [T]_{\mathcal{B}}^{\mathcal{C}}[v_i]_{\mathcal{B}} = [Tv_i]_{\mathcal{C}}.$$

Remark 1. Again, this matrix is **DEPENDENT** on the order of basis. That is the reason we emphasize the bases are ordered.

2. A compact way of writing this is

$$(Tv_1, \ldots, Tv_n) = (w_1, \ldots, w_m) \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

whose columns are, respectively, $[Tv_1]_{\mathcal{C}}, [Tv_2]_{\mathcal{C}}, \ldots, [Tv_n]_{\mathcal{C}}$.

Moreover, for any $v \in V$, we have

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$$\underbrace{(Tv_1, \ldots, Tv_n)[v]_{\mathcal{B}}}_{Tv} = \underbrace{(w_1, \ldots, w_m)}_{\mathcal{C}} \underbrace{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}}_{[Tv]_{\mathcal{C}}} [v]_{\mathcal{B}}.$$

② A direct verification is doable. Yet note that suppose

$$S_1 : \mathbb{F}^n \xrightarrow{\sim} V, \qquad S_2 : W \xrightarrow{\sim} \mathbb{F}^m$$

are the isomorphisms, then the map defined by

$$\varphi : \mathrm{Hom}(V, W) \longrightarrow \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m), \qquad T \longmapsto S_2 \circ T \circ S_1$$

is well-defined and in fact an $\mathbb{F}$-vector space isomorphism. Then recall before we have proven:

1. $\forall T \in \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m)$, $\exists! A \in M_{m\times n}(\mathbb{F})$ such that

$$Tv = Av.$$

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2. The map

$$T : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \qquad v \longmapsto Av$$

is a linear map.

Thus the map

$$\psi : \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m) \longrightarrow M_{m\times n}(\mathbb{F}), \qquad T \longmapsto A$$

is injective by ①, surjective by ②, and trivially linear (by how this A was found when we prove ①). Thus

$$\psi : \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m) \xrightarrow{\sim} M_{m\times n}(\mathbb{F}).$$

Therefore

$$\psi \circ \varphi : \mathrm{Hom}(V, W) \xrightarrow{\sim} \mathrm{Hom}(\mathbb{F}^n, \mathbb{F}^m) \xrightarrow{\sim} M_{m\times n}(\mathbb{F}).$$

③ To show this, we observe that with the isomorphism

$$S_1 : \mathbb{F}^n \to V, \qquad S_2 : W \to \mathbb{F}^m,$$

which maps

$$e_i \longmapsto v_i, \qquad w_j \longmapsto \varepsilon_j,$$

T and $S_2 \circ T \circ S_1$ have the same matrix representation in $M_{m\times n}(\mathbb{F})$:

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$$i\text{th column of } [T]_{\mathcal{B}}^{\mathcal{C}} = [Tv_i]_{\mathcal{C}},$$

whereas

$$i\text{th column of } [S_2 \circ T \circ S_1]_{\mathrm{std}_n}^{\mathrm{std}_m} = [S_2 \circ T \circ S_1(e_i)]_{\mathrm{std}_m}.$$

Here std_m denotes the standard basis of $\mathbb{F}^m$. Suppose

$$[Tv_i]_{\mathcal{C}} = \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}.$$

Then

$$Tv_i = a_1 w_1 + \cdots + a_m w_m.$$

We show this is also the coordinate vector of $S_2 \circ T \circ S_1(e_i)$:

$$\begin{aligned} S_2 \circ T \circ S_1(e_i) &= S_2 \circ T(v_i) \\ &= S_2(a_1 w_1 + a_2 w_2 + \cdots + a_m w_m) \\ &= a_1 S_2(w_1) + a_2 S_2(w_2) + \cdots + a_m S_2(w_m) \\ &= a_1 \varepsilon_1 + a_2 \varepsilon_2 + \cdots + a_m \varepsilon_m, \end{aligned}$$

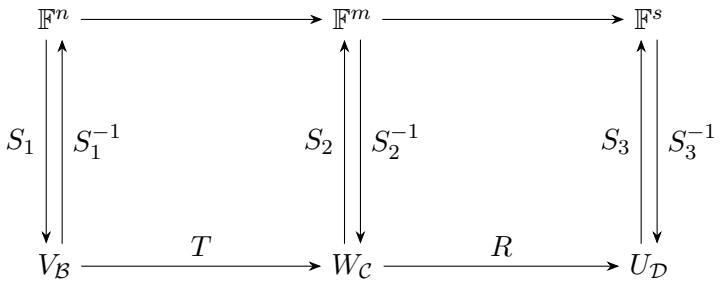
which is its std_m coordinate vector. Thus

$$i\text{th column of } [T]_{\mathcal{B}}^{\mathcal{C}} = i\text{th column of } [S_2 \circ T \circ S_1]_{\mathrm{std}_n}^{\mathrm{std}_m}.$$

Then notice we have proven ③ with $\mathbb{F}^n \to \mathbb{F}^m \to \mathbb{F}^s$. Thus, transferring everything into $\mathbb{F}^?$, we have the schematic

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diagram of linear transformations/isomorphisms:



Consequently,

$$\begin{aligned}[R\circ T]_{\mathcal{B}}^{\mathcal{D}} &= [S_3\circ(R\circ T)\circ S_1]_{\text{std}_n}^{\text{std}_s}\\ &= [(S_3\circ R\circ S_2^{-1})\circ(S_2\circ T\circ S_1)]_n^s\\ &= [S_3\circ R\circ S_2^{-1}]_m^s[S_2\circ T\circ S_1]_n^m\\ &= [R]_{\mathcal{C}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{C}}.\end{aligned}$$

□

Corollary 2.7. *Let V, W be finite-dimensional $\mathbb{F}$-vector spaces. Then*

$$\dim(\text{Hom}(V,W)) = (\dim V)(\dim W).$$

Proof. From part ② of the last theorem, we know

$$\varphi : \text{Hom}(V,W) \longrightarrow M_{m\times n}(\mathbb{F}), \qquad T \longmapsto [T]_{\mathcal{B}}^{\mathcal{C}}$$

is an

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vector space isomorphism. By another proposition we have proven, it follows that

$$\dim(\text{Hom}(V,W)) = (\dim V)(\dim W).$$

□

2.2 Projections and Computational Examples

Example 2.8. Consider

$$T : \mathbb{R}[x]_n \longrightarrow \mathbb{R}[x]_n, \qquad Tf = f'.$$

Let $\mathcal{B} = \{1, x, \ldots, x^n\}$. Then

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix}[T(1)]_{\mathcal{B}} & [T(x)]_{\mathcal{B}} & \cdots & [T(x^n)]_{\mathcal{B}}\end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 2 & 0 & \cdots & 0 \\ \vdots & \vdots & 0 & 3 & & \vdots \\ \vdots & \vdots & & 0 & \ddots & 0 \\ \vdots & \vdots & & & 0 & n \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Question. Given two bases $\mathcal{B}, \mathcal{C}$ of the same finite-dimensional $\mathbb{F}$-vector space V, how are $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{C}}^{\mathcal{C}}$ related for $T \in \mathrm{Hom}(V, V)$?

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Proof/Answer. By part ③ of the last theorem,

$$\begin{aligned} [T]_{\mathcal{C}}^{\mathcal{C}} &= [\mathrm{Id} \circ T \circ \mathrm{Id}]_{\mathcal{C}}^{\mathcal{C}} \\ &= [\mathrm{Id} \circ T]_{\mathcal{B}}^{\mathcal{C}} [\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}} \\ &= [\mathrm{Id}]_{\mathcal{B}}^{\mathcal{C}} [T]_{\mathcal{B}}^{\mathcal{B}} [\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}}. \end{aligned}$$

Furthermore, by ③ again,

$$[\mathrm{Id}]_{\mathcal{B}}^{\mathcal{C}} [\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{Id}]_{\mathcal{C}}^{\mathcal{C}} = I_n,$$

so

$$[\mathrm{Id}]_{\mathcal{B}}^{\mathcal{C}} = \left([\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}} \right)^{-1}.$$

Definition and Proposition

Let

$$[\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}} = S.$$

Then S is also the change of basis matrix:

① $$[T]_{\mathcal{C}}^{\mathcal{C}} = S^{-1} [T]_{\mathcal{B}}^{\mathcal{B}} S \qquad \text{(shown)}.$$

②

$$\begin{aligned} \mathcal{C} = \mathcal{B}S &:= \mathcal{B}[\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}} \\ &= \mathcal{B}[\mathrm{Id}(\mathcal{C})]_{\mathcal{B}} = \mathrm{Id}(\mathcal{C}) = \mathcal{C}. \end{aligned}$$

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[By $\mathcal{B}, \mathcal{C}$, we meant a hypervector:

$$T(v_1, \ldots, v_n) = (Tv_1, \ldots, Tv_n),$$

and

$$(v_1, \ldots, v_n) \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nm} \end{pmatrix}$$
$$= (a_{11}v_1 + a_{21}v_2 + \cdots + a_{n1}v_n, \ldots, a_{1m}v_1 + a_{2m}v_2 + \cdots + a_{nm}v_n).]$$

③

$$[\mathrm{Id}]_{\mathcal{C}}^{\mathcal{B}}[v]_{\mathcal{C}} = [\mathrm{Id}\, v]_{\mathcal{B}} = [v]_{\mathcal{B}}.$$

Note

How we calculate the change of basis matrix is exactly as usual:

$$\begin{aligned}[\mathrm{Id}]_{\mathcal{B}}^{\mathcal{C}} &= [\mathrm{Id}(\mathcal{B})]_{\mathcal{C}} \\ &= ([\mathrm{Id}\, w_1]_{\mathcal{C}}\ [\mathrm{Id}\, w_2]_{\mathcal{C}}\ \cdots\ [\mathrm{Id}\, w_n]_{\mathcal{C}}),\end{aligned}$$

where $\mathcal{B} = (w_1, w_2, \ldots, w_n)$.

Example 2.9. Consider

$$W = \{(x, y, z) \mid x + z = 0\}, \qquad U = \mathrm{Span}\{(1, 0, 1)\}.$$

It is easy to see

$$U + W = \mathbb{R}^3, \qquad U \cap W = \{0\}.$$

By Assignment 3, Question 3,

$$\mathbb{R}^3 = U \oplus W,$$

so $\exists$

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$$T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

projection given by

$$\underbrace{u}_{\in U} + \underbrace{w}_{\in W} = v \quad (u, w \text{ are unique by internal direct sum}), \qquad T(v) := w.$$

This is in particular called *projection on W along U*.
The plane W is spanned by

$$\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

We can check

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is a basis of $\mathbb{R}^3$. Then matrix representation:

$$T \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

$$T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

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$$T\underbrace{\begin{pmatrix}1\\0\\1\end{pmatrix}}_{\in U} = T\left(\underbrace{\begin{pmatrix}1\\0\\1\end{pmatrix}}_{\in U} + \underbrace{0}_{\in W}\right) = 0.$$

Thus

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix}1&0&0\\0&1&0\\0&0&0\end{pmatrix}.$$

What about $[T]_{\mathrm{St}}^{\mathrm{St}}$? It would be one way to compute the coordinates one by one. On the other hand, we have

$$[T]_{\mathrm{St}}^{\mathrm{St}} = [\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}}[T]_{\mathcal{B}}^{\mathcal{B}}[\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}.$$

[Trick to memorize:

$$[T \circ S]_{\mathcal{B}}^{\mathcal{D}} = [T]_{\mathcal{C}}^{\mathcal{D}}[S]_{\mathcal{B}}^{\mathcal{C}}.$$

]

One of $[\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}}$, $[\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}$ is easy

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to find:

$$\begin{aligned} Q := [\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}} &= [\mathrm{Id}(\mathcal{B})]_{\mathrm{St}} = [\mathcal{B}]_{\mathrm{St}} \\ &= \begin{pmatrix}1&0&1\\0&1&0\\-1&0&1\end{pmatrix}. \end{aligned}$$

Then by Gauss–Jordan elimination we can find

$$Q^{-1} = [\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}.$$

×

Example 2.10 (Computing high powers of matrices). Let

$$A = \begin{pmatrix}1&2&-2\\0&-1&0\\1&1&4\end{pmatrix}.$$

Consider the basis

$$\mathcal{B} = \left\{\begin{pmatrix}2\\0\\-1\end{pmatrix}, \begin{pmatrix}-1\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\1\end{pmatrix}\right\},$$

and

$$L_A : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad L_A(v) = Av.$$

Then the matrix representation of L_A with respect to $\mathcal{B}$ is

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$$D := [L_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Moreover, notice that

$$A = [L_A]_{\mathrm{St}}^{\mathrm{St}} \qquad \text{trivially.}$$

We also compute

$$Q := [\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}} = \begin{pmatrix} 2 & -1 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix},$$

and by either the Gauss–Jordan algorithm or the adjugate formula,

$$Q^{-1} = [\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}.$$

Therefore

$$\begin{aligned} A^n &= \left([L_A]_{\mathrm{St}}^{\mathrm{St}}\right)^n = (QDQ^{-1})^n \\ &= QD^nQ^{-1}. \end{aligned}$$

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$$A^n = \begin{pmatrix} 2 & -1 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2^n & 0 & 0 \\ 0 & (-1)^n & 0 \\ 0 & 0 & 3^n \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}.$$

×

3 Determinants

Objective

A function

$$\det : M_{n\times n}(\mathbb{F}) \longrightarrow \mathbb{F}$$

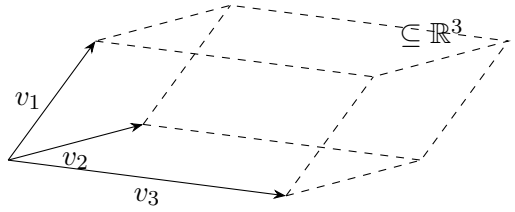
measures how much multiplication by $A \in M_{n\times n}(\mathbb{F})$.

Furthermore, if $\mathbb{F} = \mathbb{R}$ and

$$A = (v_1, \ldots, v_n) \in M_n(\mathbb{R})$$

has column vectors v_i, then $|\det A|$ will be the volume of the parallelepiped

$$P := \{a_1v_1 + a_2v_2 + \cdots + a_nv_n | a_1, \ldots, a_n \in [0, 1]\}.$$

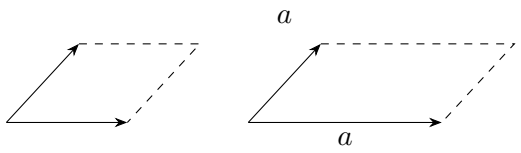


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Question. Desired properties? (Example in $\mathbb{R}^2$.)

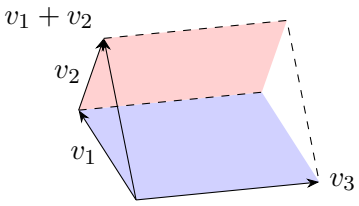
① Rescaling one vector by a rescales det by a:

$$\det(v_1, \ldots, av_i, \ldots, v_n) = a \det(v_1, \ldots, v_i, \ldots, v_n).$$



② Additive in any column:

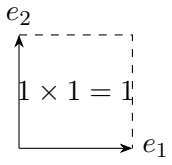
$$\det(v_1 + v_2, v_3) = A_{\text{red}} = A_{\text{blue}} + A_{\text{black}} \qquad \text{(why?)}$$



③ Two identical columns $\Longrightarrow \det = 0$:

$$\longrightarrow \qquad A = 0.$$

④ $\det(I_n) = 1$:



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Theorem 3.1. *Let $\mathbb{F}$ be a field. $\exists!$ function*

$$\det : M_n(\mathbb{F}) \longrightarrow \mathbb{F}$$

satisfying the following properties: for all $1 \leq i \leq n$, $v_1, \ldots, v_n \in \mathbb{F}^n$, $a \in \mathbb{F}$, and $v_i' \in \mathbb{F}^n$,

①

$$\det(v_1, v_2, \ldots, v_{i-1}, av_i, v_{i+1}, \ldots, v_n) = a \det(v_1, v_2, \ldots, v_n).$$

②

$$\det(v_1, v_2, \ldots, v_{i-1}, v_i + v_i', v_{i+1}, \ldots, v_n) = \det(v_1, \ldots, v_n) + \det(v_1, \ldots, v_i', \ldots, v_n).$$

③

$$\det(v_1, \ldots, v_n) = 0 \quad \text{if } \exists v_i = v_j, \text{ for } i \neq j.$$

④

$$\det(I_n) = \det(e_1, e_2, \ldots, e_n) = 1.$$

Definition 3.2. Let V be an $\mathbb{F}$-vector space. Let $n \geq 1$, $n \in \mathbb{N}$. Then

$$V^n := V \times \cdots \times V = V \oplus V \oplus \cdots \oplus V.$$

Each element in V^n is called a hypervector. A function $H : V^n \to \mathbb{F}$ is

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multilinear/n-linear if $\forall, 1 \leq i \leq n$, fixing $v_1, \ldots, v_{i-1}, v_{i+1}, v_n$, we have the map

$$v_i \longmapsto H(v_1, \ldots, v_n)$$

being linear. In other words, H is n-linear if it is linear in its every component. Furthermore, H is *alternating* if $v_i = v_j$ for some $i \neq j$ implies

$$H(v_1, \ldots, v_n) = 0.$$

Example 3.3. $\mathbb{R}$-inner products defined on $\mathbb{R}$-vector spaces are, by definition, bilinear. ×

Example 3.4. The map

$$H : \mathbb{F}^2 \times \mathbb{F}^2 \longrightarrow \mathbb{F}$$

defined by

$$H\left(\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}\right) = ad - bc$$

is bilinear and alternating. ×

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Proposition 3.5. *Let V be an $\mathbb{F}$-vector space, $H : V^n \to \mathbb{F}$ be n-linear and alternating. Then $\forall v_1, \ldots, v_n \in V$, $1 \leq i < j \leq n$,*

$$H(v_1, \ldots, \underbrace{v_j}_{ith}, \ldots, \underbrace{v_i}_{jth}, \ldots, v_n) = -H(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_n).$$

Proof. Compute

$$\begin{aligned}
&H(v_1,\dots,v_j,\dots,v_i,\dots,v_n)+H(v_1,\dots,v_i,\dots,v_j,\dots,v_n)\\
&=H(v_1,\dots,v_j,\dots,v_i,\dots,v_n)+H(v_1,\dots,v_i,\dots,v_j,\dots,v_n)\\
&\quad+\underbrace{H(v_1,\dots,v_j,\dots,v_j,\dots,v_n)}_{0}+\underbrace{H(v_1,\dots,v_i,\dots,v_i,\dots,v_n)}_{0}\\
&=H(v_1,\dots,v_j+v_i,\dots,v_i,\dots,v_n)\\
&\quad+H(v_1,\dots,v_j+v_i,\dots,v_j,\dots,v_n)\qquad\text{(linearity)}\\
&=H(v_1,\dots,v_j+v_i,\dots,v_j+v_i,\dots,v_n)=0\qquad\text{(alternating).}
\end{aligned}$$

□

Note

(Long Discussion.) Now observe that the det is n-

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linear and alternating. That said, we can expand in each column inductively and get

$$\begin{aligned}
&\det\begin{pmatrix}a_{11}&\cdots&a_{1n}\\ \vdots&&\vdots\\ a_{n1}&\cdots&a_{nn}\end{pmatrix}\\
&=\det\begin{pmatrix}a_{11}&\cdots&a_{1n}\\ 0&&\vdots\\ \vdots&&\vdots\\ 0&\cdots&a_{nn}\end{pmatrix}+\det\begin{pmatrix}0&a_{12}&\cdots&a_{1n}\\ a_{21}&a_{22}&\cdots&\vdots\\ 0&\vdots&&\vdots\\ \vdots&\vdots&&\vdots\\ 0&a_{n2}&\cdots&a_{nn}\end{pmatrix}+\cdots.
\end{aligned}$$

Keep separating the other columns like this, going from column 1 to column n.

Eventually the matrices are going to look like

$$\begin{pmatrix}0&a_{22}&0&\cdots\\ a_{21}&0&0&\cdots\\ 0&0&a_{33}&\\ \vdots&\vdots&&\ddots\end{pmatrix},$$

where there are exactly one entry

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per column, and we are summing their determinants. Then by linearity we can see that

$$\det(a_{i_1 1}e_{i_1},a_{i_2 2}e_{i_2},\dots,a_{i_n n}e_{i_n})=a_{i_1 1}a_{i_2 2}\cdots a_{i_n n}\det(e_{i_1},e_{i_2},\dots,e_{i_n}).$$

Thereby, formally writing out the sum, we have

$$\det\begin{pmatrix}a_{11}&\cdots&a_{1n}\\ \vdots&&\vdots\\ a_{n1}&\cdots&a_{nn}\end{pmatrix}=\sum_{i_1=1}^{n}\sum_{i_2=1}^{n}\cdots\sum_{i_n=1}^{n}a_{i_1 1}a_{i_2 2}\cdots a_{i_n n}\det(e_{i_1},e_{i_2},\dots,e_{i_n}).$$

Furthermore, when $i_k = i_\ell$, $k \neq \ell$,

$$e_{i_k} = e_{i_\ell} \quad \Longrightarrow \quad \det(e_{i_1}, e_{i_2}, \ldots, e_{i_n}) = 0$$

by the alternating property. Thus

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removing the terms which are zero, the remaining terms are exactly those indexed by permutations $\sigma \in S_n$. Thus

$$\det((a_{ij})_n) = \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n} \det(e_{\sigma(1)}, e_{\sigma(2)}, \ldots, e_{\sigma(n)}).$$

This in fact shows the uniqueness of

$$\det : (\mathbb{F}^n)^n \longrightarrow \mathbb{F},$$

as we now have a closed form of determinants.

Question: What is $\det(e_{i_1}, e_{i_2}, \ldots, e_{i_n})$?

By last proposition, if we rearrange $i_1, i_2, \ldots, i_n$ to get back $\{1, 2, \ldots, n\}$, we just need to swap e_{i_k}'s. Thus

$$\det(e_{i_1}, \ldots, e_{i_n}) = \pm \det(e_1, e_2, \ldots, e_n) = \pm 1.$$

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3.1 Permutations and the Sign Map

Recall

(MATH 245) S_n = symmetric group on n elements

$$= \left\{ \sigma : \{1, 2, \ldots, n\} \xrightarrow{\text{bijection}} \{1, 2, \ldots, n\} \right\}.$$

In particular, $\sigma = (ij)$ in cycle notation is a transposition. Every $g \in S_n$ can be rewritten as product of transpositions:

$$\begin{aligned}(a_1, a_2, \ldots, a_k) &= (a_1, a_k)(a_1, a_{k-1}) \cdots (a_1, a_2) \\ &= (a_{k-1}, a_k)(a_{k-2}, a_k) \cdots (a_1, a_k).\end{aligned}$$

In fact, we have the following:

Theorem 3.6. *There exists a unique group homomorphism*

$$\operatorname{sgn} : S_n \longrightarrow \{\pm 1\}$$

such that $\operatorname{sgn}(\sigma) = -1$ *for* σ *any transposition.*

As this theorem suggests, each $g \in S_n$ has

$$g = \sigma_1 \sigma_2 \cdots \sigma_s$$

(product of

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transpositions), and thereby

$$\boxed{\operatorname{sgn}(g) = \operatorname{sgn}(\sigma_1)\operatorname{sgn}(\sigma_2)\cdots\operatorname{sgn}(\sigma_s) = (-1)^s.}$$

Example

$$\begin{aligned} g &= (1\,2\,5)(3\,4)(6\,7\,8\,10\,9) \\ &= (1\,5)(1\,2)(3\,4)(6\,9)(6\,10)(6\,8)(6\,7), \end{aligned}$$

so

$$\begin{aligned} \operatorname{sign}(g) &= (-1)^{3-1}(-1)^{2-1}(-1)^{5-1} \\ &= (-1)^7 = -1. \end{aligned}$$

Here

$$\#\text{ elements in cycle} - 1 = \#\text{ transpositions it splits into.}$$

Hence g is an odd permutation.

Continued: $\det(e_{i_1}, e_{i_2}, \ldots, e_{i_n}) = \pm 1$. Each swap on $(1, 2, 3, \ldots, n)$ contributes a factor of -1. How many times of swaps is exactly the # of transpositions there are if we decompose $(i_1, i_2, \ldots, i_k)$. Therefore

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$$\det(e_{i_1}, e_{i_2}, \ldots, e_{i_n}) = \operatorname{sign}(i_1, i_2, \ldots, i_n).$$

[$(i_1, i_2, \ldots, i_n)$ here is not in cycle notation, but in the sense $\left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ i_1 & i_2 & \cdots & i_n \end{smallmatrix}\right)$.]

Example

$n = 2$:

σ	$\operatorname{sgn}(\sigma)$	$a_{\sigma(1)1}a_{\sigma(2)2}$
1	1	$a_{11}a_{22}$
$(1\,2)$	-1	$a_{21}a_{12}$

Thus

$$\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{21}a_{12}.$$

$n = 3$:

σ	$\operatorname{sgn}(\sigma)$	$a_{\sigma(1)1}a_{\sigma(2)2}a_{\sigma(3)3}$
1	1	$a_{11}a_{22}a_{33}$
$(1\,2)$	-1	$a_{21}a_{12}a_{33}$
$(1\,3)$	-1	$a_{31}a_{22}a_{13}$
$(2\,3)$	-1	$a_{11}a_{32}a_{23}$
$(1\,3\,2)$	1	$a_{31}a_{12}a_{23}$
$(1\,2\,3)$	1	$a_{21}a_{32}a_{13}$

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Proof (of the theorem). By previous discussion, we have seen the probable way of defining det is

$$\det(A) = \sum_{g \in S_n} \operatorname{sgn}(g) a_{g(1)1} \cdots a_{g(n)n}.$$

We now show that it is

1) n-linear,
2) alternating,
3) $\det(I_n) = 1$.

1) Let $v_1, \ldots, v_n \in \mathbb{F}^n$, $i \in \{1, \ldots, n\}$, $v_i' \in \mathbb{F}^n$, and $\alpha \in \mathbb{F}$. Denote

$$(v_1, \ldots, v_n) = A, \qquad (v_1, \ldots, v_i', \ldots, v_n) = B, \qquad (v_1, \ldots, \alpha v_i + v_i', \ldots, v_n) = C.$$

Then

$$\det(C) = \sum_{g \in S_n} \operatorname{sgn}(g) c_{g(1)1} \cdots c_{g(n)n},$$

where

$$c_{g(j),j} = \begin{cases} (\alpha v_i + v_i')_{g(j)}, & \text{if } j = i, \\ (v_j)_{g(j)} = a_{g(j),j} = b_{g(j),j}, & \text{if } j \neq i. \end{cases}$$

Furthermore,

$$(\alpha v_i + v_i')_{g(j)} = \alpha a_{g(j),j} + b_{g(j),j}.$$

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Thus

$$\begin{aligned} \det C &= \sum_{g \in S_n} \operatorname{sgn}(g) c_{g(1)1} \cdots (\alpha a_{g(i),i} + b_{g(i),i}) \cdots c_{g(n)n} \\ &= \sum_{g \in S_n} \operatorname{sgn}(g) a_{g(1)1} \cdots \alpha a_{g(i),i} \cdots a_{g(n)n} \\ &\qquad + \sum_{g \in S_n} \operatorname{sgn}(g) b_{g(1)1} \cdots b_{g(i),i} \cdots b_{g(n)n} \\ &= \alpha \det A + \det B. \end{aligned}$$

2) Let $i, j \in \{1, 2, 3, \ldots, n\}$, $i < j$ without loss of generality. Suppose $v_i = v_j$. For $\sigma \in S_n$, set

$$t_\sigma = \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n}.$$

The map $\sigma \mapsto \sigma(ij)$ is a fixed-point-free involution of S_n.

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Because columns i and j are equal,

$$t_{\sigma(ij)} = -t_\sigma.$$

Therefore the terms in the determinant sum cancel in pairs:

$$\det A = \sum_{\sigma \in S_n} t_\sigma = 0.$$

This pairing argument is valid in every characteristic, including characteristic 2.

3) The only permutation $g \in S_n$ that gives a nonzero term in $\det(I_n)$ is the identity 1:

$$\begin{array}{ccccc} g(1) & \longmapsto & 1 & & \\ g(2) & \longmapsto & 1 & & \\ & & \ddots & \ddots & \\ g(n) & \longmapsto & & & 1 \end{array}$$

Thus

$$\det A = \sum_{g \in S_n} \operatorname{sgn}(g) a_{g(1)1} \cdots a_{g(n)n} = (-1)^0 a_{11} a_{22} \cdots a_{nn} = 1.$$

□

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Theorem 3.7. *Let $H : M_n(\mathbb{F}) \longrightarrow \mathbb{F}$ be n-linear and alternating. Then for*

$$\alpha = H(I_n),$$

we have

$$H = \alpha \det.$$

(Proof is similar to our discussion.)

Proof. Let $A = (a_{ij}) \in M_n(\mathbb{F})$. By n-linearity of H,

$$\begin{aligned} H(A) &= H\left(\sum_{i_1=1}^{n} a_{i_1 1} e_{i_1}, \sum_{i_2=1}^{n} a_{i_2 2} e_{i_2}, \ldots, \sum_{i_n=1}^{n} a_{i_n n} e_{i_n} \right) \\ &= \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} H(e_{i_1}, e_{i_2}, \ldots, e_{i_n}). \end{aligned}$$

By the alternating property, since H is alternating, if $i_\alpha = i_\beta$, for $\alpha \neq \beta$, then

$$H(e_{i_1}, \ldots, e_{i_\alpha}, \ldots, e_{i_\beta}, \ldots, e_{i_n}) = 0.$$

Thus the sum can be simplified to

$$\sum_{g \in S_n} a_{g(1)1} a_{g(2)2} \cdots a_{g(n)n} H(e_{g(1)}, e_{g(2)}, \ldots, e_{g(n)}).$$

Suppose $g = \tau_1 \tau_2 \cdots \tau_k$, where τ_i's are transpositions.

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Now we apply

$$g^{-1} = \tau_k^{-1} \tau_{k-1}^{-1} \cdots \tau_1^{-1} = \tau_k \tau_{k-1} \cdots \tau_1$$

to $H(e_{g(1)}, e_{g(2)}, \ldots, e_{g(n)})$ to swap the columns (each time multiplying τ_i to all the indices, only two would be interchanged, since τ is a transposition, so it is indeed a "swap" between columns). Then

$$\begin{aligned} H(e_{g(1)}, e_{g(2)}, \ldots, e_{g(n)}) &= (-1) H(e_{\tau_1 g(1)}, e_{\tau_1 g(2)}, \ldots, e_{\tau_1 g(n)}) \\ &= (-1)^2 H(e_{\tau_2 \tau_1 g(1)}, e_{\tau_2 \tau_1 g(2)}, \ldots, e_{\tau_2 \tau_1 g(n)}) \\ &\vdots \end{aligned}$$

$$= (-1)^k H(e_{g^{-1}g(1)}, e_{g^{-1}g(2)}, \ldots, e_{g^{-1}g(n)}).$$

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Therefore

$$H(e_{g(1)}, \ldots, e_{g(n)}) = \operatorname{sgn}(g) H(e_1, \ldots, e_n) = \operatorname{sgn}(g)\alpha.$$

Thus

$$\begin{aligned} H(A) &= \sum_{g \in S_n} a_{g(1)1} a_{g(2)2} \cdots a_{g(n)n} H(e_{g(1)}, \ldots, e_{g(n)}) \\ &= \sum_{g \in S_n} \alpha \operatorname{sgn}(g) a_{g(1)1} a_{g(2)2} \cdots a_{g(n)n} \\ &= \alpha \det A. \end{aligned}$$

□

Proposition 3.8 (Homework 4).

$$\det(A) = \det(A^t).$$

Proposition 3.9 (Homework 4). *If $A \in M_n(\mathbb{F})$ is upper triangular / lower triangular, then*

$$\det(A) = \prod_{i=1}^{n} a_{ii}.$$

3.2 Computing Determinants

Remark. Along with properties with det, we can compute $\det A$ in an easier way: first use elementary row operations to

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reduce A to row echelon form, then we have an upper triangular matrix.

Example

Calculate $\det(A)$ for

$$A = \begin{pmatrix} 2 & 4 & 0 & 4 \\ -1 & -2 & -5 & 2 \\ 0 & 1 & 3 & -3 \\ 4 & 6 & -6 & 6 \end{pmatrix} \in M_4(\mathbb{Q}).$$

Then

$$A \xrightarrow{R_2 + \frac{1}{2}R_1} \begin{pmatrix} 2 & 4 & 0 & 4 \\ 0 & 0 & -5 & 4 \\ 0 & 1 & 3 & -3 \\ 4 & 6 & -6 & 6 \end{pmatrix}$$

$$\xrightarrow[R_2\leftrightarrow R_3]{R_4+2R_3} \begin{pmatrix} 2 & 4 & 0 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & -5 & 4 \\ 4 & 8 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_4+(-2)R_1} \begin{pmatrix} 2 & 4 & 0 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & -5 & 4 \\ 0 & 0 & 0 & -8 \end{pmatrix}.$$

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Since $\det(A) = \det(A^t)$, swapping rows introduces a $-$factor of -1. Adding scalar multiple of a row to another row makes no difference:

$$\begin{aligned} \det \begin{pmatrix} \\ r_1 \\ \vdots \\ r_i + \alpha r_j \\ \vdots \\ r_n \\ \\ \end{pmatrix} &= \det(r_1, \ldots, r_i + \alpha r_j, \ldots, r_n) \\ &= \det(r_1, \ldots, r_n) + \det(r_1, \ldots, \alpha r_j, \ldots, r_j, \ldots, r_n) \\ &= \det(r_1, \ldots, r_n) + \alpha \det(r_1, \ldots, r_j, \ldots, r_j, \ldots, r_n) \\ &= \det(r_1, \ldots, r_n) + 0 = \det \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}. \end{aligned}$$

Thus, swapping two rows,

$$\det(A) = (-1) \det \begin{pmatrix} 2 & 4 & 0 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & -5 & 4 \\ 0 & 0 & 0 & -8 \end{pmatrix} = -80.$$

Theorem 3.10. *For all $A, B \in M_n(\mathbb{F})$,*

$$\det(AB) = \det(A) \det(B).$$

Proof.
[There are a lot of ways of proving

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this (for example, by Laplace expansion along multiple columns or by row reducing A through left multiplication by elementary matrices), we will use another approach.]
For vectors

$$v = (a_1, \ldots, a_n), \qquad w = (b_1, \ldots, b_n),$$

define

$$\langle v, w \rangle := \sum_{i=1}^{n} a_i b_i.$$

The map

$$\langle \cdot, \cdot \rangle : \mathbb{F}^n \times \mathbb{F}^n \longrightarrow \mathbb{F}$$

is bilinear, that is, $\langle v, \cdot \rangle$ and $\langle \cdot, w \rangle$ are both linear in their respective component. Furthermore, notice that

$$AB = \begin{pmatrix} \langle v_1, w_1 \rangle & \langle v_1, w_2 \rangle & \cdots & \langle v_1, w_n \rangle \\ \langle v_2, w_1 \rangle & \langle v_2, w_2 \rangle & \cdots & \langle v_2, w_n \rangle \\ \vdots & \vdots & & \vdots \\ \langle v_n, w_1 \rangle & \langle v_n, w_2 \rangle & \cdots & \langle v_n, w_n \rangle \end{pmatrix}$$

if

$$A = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \qquad \text{and} \qquad B = (w_1, w_2, \ldots, w_n).$$

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Fix $A \in M_n(\mathbb{F})$ with row vectors $v_1, v_2, \ldots, v_n$. Define

$$H : M_n(\mathbb{F}) \longrightarrow \mathbb{F} \qquad \text{via} \qquad H(B) := \det(AB).$$

We claim: H is n-linear and alternating.
Let $w_1, \ldots, w_n \in \mathbb{F}^n$, $j \in \{1, 2, \ldots, n\}$, $\alpha \in \mathbb{F}$, $w_j' \in \mathbb{F}^n$. Then

$$\begin{aligned} &H(w_1, \ldots, w_j + \alpha w_j', \ldots, w_n) \\ &= \det \begin{pmatrix} \vdots & & \vdots & \vdots \\ \cdots & \langle v_1, w_j + \alpha w_j' \rangle & \cdots & \\ \cdots & \langle v_2, w_j + \alpha w_j' \rangle & \cdots & \\ & \vdots & & \\ \cdots & \langle v_n, w_j + \alpha w_j' \rangle & \cdots & \end{pmatrix} \\ &= \det \begin{pmatrix} \vdots & & \vdots \\ \cdots & \langle v_1, w_j \rangle & \cdots \\ \cdots & \langle v_2, w_j \rangle & \cdots \\ & \vdots & \\ \cdots & \langle v_n, w_j \rangle & \cdots \end{pmatrix} + \det \begin{pmatrix} \vdots & & \vdots \\ \cdots & \alpha \langle v_1, w_j' \rangle & \cdots \\ \cdots & \alpha \langle v_2, w_j' \rangle & \cdots \\ & \vdots & \\ \cdots & \alpha \langle v_n, w_j' \rangle & \cdots \end{pmatrix}, \end{aligned}$$

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thus H is n-linear.
If $w_i = w_j$, then

$$\langle v_k, w_i \rangle = \langle v_k, w_j \rangle, \qquad k = 1, 2, \ldots, n.$$

Then

$$H(w_1, \ldots, w_n) = \det \begin{pmatrix} \langle v_1, w_i \rangle & \cdots & \langle v_1, w_j \rangle \\ \langle v_2, w_i \rangle & \cdots & \langle v_2, w_j \rangle \\ \vdots & & \vdots \\ \langle v_n, w_i \rangle & \cdots & \langle v_n, w_j \rangle \end{pmatrix} = 0.$$

By the very last theorem we proved,

$$H = [H(I_n)] \det .$$

Moreover,

$$H(I_n) = \det(AI_n) = \det A.$$

Therefore

$$H(B) = \det(AB) = \det(A)\det(B).$$

□

Theorem 3.11. *Let $A = (v_1, v_2, \dots, v_n) \in M_n(\mathbb{F})$. Then the following are equivalent:*
(1) A is invertible;

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(2) $\det(A) \neq 0$;
(3) $\{v_1, v_2, \dots, v_n\}$ is a basis of $\mathbb{F}^n$.
Moreover, if these hold, then

$$\det(A^{-1}) = \det(A)^{-1}.$$

Proof. (**1** $\Rightarrow$ **2**) If there exists $B \in M_n(\mathbb{F})$, then

$$AB = BA = I_n \implies \det(A)\det(B) = 1 \implies \det A \neq 0.$$

(**2** $\Rightarrow$ **3**) **Claim.** $\{v_1, v_2, \dots, v_n\}$ is linearly independent. Suppose otherwise, then there exists i such that

$$v_i = a_1v_1 + a_2v_2 + \cdots + a_{i-1}v_{i-1} + a_{i+1}v_{i+1} + \cdots + a_nv_n.$$

Then

$$\begin{aligned}\det(v_1, \dots, v_n) &= \det(v_1, \dots, a_1v_1 + \cdots + a_nv_n, \dots, v_n) \\ &= \sum_{\substack{j=1 \\ j\neq i}}^{n} \det(v_1, \dots, v_j, \dots, v_n) = \sum 0 = 0,\end{aligned}$$

where v_j is in the ith entry.

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This is a contradiction. Thus $\{v_1, \dots, v_n\}$ is independent. Since $\dim \mathbb{F}^n = n$ and $\{v_1, \dots, v_n\}$ has n vectors, it is a basis.
(**3** $\Rightarrow$ **1**) If $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis of $\mathbb{F}^n$, then

$$A = [\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}},$$

and furthermore, by a proposition proven before,

$$\begin{aligned}[\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}}[\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}} &= [\mathrm{Id}]_{\mathrm{St}}^{\mathrm{St}} = I_n, \\ [\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}[\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}} &= [\mathrm{Id}]_{\mathcal{B}}^{\mathcal{B}} = I_n.\end{aligned}$$

□

Corollary 3.12. *If $A, Q \in M_{n\times n}(\mathbb{F})$, with Q invertible, then*

$$\det(QAQ^{-1}) = \det(A).$$

Proof.

$$\begin{aligned}\det(QAQ^{-1}) &= \det(Q)\det(A)\det(Q^{-1}) \\ &= \det(Q)\det(A)\det(Q)^{-1} = \det(A).\end{aligned}$$

□

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3.3 Minors and Determinants of Operators

Consequence

We can define the determinant $\det(T)$ for a linear transformation

$$T : V \longrightarrow V$$

on a finite-dimensional vector space V, independently of the choice of basis of V. Let $\mathcal{B}$ be any basis of V. Define

$$\det(T) := \det([T]_{\mathcal{B}}^{\mathcal{B}}).$$

This is well-defined: if $\mathcal{E}$ is another basis of V, then

$$\begin{aligned}\det([T]_{\mathcal{E}}^{\mathcal{E}}) &= \det([\mathrm{Id}]_{\mathcal{B}}^{\mathcal{E}}[T]_{\mathcal{B}}^{\mathcal{B}}[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}}) \\ &= \det([T]_{\mathcal{B}}^{\mathcal{B}}).\end{aligned}$$

We now prove the Laplace expansion theorem.

[Note that we proved this once in Assignment 4, Question 4. We use a slightly different notation here but the step is exactly the same.]

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Definition 3.13. Let $A \in M_n(\mathbb{F})$, $1 \le i, j \le n$. The (i,j)-minor of A, denoted A_{ij}, is the $(n-1) \times (n-1)$ matrix obtained by deleting the ith row and j^{th} column from A.

Lemma 3.14. *For all* $1 \le i, j \le n$,

$$\sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{k \neq j} a_{\sigma(k),k} = (-1)^{i+j} \det(A_{ij}).$$

Proof. Write

$$A_{ij} = (b_{\ell m})_{1 \le \ell, m \le n-1},$$

where

$$b_{\ell m} = \begin{cases} a_{\ell m}, & \text{if } \ell < i, m < j, \\ a_{\ell+1,m}, & \text{if } \ell \ge i, m < j, \\ a_{\ell,m+1}, & \text{if } \ell < i, m \ge j, \\ a_{\ell+1,m+1}, & \text{if } \ell \ge i, m \ge j. \end{cases}$$

Let

$$\rho = (i\ i+1\ \cdots\ n), \qquad \tau = (j\ j+1\ \cdots\ n)$$

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be permutations in S_n. Then notice that

$$\boxed{b_{\ell m} = a_{\rho(\ell),\tau(m)}}, \qquad \forall\ 1 \le \ell, m \le n-1.$$

Furthermore,

$$\tau^{-1}(\{1,2,\dots,n\} \setminus \{j\}) = \{1,2,\dots,n-1\},$$
$$\rho^{-1}(\{1,2,\dots,n\} \setminus \{i\}) = \{1,2,\dots,n-1\}.$$

Then compute

$$\sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{k \neq j} a_{\sigma(k),k}$$
$$= \sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{k \neq j} b_{\rho^{-1}\sigma(k),\tau^{-1}(k)}.$$

Since

$$\tau^{-1}(\{1,2,\dots,n\} \setminus \{j\}) = \{1,2,\dots,n-1\},$$

we rewrite the product by mapping $k \xrightarrow{\tau^{-1}} \tau^{-1}(k) = s$. We have $\tau(s) = k$, and thus

$$= \sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{s=1}^{n-1} b_{\rho^{-1}\sigma\tau(s),\tau^{-1}(\tau(s))}$$

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$$= \sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{s=1}^{n-1} b_{\rho^{-1}\sigma\tau(s),s}.$$

Define

$$\rho^{-1}\sigma\tau = g.$$

Notice that $g(n) = n$; conversely, for every $g \in S_n$ that fixes n, $\rho g \tau^{-1} \in S_n$ sends $j \mapsto i$. Thus

$$\sigma \quad \Longleftrightarrow \quad g = \rho^{-1}\sigma\tau,$$

and we can rewrite the sum as

$$\sum_{\substack{g \in S_n \\ g(n)=n}} \operatorname{sgn}(\rho g \tau^{-1}) \prod_{s=1}^{n-1} b_{g(s),s}$$
$$= \operatorname{sgn}(\rho)\operatorname{sgn}(\tau^{-1}) \sum_{\substack{g \in S_n \\ g(n)=n}} \operatorname{sgn}(g) \prod_{s=1}^{n-1} b_{g(s),s}$$
$$= (-1)^{n-i}\big((-1)^{n-j}\big)^{-1} \det(A_{ij})$$
$$= (-1)^{i+j} \det(A_{ij}).$$

□

3.4 Laplace Expansion and the Adjugate

Theorem 3.15 (Laplace). *Let $A = (a_{ij}) \in M_n(\mathbb{F})$.*

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1. *For every $1 \le j \le n$,*
$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}),$$
and if $\ell \neq j$, then
$$\sum_{i=1}^{n} (-1)^{i+\ell} a_{ij} \det(A_{i\ell}) = 0.$$
2. *For every $1 \le i \le n$,*
$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}),$$
and if $\ell \neq i$, then
$$\sum_{j=1}^{n} (-1)^{\ell+j} a_{ij} \det(A_{\ell j}) = 0.$$

Proof. Note that 2 follows from 1 and that $\det(A) = \det(A^t)$. Let $1 \le j \le n$. Then

$$\begin{aligned}
\det(A) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{k=1}^{n} a_{\sigma(k),k} \\
&= \sum_{i=1}^{n} \sum_{\substack{\sigma \in S_n \\ \sigma(j)=i}} \operatorname{sgn}(\sigma) \prod_{k=1}^{n} a_{\sigma(k),k} \\
&= \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det(A_{ij}) \qquad \text{by lemma.}
\end{aligned}$$

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Furthermore, consider the matrix B obtained from A by replacing its ℓth column by its jth column:

$$B = \begin{pmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}.$$

Then

$$\begin{aligned}
\det(B) &= \sum_{i=1}^{n} (-1)^{i+\ell} b_{i\ell} \det(B_{i\ell}) \\
&= \sum_{i=1}^{n} (-1)^{i+\ell} a_{ij} \det(A_{i\ell}).
\end{aligned}$$

Yet B has two identical columns, thus

$$\sum_{i=1}^{n} (-1)^{i+\ell} a_{ij} \det(A_{i\ell}) = \det(B) = 0$$

by the alternating property. □

Definition 3.16. Let $A = (a_{ij}) \in M_n(\mathbb{F})$. The adjoint of A, denoted $\mathrm{Adj}(A)$, is the matrix whose (i,j)-entry is

$$(-1)^{i+j} \det(A_{ji}).$$

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Example

If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{F}),$$

then

$$\mathrm{Adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Example

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \in M_3(\mathbb{Q}),$$

then

$$\begin{aligned} \mathrm{Adj}(A) &= \begin{pmatrix} \det(A_{11}) & -\det(A_{21}) & \det(A_{31}) \\ -\det(A_{12}) & \det(A_{22}) & -\det(A_{32}) \\ \det(A_{13}) & -\det(A_{23}) & \det(A_{33}) \end{pmatrix} \\ &= \begin{pmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{pmatrix}. \end{aligned}$$

Theorem 3.17. *For all $A \in M_n(\mathbb{F})$,*

$$A \cdot \mathrm{Adj}(A) = \mathrm{Adj}(A) \cdot A = \det(A) \cdot I_n.$$

If $A \in GL_n(\mathbb{F})$, then

$$A^{-1} = \det(A)^{-1} \mathrm{Adj}(A).$$

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Proof. By direct computation through previous theorem (Laplace). ☒

Example

If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{F}),$$

then

$$A\operatorname{Adj}(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}.$$

⊠

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4 System of Linear Equations

Proposition

Consider a system of linear equations

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1, \\ \qquad\qquad\qquad\qquad\quad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m, \end{cases}$$

and let $A = (a_{ij}) \in M_{m\times n}(\mathbb{F})$.

1. A solution exists in $\mathbb{F}^n$ if and only if

$$\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \operatorname{im}(L_A), \qquad [L_A : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \quad x \longmapsto Ax],$$

   if and only if

$$\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \operatorname{Span}\left\{ \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix}, \ldots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \right\}.$$

2. The set of solutions is either $\varnothing$ or is the affine coset

$$y + \ker(L_A) \subseteq \mathbb{F}^n$$

   for any $y \in \mathbb{F}^n$ satisfying

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$$Ay = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

   Equivalently, all solutions represent one class in $\mathbb{F}^n/\ker(L_A)$.

In particular, if $\ker(L_A) = \{0\}$, that is, the homogeneous system has only the trivial solution, then the system has at most one solution.

Proof. Note that

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n \text{ solves the system } \iff A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

$$\iff L_A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \iff \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \in \operatorname{im}(L_A) = \operatorname{Span}\left\{ \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix}, \dots, \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} \right\}.$$

This shows part 1.

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To show part 2, note that if $x, y \in \mathbb{F}^n$ solve the system, then

$$A(x-y) = Ax - Ay = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} - \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Thus

$$x - y \in \ker(L_A) \quad \Longrightarrow \quad \text{SolutionSet}(A) \subseteq y + \ker(L_A).$$

Now let $z \in \ker(L_A)$. Then

$$A(y+z) = Ay + Az = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix},$$

so $y + z = x \in \text{SolutionSet}(A)$. Hence

$$y + \ker(L_A) \subseteq \text{SolutionSet}(A), \qquad \text{SolutionSet}(A) = y + \ker(L_A).$$

□

Corollary

If $n > m$ and $b_1 = b_2 = \cdots = b_m = 0$, then there exists a nontrivial solution (that is, nonzero).

Proof. By the dimension theorem,

$$\dim \ker(L_A) = n - \dim \operatorname{im}(L_A).$$

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$$\dim \ker(L_A) \geq n - m > 0.$$

Thus $|\mathcal{B}_{\ker(L_A)}| > 0$, so there is at least one nonzero element in $\ker(L_A)$. □

4.1 Row Spaces, Column Spaces, and Rank

Definition

The *column space* of $A \in M_{m\times n}(\mathbb{F})$ is the span of its columns. The dimension of the column space is called the *column rank* of A, denoted $\mathrm{rk}_c(A)$. Similarly, the *row space* of A is the span of its rows, and its dimension is called the *row rank* of A, denoted $\mathrm{rk}_r(A)$. ⊠

Note

$$\mathrm{Col}(A) \leq \mathbb{F}^m, \qquad \mathrm{row}(A) \leq \mathbb{F}^n.$$

⊠

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Example

Let

$$A = \begin{pmatrix} 1 & 2 & 3 & -1 \\ 3 & 4 & 7 & -3 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Since

$$\begin{pmatrix} 3 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}, \qquad \begin{pmatrix} -1 \\ -3 \\ 0 \end{pmatrix} = -\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix},$$

we have

$$\mathrm{Col}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} \right\}.$$

Furthermore, they are linearly independent (either by direct verification or by observing that two vectors are dependent if and only if one is a scalar multiple of the other). Thus

$$\mathrm{rk}_c(A) = \dim \mathrm{Col}(A) = 2.$$

In terms of the row space, since

$$(3, 4, 7, -3) = 3(1, 2, 3, -1) - 2(0, 1, 1, 0),$$

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it follows that

$$\mathrm{row}(A) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ -1 \end{pmatrix}^t, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}^t \right\},$$

and they are independent. Thus $\mathrm{rk}_r(A) = 2$. ⊠

Notation

Let $A \in M_{m\times n}(\mathbb{F})$, and let $R_1, \dots, R_m$ be the rows of A. For

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n, \qquad R = (r_1, \dots, r_n) \in \mathbb{F}^n,$$

define

$$\langle R, x\rangle = \sum_{j=1}^{n} r_j x_j.$$

Then

$$\begin{aligned} Ax = 0_{\mathbb{F}^m} &\iff \langle R_i, x\rangle = 0, \quad i = 1, 2, \dots, m \\ &\iff \langle R, x\rangle = 0, \quad \forall R \in \mathrm{row}(A), \end{aligned}$$

by expanding R into the R_i and using bilinearity of $\langle\cdot,\cdot\rangle$. ⊠

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4.2 Row Equivalence

Proposition

A homogeneous system of m equations in n unknowns has the same set of solutions as another system with m' equations in n unknowns, where $m' \le n$.

Proof. Let $A = (a_{ij})$. Let $\{S_1, \dots, S_{m'}\}$ be a basis of

$$\mathrm{row}(A) = \mathrm{Span}\{R_1, \dots, R_m\}.$$

Consider

$$S := \begin{pmatrix} S_1 \\ \vdots \\ S_{m'} \end{pmatrix} \in M_{m'\times n}(\mathbb{F}).$$

Then x is a solution to S if and only if

$$\langle S_1, x\rangle = \cdots = \langle S_{m'}, x\rangle = 0.$$

Since $\{S_1, \dots, S_{m'}\}$ is a basis of $\mathrm{Span}\{R_1, \dots, R_m\}$, this is equivalent to

$$\langle R_1, x\rangle = \cdots = \langle R_m, x\rangle = 0,$$

which is equivalent to x being a solution to A. □

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$$Ax = 0 \xrightarrow{\text{elementary row operation}} A'x = 0.$$

Proposition

Two homogeneous systems related by a sequence of elementary row operations have the same solution.

Proof. If $R_1, \dots, R_m$ and $R'_1, \dots, R'_m$ are rows of A, A' respectively, then notice that

$$\text{Span}\{R_1, \dots, R_m\} = \text{Span}\{R'_1, \dots, R'_m\},$$

as all elementary row operations are linear combinations and reversible (that is, if we get $\{R'_1, \dots, R'_m\}$ from $\{R_1, \dots, R_m\}$ by a sequence of elementary row operations, then by the exact reversed set of elementary row operations we get $\{R_1, \dots, R_m\}$ as linear combinations of $\{R'_1, \dots, R'_m\}$). Then, as before,

$$a \in \mathbb{F}^n \text{ solves } A \iff \langle R_i, x \rangle = 0 \ \forall i$$

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$$\langle R'_i, x \rangle = 0 \ \forall i \iff x \text{ solves } A'.$$

□

Remark

It is trivial to see that the relation defined by

$$A \sim B \iff B \text{ can be obtained from } A \text{ by a sequence of row operations}$$

is an equivalence relation, that is, it satisfies:

1. reflexivity,
2. symmetry,
3. transitivity.

⊠

Definition

We say A and B are *row equivalent* in this case. ⊠

Definition

A matrix $A \in M_{m \times n}(\mathbb{F})$ is in *reduced row echelon form* if

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4.3 Reduced Row-Echelon Form

the following hold:

1. Any nonzero row precedes any zero rows.
2. The first nonzero entry in any nonzero row is a 1 (call this a *leading 1/pivot*) and is the only nonzero entry in its column.
3. The first nonzero entry in each row is to the right of the first nonzero entry in any preceding row.

⊠

Example

$$\begin{pmatrix} 1 & 2 & 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 0 & 7 & 0 \\ 0 & 0 & 0 & 1 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is in reduced row echelon form. ⊠

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Example

$$\begin{pmatrix} 1 & 2 & 1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

is not in reduced row echelon form. ⊠

Theorem

Any matrix is row equivalent to a matrix in reduced row echelon form.

Proof. This only serves as a step-by-step guide.

If $A = 0$, it is already in reduced row echelon form. Otherwise proceed as follows.

Step 1. Let j_1 be the smallest index such that the j_1th column of A is nonzero. Perform row operations to get

$$\left(\begin{array}{cccc|cccc} 0 & \cdots & 0 & 1 & * & * & \cdots & * \\ 0 & \cdots & 0 & 0 & * & * & \cdots & * \\ \vdots & & \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & * & * & \cdots & * \end{array}\right), \qquad \text{the displayed pivot column is } j_1.$$

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Step 2. Go into the right-bottom submatrix:

$$\left(\begin{array}{cccc|cccc} 0 & \cdots & 0 & 1 & * & * & \cdots & * \\ 0 & \cdots & 0 & 0 & * & * & \cdots & * \\ \vdots & & \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & * & * & \cdots & * \end{array}\right).$$

Repeat Step 1 in the submatrix. The leading 1 found will be the next leading 1 in our original matrix. Then clear the entries above it and below it to make it a valid pivot. Repeat Step 2 until we get a matrix in reduced row echelon form. □

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Example

$$\begin{pmatrix} 3 & 2 & 0 \\ 1 & 1 & 1 \\ 2 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 1 & -1 \end{pmatrix} \xrightarrow[,R_3+(-2)R_1,]{,R_2+(-3)R_1,} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -3 \\ 0 & -1 & -3 \end{pmatrix}$$

$$\xrightarrow[,R_3+R_2,]{,(-1)R_2,} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1+(-1)R_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{(reduced row-echelon form).}$$

⊠

Notation

For a matrix A, write $\mathcal{R}(A)$ for its unique reduced row-echelon form.

Theorem

Two matrices in $M_{m\times n}(\mathbb{F})$ in reduced row echelon form with the same row space are equal.

First proof. We proceed by induction on the number of columns n.

Base case. If $n = 1$, then our matrices are column vectors in $\mathbb{F}^m$.

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Since they are matrices in reduced row echelon form, having one column leaves two possibilities:

1. $$\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

   Then $\text{Row}(A) = \text{Row}(B)$ if and only if

   $$A = B = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

2. $$\begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

   Then $\text{Row}(A) = \text{Row}(B)$ if and only if

   $$A = B = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Inductive step. Suppose it holds for $n-1 \in \mathbb{N}$. Then we show this for $n \in \mathbb{N}$. By removing a column, the row space becomes a subspace of $\mathbb{F}^{n-1}$. Furthermore, notice that chopping off the nth component does not change that

$$\text{Row}(A') = \text{Row}(B'),$$

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where

$$A' = A - n\text{th column}, \qquad B' = B - n\text{th column}.$$

Moreover, A' and B' are still matrices in reduced row echelon form. Then, by the induction hypothesis,

$$A' = B'.$$

Now we consider the nth column of A, B and proceed by a contradiction.

Case 1. The nth column contains a pivot. This case is trivial, because we know the column will look like

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

So if B does not have a leading 1 in its nth column, then $\text{Row}(B)$ would not contain

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the vector $(0, 0, \ldots, 0, 1)$. Thus this contradicts

$$\text{Row}(A) = \text{Row}(B).$$

Case 2. If the nth column of A does not contain a pivot, this case is slightly less trivial. Note that, in the nth column, every row with a nonzero element must have a pivot in the same row in A; otherwise this nonzero element would be a pivot, and then this column would have a pivot, a contradiction to our assumption.

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This implies that if, in A, the first k rows are nonzero (thus have pivots), and the rest of the rows are zero rows, then the first k rows in B must be nonzero and the rest must be zero rows. Both A and B should look like

$$\left(\begin{array}{ccccccc|c} 1 & * & 0 & * & \cdots & 0 & * & \\ 0 & 1 & * & * & \cdots & 0 & * & * \\ \vdots & & \ddots & & & \vdots & \vdots & \\ 0 & \cdots & 0 & 1 & * & \cdots & * & \vdots \\ \hline & & 0 & \cdots & 0 & & & 0 \\ & & \vdots & & \vdots & & & \vdots \\ & & 0 & \cdots & 0 & & & 0 \end{array}\right).$$

Then suppose the nth column of A is not the nth column of B. Then there is an ith entry at which they are not equal: $a \neq b$. Consider the ith rows of A, B:

$$\alpha = (0, \ldots, 0, 1, *, \ldots, *, 0, *, \ldots, *, a), \qquad \beta = (0, \ldots, 0, 1, *, \ldots, *, 0, *, \ldots, *, b).$$

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Notice that $\alpha \in \mathrm{Row}(A)$, yet $\alpha \notin \mathrm{Row}(B)$. If $\alpha \in \mathrm{Row}(B)$, then

$$\alpha = \sum_{j=1}^{m} \lambda_j (\text{row } j \text{ of } B).$$

Since the last $m - k$ rows are zero,

$$\alpha = \sum_{j=1}^{k} \lambda_j (\text{row } j \text{ of } B).$$

Moreover, each row among the first k rows has a pivot, so $\lambda_j = 0$ for $j \neq i$. Otherwise we would have an entry in α where there is a pivot in the same column. Then A is not in reduced row echelon form, since every pivot column should have the form

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Therefore

$$\alpha = \lambda_i (\text{row } i \text{ of } B) = \lambda_i \beta.$$

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Thus

$$(0, \ldots, 0, 1, *, \ldots, *, 0, *, \ldots, *, a) = (0, \ldots, 0, \lambda_i, *, \ldots, *, 0, *, \ldots, *, \lambda_i b),$$

so $\lambda_i = 1$ and hence $a = b$, a contradiction. Therefore, the nth column of A is the nth column of B. Along with $A' = B'$, this gives $A = B$.

Second proof. Let $A, B \in M_{m \times n}(\mathbb{F})$ be matrices in reduced row echelon form. Write

$$A = \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \qquad B = \begin{pmatrix} S_1 \\ S_2 \\ \vdots \\ S_u \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \qquad 0 \leq t, u \leq m,$$

where $R_i, S_\ell \in \mathbb{F}^n \setminus \{0\}$. Furthermore, each R_i, S_ℓ contains a pivot; suppose the pivot in R_i is in its (i, j_i) entry and the pivot in S_ℓ is in its

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(ℓ, k_ℓ) entry.

Suppose $\mathrm{Row}(A) = \mathrm{Row}(B)$. We want to show $t = u$ and $R_i = S_i$ for $i = 1, 2, \ldots, t$.

First notice that $\{R_1, \ldots, R_t\}$ is a basis of $\mathrm{Row}(A)$. It is clearly spanning. Suppose $\alpha_1, \ldots, \alpha_t \in \mathbb{F}$ satisfy

$$\alpha_1 R_1 + \cdots + \alpha_t R_t = 0.$$

Furthermore,

$$\alpha_1 R_1 + \cdots + \alpha_t R_t = (\ldots, \alpha_1, \ldots, \alpha_2, \ldots, \alpha_t, \ldots),$$

where the displayed entries are in columns $j_1, j_2, \ldots, j_t$. Since A is in reduced row echelon form, these columns should only have their pivots being nonzero. Thus

$$\alpha_1 = \alpha_2 = \cdots = \alpha_t = 0,$$

so $\{R_1, R_2, \ldots, R_t\}$ is linearly independent.

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Symmetrically, $S_1, \ldots, S_u$ is a basis of $\mathrm{Row}(B)$. Since $\mathrm{Row}(A) = \mathrm{Row}(B)$, it follows that

$$t = u = \dim \mathrm{Row}(A).$$

Now we prove by descending induction that

$$R_i = S_i \qquad \text{for } i = t, t-1, \ldots, 1.$$

Fix $1 \le i \le t$ and suppose that $R_p = S_p$ for every $p > i$. For the base case $i = t$, this higher-index assumption is empty. Since $R_i \in \mathrm{Row}(B)$, we have

$$R_i = \sum_{\ell=1}^{t} \alpha_\ell S_\ell.$$

Furthermore, R_i has a pivot in column j_i and zero entries in every other pivot column of A. By the induction hypothesis, for $p > i$ we have $S_p = R_p$ and hence $k_p = j_p$. Therefore

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$$\alpha_{i+1} = \alpha_{i+2} = \cdots = \alpha_t = 0;$$

the coefficients can be read from those pivot columns. Then

$$R_i = \sum_{\ell=1}^{i} \alpha_\ell S_\ell.$$

Case 1. $j_i > k_i$ (the ith pivot in B comes before the ith pivot in A). Then

$$\sum_{\ell=1}^{i} \alpha_\ell S_\ell \neq R_i,$$

a contradiction.

Case 2. $j_i < k_i$ (the ith pivot in B comes after the ith pivot in A).

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By a symmetric argument applied to

$$S_i = \sum_{\ell=1}^{i} \beta_\ell R_\ell,$$

we get a contradiction as in Case 1.

Case 3. $j_i = k_i$. Then $\alpha_1, \ldots, \alpha_{i-1} = 0$, so that entries before k_i are zero as in R_i. Thereby

$$R_i = \alpha_i S_i.$$

Furthermore, $\alpha_i = 1$, since both R_i and S_i have a pivot in the j_ith entry. Thus $R_i = S_i$.

By induction, $R_i = S_i$ for all i. □

Corollary

Every $A \in M_{m\times n}(\mathbb{F})$ is row equivalent to a unique matrix in reduced row echelon form.

Proof. Note that row operations preserve

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row space. Now suppose $A \sim A'$ and $A \sim B'$, where A' and B' are in reduced row echelon form. Then

$$\mathrm{Row}(A) = \mathrm{Row}(A') = \mathrm{Row}(B'),$$

so $A' = B'$ by the last theorem. □

Corollary

If $A, B \in M_{m\times n}(\mathbb{F})$ and $\mathrm{Row}(A) = \mathrm{Row}(B)$, then

$$A \sim B \qquad \text{(row equivalent)}.$$

Proof. By the previous corollary,

$$A \sim \mathcal{R}(A), \qquad B \sim \mathcal{R}(B).$$

Also,

$$\mathrm{Row}(\mathcal{R}(A)) = \mathrm{Row}(A), \qquad \mathrm{Row}(\mathcal{R}(B)) = \mathrm{Row}(B).$$

Thus

$$\mathcal{R}(A) = \mathcal{R}(B),$$

and hence

$$A \sim \mathcal{R}(A) \sim B,$$

using reversed row operations in the second equivalence. □

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Theorem

For every $A \in M_{m\times n}(\mathbb{F})$,

$$\mathrm{rk}_r(A) = \mathrm{rk}_c(A).$$

Proof. Do row operations and turn A into its reduced row echelon form A'. Since row space is preserved by row operations,

$$\mathrm{Row}(A) = \mathrm{Row}(A') \implies \dim \mathrm{Row}(A) = \dim \mathrm{Row}(A')$$
$$\implies \mathrm{rk}_r(A) = \mathrm{rk}_r(A') = \#\{\text{pivots}\}.$$

(The second equality was shown in the proof of another theorem.)

Now notice that if

$$L_A : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \qquad x \longmapsto Ax,$$

then $\mathrm{Col}(A) = \mathrm{im}(L_A)$. Hence

$$\dim \mathrm{Col}(A) = \dim \mathrm{im}(L_A) = n - \dim \ker(L_A) = n - \dim(\text{space of solutions of } A),$$

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Similarly,

$$\dim \mathrm{Col}(A') = n - \dim(\text{space of solutions of } A').$$

Lastly, by a proposition we proved before, two row-equivalent matrices have the same space of solutions. Write

$$\mathcal{S}(A) := \{x \in \mathbb{F}^n : Ax = 0\}$$

for the solution space of the homogeneous system associated with A. Therefore

$$\mathcal{S}(A) = \mathcal{S}(A'),$$

and

$$\mathrm{rk}_c(A) = \dim \mathrm{Col}(A) = n - \dim \mathcal{S}(A) = n - \dim \mathcal{S}(A') = \dim \mathrm{Col}(A') = \mathrm{rk}_c(A').$$

It is trivial to see from the fact that A' is in reduced row echelon form, we have

$$\mathrm{rk}_c(A') = \#\{\text{pivots in } A'\}.$$

Thus

$$\mathrm{rk}_c(A) = \#\{\text{pivots in } A'\} = \mathrm{rk}_r(A).$$

□

Definition

We define

$$\mathrm{rk}(A) := \mathrm{rk}_r(A) = \mathrm{rk}_c(A)$$

as the

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4.4 Elementary Matrices and Inversion

rank of A. ×

Definition

An *elementary matrix* is a square matrix of one of the following types:

1. A diagonal matrix equal to I_n except for a nonzero scalar $\lambda \in \mathbb{F}^\times$ in the ith diagonal position:

$$\begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

2. A matrix equal to I_n except that the kth and ℓth rows are swapped:

$$\begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & \cdots & 1 & & \\ & & \vdots & & \vdots & & \\ & & 1 & \cdots & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix}.$$

3. A matrix equal to I_n with an additional entry $\lambda \in \mathbb{F}$ in the (i, j) position, where $i \neq j$:

$$\begin{pmatrix} 1 & & & & \\ & \ddots & & \lambda & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

×

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Remark

Those are the matrices one obtains by performing one row operation on I_n.

1. Furthermore, left-multiplying an elementary matrix of type 1 is equivalent as multiplying a scalar multiple λ to the ith row.
2. ... of type 2 is equivalent to swapping the kth and ℓth rows.
3. ... of type 3 is equivalent to adding a λ multiple of row j to row i.

Thus performing a row operation $\cong$ left-multiplying an elementary matrix. Moreover, elementary matrices are invertible. Thus if

$$B = E_k \cdots E_1$$

is a product of elementary matrices, then we have that

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$$A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix} \iff BA\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = B\begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix},$$

since we can simply multiply B and B^{-1}.

As well, we can write and solve

$$A\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix} = I_n,$$

which is equivalent to solving

$$A'\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix} = E_k \cdots E_1 I_n,$$

where $A' = \mathcal{R}(A)$ and $A' = E_k \cdots E_1 A$. Thus, if $A' = I_n$, we found

$$A^{-1} = \begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix} = E_k \cdots E_1,$$

since

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we started from

$$A\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix} = I_n.$$

□

Remark on the Remark

By the last remark, the easier way of solving linear systems is to perform elementary row operations on the *augmented matrix*

$$(A \mid b) \longrightarrow (A' \mid b'),$$

and when $A' = \mathcal{R}(A)$, this system is much easier to solve. The reason they have the same solution is again

$$Ax = b \iff (E_k \cdots E_1)Ax = (E_k \cdots E_1)b \iff A'x = b'.$$

Likewise, we perform elementary row operations on

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$$(A \mid I_n)$$

to find the inverse of an invertible square matrix A, since

$$AB = I_n \iff (E_k \cdots E_1 A)B = E_k \cdots E_1.$$

Yet A is invertible, so

$$\mathcal{R}(A) = E_k \cdots E_1 A = I_n \implies B = E_k \cdots E_1.$$

Thus, to find $B = A^{-1}$, perform the same set of elementary row operations used to reduce A to I_n; this yields A^{-1}. ×

Example

Let

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 3 & 0 \\ 3 & 2 & 5 & 1 \\ -4 & -1 & -3 & -1 \end{pmatrix}.$$

To find its inverse, start with

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 0 & 1 & 0 & 0 \\ 3 & 2 & 5 & 1 & 0 & 0 & 1 & 0 \\ -4 & -1 & -3 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[,R_3+(-3)R_1,\ R_4+4R_1,]{,R_2+(-2)R_1,}$$

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$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & -2 & 1 & 0 & 0 \\ 0 & 2 & 2 & -2 & -3 & 0 & 1 & 0 \\ 0 & -1 & 1 & 3 & 4 & 0 & 0 & 1 \end{array}\right) \xrightarrow[,R_4+R_2,]{,R_3+(-2)R_2,} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & -2 & 1 & 0 \\ 0 & 0 & 2 & 1 & 2 & 1 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[,R_4\mapsto\frac{1}{2}R_4,\ R_3\mapsto\frac{1}{2}R_3,]{,R_3\leftrightarrow R_4,} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} & 0 \end{array}\right)$$

$$\xrightarrow[,R_2+2R_4,\ R_1+(-1)R_4,]{,R_3+(-\frac{1}{2})R_4,} \left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 1 & 1 & 0 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & \frac{3}{4} & 1 & -\frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} & 0 \end{array}\right)$$

$$\xrightarrow[,R_1+(-1)R_3,]{,R_2+(-1)R_3,} \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -\frac{1}{4} & 0 & -\frac{1}{4} & -\frac{1}{2} \\ 0 & 1 & 0 & 0 & -\frac{7}{4} & -2 & \frac{5}{4} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 & \frac{3}{4} & 1 & -\frac{1}{4} & \frac{1}{2} \\ 0 & 0 & 0 & 1 & \frac{1}{2} & -1 & \frac{1}{2} & 0 \end{array}\right).$$

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4.5 Equivalent Criteria for Invertibility

Remark

Up till now, the following statements are equivalent: for $A \in M_n(\mathbb{F})$,

1. A is invertible (and definition).
2. $L_A : \mathbb{F}^n \to \mathbb{F}^n$ is an isomorphism.
3. $\ker(L_A) = \{0\}$.
4. $\mathrm{im}(L_A) = \mathbb{F}^n$.
5. $\det(A) \neq 0$.
6. $\mathrm{rk}(A) = n$.
7. $\mathcal{R}(A) = I_n$.
8. The system $Ax = 0$ has a unique trivial solution.
9. The system $Ax = b$ has a unique solution.
10. L_A maps a basis of $\mathbb{F}^n$ to a basis of $\mathbb{F}^n$.
11. The column and row vectors of A form a basis

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of $\mathbb{F}^n$.

□

5 Dual Spaces

Definition

Let V be a finite-dimensional $\mathbb{F}$-vector space. The *dual space* of V is the $\mathbb{F}$-vector space

$$\mathrm{Hom}(V, \mathbb{F}),$$

which we denote by V^*. Elements of V^* are usually called *linear functionals*. □

Proposition

$$\dim(V^*) = \dim(V).$$

Proof.

$$\dim(V^*) = \dim \mathrm{Hom}(V, \mathbb{F}) = \dim(V) \cdot \dim(\mathbb{F}) = \dim(V).$$

□

Proposition

Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis of a finite-dimensional $\mathbb{F}$-vector space V. Then there exists a unique basis

$$\mathcal{B}^* = \{f_1, \ldots, f_n\}$$

of V^* such that

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$$f_i(v_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

This $\mathcal{B}^*$ is called the *dual basis* of $\mathcal{B}$ in V^*.

Proof. By a proposition we have shown previously, a linear transformation is uniquely determined by how it acts on the set of basis vectors. Therefore uniqueness is proven as well as existence. It is left to show that this is a basis, which showing independence would suffice as

$$\dim(V^*) = \dim(V) = n = |\mathcal{B}^*|.$$

Suppose

$$\alpha_1 f_1 + \cdots + \alpha_n f_n = 0.$$

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Applying this to $v_1, v_2, \ldots, v_n$, we get

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$$

right away. Thus it is a basis. □

Corollary

For every $v \in V$,

$$v = f_1(v)v_1 + \cdots + f_n(v)v_n.$$

Proof. We know

$$v = \sum_{j=1}^{n} c_j v_j$$

since $\{v_1, \ldots, v_n\}$ is a basis. Applying $f_1, f_2, \ldots, f_n$ to v gives

$$f_i(v) = f_i\left(\sum_{j=1}^{n} c_j v_j\right) = \sum_{j=1}^{n} c_j f_i(v_j) = c_i, \qquad i = 1, \ldots, n.$$

Replacing c_i with $f_i(v)$ gives the desired formula. □

Corollary

For every $v \in V \setminus \{0\}$, there exists $f \in V^*$ such that $f(v) \neq 0$.

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Proof. Extend $\{v\}$ to a basis of V; then f_1 in the dual basis sends v to 1. □

Example

Let $V = \mathbb{F}^n$ and let

$$\mathcal{B} = \mathrm{St} = \{e_1, \ldots, e_n\}.$$

Then $\mathrm{St}^* = \{f_1, \dots, f_n\}$ satisfies

$$f_i \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_i.$$

Thus if

$$f = \sum_{i=1}^{n} a_i f_i \in V^*,$$

it can be identified with the row vector $(a_1, \dots, a_n)$, since for every

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n,$$

$$f(x) = \sum_{i=1}^{n} a_i f_i(x) = \sum_{i=1}^{n} a_i x_i = (a_1, \dots, a_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

□

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Example

Let $V = \mathbb{R}[x]_n$ and

$$\mathcal{B} = \{1, x, \dots, x^n\}.$$

Then $\mathcal{B}^* = \{f_0, f_1, \dots, f_n\}$ satisfies

$$f_k \left(\sum_{j=0}^{n} a_j x^j \right) = a_k.$$

Thus

$$f_k(g(x)) = \frac{1}{k!} \frac{d^k g}{dx^k}(0).$$

The elements in the dual space can be identified as differential operators. By a corollary,

$$g(x) = \sum_{k=0}^{n} \frac{1}{k!} \frac{d^k g}{dx^k}(0) x^k.$$

This is Taylor's formula for the polynomial $g(x)$. □

Theorem

There is a natural isomorphism

$$V \cong V^{**} = (V^*)^*,$$

the *double dual* of V.

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5.1 The Double Dual

Remark

Recall that a finite-dimensional vector space V and another vector space W are isomorphic if and only if

$$\dim V = \dim W.$$

Furthermore, general isomorphisms, including the isomorphism mapping

$$V \longrightarrow V^*, \qquad \mathcal{B} \longmapsto \mathcal{B}^* \quad \text{(dual basis)},$$

depend on the choice of $\mathcal{B}$. Yet we claim that the canonical isomorphism

$$\varphi : V \longrightarrow V^* \longrightarrow V^{**}, \qquad \mathcal{B} \longmapsto \mathcal{B}^* \longmapsto (\mathcal{B}^*)^*,$$

is independent of the choice of $\mathcal{B}$. ☒

Proof. One standard way of proving this is to define $f_v : V^* \to \mathbb{F}$ explicitly

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for every $v \in V$, without relying on any basis. Yet I would prefer the following.

Let $\mathcal{B}$ be a basis of V. We exceptionally denote

$$\mathcal{B} = \{v_1, v_2, \dots, v_n\}, \quad \mathcal{B}^* = \{v_1^*, v_2^*, \dots, v_n^*\}, \quad \mathcal{B}^{**} = \{v_1^{**}, v_2^{**}, \dots, v_n^{**}\}$$

as the successive dual bases. By the proposition about dual bases, there are isomorphisms

$$V \xrightarrow{\sim} V^* \xrightarrow{\sim} V^{**}, \qquad v_i \longmapsto v_i^* \longmapsto v_i^{**}.$$

Call this map $\varphi : V \xrightarrow{\sim} V^{**}$. We show that φ is independent of the choice of basis. Since $\mathcal{B}$ was

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arbitrarily chosen, we just need to show that $\varphi(v)$ is fixed in V^{**}.

Let $f \in V^*$. Suppose

$$v = \sum_{i=1}^{n} a_i v_i, \qquad f = \sum_{j=1}^{n} b_j v_j^*.$$

Then

$$(\varphi(v))(f) = \underbrace{\left(\varphi\left(\sum_{i=1}^{n} a_i v_i\right)\right)}_{\in (V^*)^*} \left(\underbrace{\sum_{j=1}^{n} b_j v_j^*}_{\in V^*}\right)$$

This is a map, not multiplication.

$$= \left(\sum_{i=1}^{n} a_i v_i^{**}\right)\left(\sum_{j=1}^{n} b_j v_j^*\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j v_i^{**}(v_j^*)$$

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$$= \sum_{i=1}^{n} a_i b_i.$$

On the other hand, recalling the other property of the dual basis,

$$v = \sum_i v_i^*(v) v_i = \sum_i a_i v_i,$$

so

$$\sum_{i=1}^{n} v_i^*(v) b_i = \left(\sum_{i=1}^{n} b_i v_i^* \right)(v) = f(v).$$

Thus

$$(\varphi(v))(f) = f(v), \qquad \forall f \in V^*, \ \forall v \in V.$$

The choice of $\mathcal{B}$ does not change this map, because this equation defines a unique map in V^{**}. □

Note

When V is infinite-dimensional, the algebraic double dual explodes, and

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$$V^{**} \ncong V, \qquad \dim(V^{**}) > \dim(V).$$

We add additional structure to it (continuity), which is essentially functional analysis (study of linear functionals). For instance, the Riesz representation theorem guarantees this for Hilbert spaces (18.102). ×

5.2 Annihilators

Definition

Let V be a finite-dimensional $\mathbb{F}$-vector space and $U \leq V$. The *annihilator* of U in V^* is

$$U^\perp := \{ f \in V^* \mid f(u) = 0, \ \forall u \in U \}.$$

×

Proposition

1. $U^\perp \leq V^*$.
2. If $W \leq U \leq V$, then $U^\perp \leq W^\perp$.
3. $\dim(U^\perp) = \dim(V) - \dim(U)$.
4. Given the natural isomorphism $\varphi : V \to V^{**}$,

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$$\varphi(U) = U^{\perp\perp} := (U^\perp)^\perp.$$

Proof.

1. Clearly $0_{V^*} \in U^\perp$. If $f, g \in U^\perp$, then for every $u \in U$,

$$f(u) = g(u) = 0 \implies (f+g)(u) = 0 \implies f + g \in U^\perp.$$

Closure under scalar multiplication is an exercise.

2. Annihilators of U always annihilate W, so $U^\perp \subseteq W^\perp$. Since they are both subspaces of V^*, it follows that $U^\perp \leq W^\perp$.

3. Suppose

$$\dim(U) = m, \qquad \dim(V) = n + m.$$

Take $\mathcal{B} = \{u_1, u_2, \ldots, u_m\}$, a basis of U. By the Steinitz exchange lemma,

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we can extend it to a basis of V,

$$\mathcal{C} := \{u_1, \ldots, u_m, v_1, \ldots, v_n\}.$$

Consider the dual basis

$$\mathcal{C}^* := \{f_1, \ldots, f_m, g_1, \ldots, g_n\}.$$

We claim

$$U^\perp = \operatorname{Span}\{g_1, g_2, \ldots, g_n\}.$$

By the definition of a dual basis,

$$\operatorname{Span}\{g_1, g_2, \ldots, g_n\} \subseteq U^\perp,$$

as every g_i annihilates the basis vectors in $\mathcal{B}$. Thus for

$$g = \sum_i a_i g_i \in \operatorname{Span}\{g_1, \ldots, g_n\}, \qquad u = \sum_j b_j u_j,$$

we have

$$g(u) = \left(\sum_i a_i g_i\right)\left(\sum_j b_j u_j\right) = \sum_{i,j} a_i b_j g_i(u_j) = 0.$$

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Suppose $\operatorname{Span}\{g_1, g_2, \ldots, g_n\} \neq U^\perp$. Then there exists

$$f \in U^\perp \setminus \operatorname{Span}\{g_1, \ldots, g_n\}.$$

Since $\mathcal{C}^*$ is a basis,

$$f = \sum_{i=1}^{m} \lambda_i f_i + \sum_{j=1}^{n} \alpha_j g_j.$$

At least one of the λ_i must be nonzero, since $f \notin \operatorname{Span}\{g_i\}$. Yet then $f \notin U^\perp$: if $\lambda_s \neq 0$, where $s \in \{1, \ldots, m\}$, then

$$f_s(u_s) = 1, \qquad f(u_s) = \lambda_s \neq 0, \qquad u_s \in U,$$

a contradiction. Therefore

$$\operatorname{Span}\{g_1, \ldots, g_n\} = U^\perp,$$

so

$$\dim(U^\perp) = n = \dim(V) - \dim(U).$$

4. Recall that the natural isomorphism satisfies

$$v \longmapsto v^{**}, \qquad v^{**}(f) = f(v), \quad \forall f.$$

First we claim $\varphi(U) \subseteq (U^\perp)^\perp$. Let $u \in U$.

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Let $f \in U^\perp$. Then

$$u^{**}(f) = f(u) = 0,$$

so $u^{**} = \varphi(u) \in (U^\perp)^\perp$.

Therefore

$$\varphi(U) \subseteq (U^\perp)^\perp.$$

Furthermore, $\varphi(U) \leq (U^\perp)^\perp$, since $\varphi(U)$ is the image of an isomorphism (thus a subspace), and $(U^\perp)^\perp$ is a subspace by part 1. By part 3,

$$\dim(U^{\perp\perp}) = \dim(V^*) - \dim(U^\perp) = \dim(V) - (\dim(V) - \dim(U)) = \dim(U).$$

Also, by a proposition long ago,

$$\varphi(U) \leq U^{\perp\perp}, \qquad \dim \varphi(U) = \dim(U) = \dim(U^{\perp\perp}),$$

so

$$\varphi(U) = U^{\perp\perp}.$$

□

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Example

Let

$$U = \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\} \leq \mathbb{F}^3.$$

For every $f \in (\mathbb{F}^3)^*$,

$$\begin{aligned} f(x, y, z) &= f\big(x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)\big) \\ &= f(1, 0, 0)x + f(0, 1, 0)y + f(0, 0, 1)z. \end{aligned}$$

Then $f \in U^\perp$ if and only if

$$ax + by + cz = 0, \qquad \forall (x, y, z) \in U.$$

For instance, the functional

$$f(x, y, z) := x + y + z$$

is in $U^\perp$. More generally, every functional of the form

$$f(x, y, z) := ax + ay + az, \qquad a \in \mathbb{F},$$

is in $U^\perp$. On the other hand,

$$\dim(U^\perp) = \dim(\mathbb{F}^3) - \dim(U) = 3 - 2 = 1.$$

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Thus, under the row-vector identification,

$$U^\perp \cong \text{Span}\{(1,1,1)\}.$$

×

Remark

By the proof of the last proposition, part 3, if $U \leq V$, $\{u_1, \ldots, u_m\}$ is a basis of U, and it is extended to a basis with $\{v_1, \ldots, v_n\}$, then

$$\{v_1^*, \ldots, v_n^*\}$$

is a basis of $U^\perp$. ×

Proposition

Let $U, W \leq V$ with $\dim(V) < \infty$. Then

$$(U+W)^\perp = U^\perp \cap W^\perp, \qquad (U \cap W)^\perp = U^\perp + W^\perp.$$

Proof. The first identity is trivial from the definition. An exercise is to show the second using part 4 of the last proposition instead of using the definition. □

Proposition

Let $U \leq V$ and $\dim V < \infty$. There exists a natural

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isomorphism

$$U^* \cong V^*/U^\perp.$$

Proof. Consider the natural isomorphism

$$\begin{aligned} \varphi : V^*/U^\perp &\longrightarrow U^*, \\ f + U^\perp &\longmapsto f|_U, \end{aligned}$$

where $f|_U$ is f restricted to U.
φ is well-defined: suppose

$$f + U^\perp = g + U^\perp, \qquad f, g \in V^*.$$

Then

$$f - g \in U^\perp \implies (f-g)|_U = 0 \implies f|_U = g|_U \implies \varphi(f+U^\perp) = \varphi(g+U^\perp).$$

φ is linear: let $\lambda \in \mathbb{F}$, $f, g \in V^*$. Then

$$\begin{aligned} \varphi(\lambda(f+U^\perp) + (g+U^\perp)) &= \varphi((\lambda f + g) + U^\perp) \\ &= (\lambda f + g)|_U \\ &= \lambda(f|_U) + g|_U. \end{aligned}$$

φ is injective: let $f + U^\perp \in V^*/U^\perp$ such that

$$\varphi(f + U^\perp) = 0_U.$$

Thus $f|_U = 0_U$.

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Hence

$$f \in U^\perp \implies f + U^\perp = U^\perp \qquad \text{(zero element in } V^*/U^\perp\text{)}.$$

φ is surjective: by the second-last proposition, a proposition proved long ago, and the dimension theorem,

$$\begin{aligned}\dim(V^*/U^\perp) &= \dim(V^*) - \dim(U^\perp) \\ &= \dim(V) - \dim(V) + \dim(U) \\ &= \dim(U),\end{aligned}$$

and

$$\begin{aligned}\dim(\operatorname{im}\varphi) &= \dim(V^*/U^\perp) - \dim(\ker\varphi) \\ &= \dim(U) - 0 \\ &= \dim(U^*).\end{aligned}$$

Thus

$$\operatorname{im}\varphi = U^*,$$

so φ is surjective. Therefore

$$V^*/U^\perp \cong U^*.$$

□

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we have

$$(V/U)^* \cong U^\perp.$$

Proof. Define

$$\begin{aligned}\psi : U^\perp &\longrightarrow (V/U)^*, \\ f &\longmapsto \psi(f),\end{aligned} \qquad \psi(f) : v + U \longmapsto f(v).$$

This map is well-defined: if $v + U = v' + U$, then $v - v' \in U$, so $f(v - v') = 0$ and hence $f(v) = f(v')$. It is linear. If $\psi(f) = 0$, then $f(v) = 0$ for every $v \in V$, so $f = 0$; thus ψ is injective. Finally,

$$\dim U^\perp = \dim V - \dim U = \dim(V/U) = \dim(V/U)^*.$$

Therefore ψ is surjective and hence is the desired natural isomorphism. □

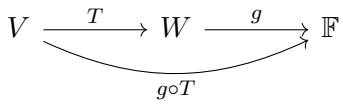
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5.3 Dual Maps

Definition

Let $T : V \to W$ be a linear transformation between finite-dimensional $\mathbb{F}$-vector spaces. The *dual transformation*, also called the *transpose of* T, is the map

$$\begin{aligned}T^* : W^* &\longrightarrow V^*, \\ g &\longmapsto g \circ T.\end{aligned}$$



Proposition: Assignment 6

T^* is linear. Furthermore, if $\mathcal{B}, \mathcal{C}$ are ordered bases of V, W, then

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$$[T^*]_{\mathcal{C}^*}^{\mathcal{B}^*} = \left([T]_{\mathcal{B}}^{\mathcal{C}}\right)^t.$$

Proposition

Let $T : V \to W$ be a linear transformation between finite-dimensional $\mathbb{F}$-vector spaces. Then

$$\operatorname{im}(T^*) = (\ker T)^{\perp}.$$

Proof. A basis extension argument. Let

$$\{v_1, \ldots, v_r\}$$

be a basis of $\ker(T)$. Expand it to a basis $\mathcal{B}$ of the vector space V, say

$$\mathcal{B} = \{v_1, \ldots, v_r, v_{r+1}, \ldots, v_n\}.$$

Again, by part (3) of a proposition 3/4 propositions ago, we have that

$$\{v_{r+1}^*, \ldots, v_n^*\}$$

is a basis of $(\ker T)^{\perp}$. Thus it is left to show that

$$\operatorname{im}(T^*) = \operatorname{Span}\{v_{r+1}^*, \ldots, v_n^*\}.$$

Let $g \in \operatorname{im}(T^*) \subseteq V^*$. Since they are all elements in V^*,

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it makes sense to look at how they act upon a vector in V. Let $v \in V$. Then

$$v = \sum_{i=1}^{n} a_i v_i.$$

Since $g \in \operatorname{im}(T^*)$, there exists $f \in W^*$ such that

$$T^*(f) = f \circ T = g.$$

Hence

$$\begin{aligned} f \circ T(v) &= f \circ T\left(\sum_{i=1}^{n} a_i v_i\right) \\ &= \sum_{i=1}^{r} a_i f(Tv_i) + \sum_{i=r+1}^{n} a_i f(Tv_i) \\ &= \sum_{i=r+1}^{n} a_i f(Tv_i) \qquad (\text{since } v_1, \ldots, v_r \in \ker T) \\ &= \sum_{i=r+1}^{n} f(Tv_i) v_i^*(v) \\ &= \left(\sum_{i=r+1}^{n} f(Tv_i) v_i^*\right)(v). \end{aligned}$$

Therefore

$$T^*(f) = f \circ T = \sum_{i=r+1}^{n} f(Tv_i) v_i^*,$$

and

$$\operatorname{im}(T^*) \subseteq \operatorname{Span}\{v_{r+1}^*, \ldots, v_n^*\} = (\ker T)^{\perp}.$$

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For the other containment,

$$(\ker T)^{\perp} \subseteq \operatorname{im}(T^*),$$

first notice that

$$\begin{aligned} \operatorname{im} T &= T(V) = T(\operatorname{Span}\{v_1, \ldots, v_n\}) \\ &= \operatorname{Span}\{Tv_1, \ldots, Tv_n\} \qquad (\text{linearity of } T) \\ &= \operatorname{Span}\{Tv_1, \ldots, Tv_r, Tv_{r+1}, \ldots, Tv_n\} \\ &= \operatorname{Span}\{Tv_{r+1}, \ldots, Tv_n\}, \end{aligned}$$

since $Tv_1 = \cdots = Tv_r = 0$. Thus

$$\{Tv_{r+1}, \ldots, Tv_n\}$$

spans $\operatorname{im} T$. Furthermore, by the dimension theorem,

$$\begin{aligned} \dim(\operatorname{im} T) &= \dim(V) - \dim(\ker T) \\ &= n - r = |\{Tv_{r+1}, \ldots, Tv_n\}|. \end{aligned}$$

Hence $\{Tv_{r+1}, \ldots, Tv_n\}$ is a basis of $\operatorname{im}(T)$, and we can extend it to a basis

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of W:

$$\{Tv_{r+1}, \ldots, Tv_n, w_1, \ldots, w_s\}.$$

Now let

$$\sum_{i=r+1}^{n} a_i v_i^* \in (\ker T)^{\perp}.$$

By a proposition proved long ago, we can get a well-defined $f \in W^*$ by defining how it acts on the basis elements:

$$f \circ T(v_{r+1}) = a_{r+1}, \ldots, \quad f \circ T(v_n) = a_n,$$

and

$$f(w_1) = \cdots = f(w_s) = 0.$$

This gives a well-defined $f \in W^*$. Moreover,

$$T^*(f) = \sum_{i=r+1}^{n} f \circ T(v_i) v_i^* = \sum_{i=r+1}^{n} a_i v_i^*.$$

Therefore

$$(\ker T)^\perp \subseteq \operatorname{im}(T^*),$$

and hence

$$(\ker T)^\perp = \operatorname{im}(T^*).$$

Alternative. we can argue that

$$\dim(\operatorname{im} T^*) = n - r.$$

□

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We present a new proof to a theorem we proved before with the new machineries we have:

Theorem

$$\forall A \in M_{m\times n}(\mathbb{F}), \qquad \operatorname{rank}_C(A) = \operatorname{rank}_r(A).$$

Proof. By propositions we seen so far, we have

$$L_A^* : (\mathbb{F}^m)^* \longrightarrow (\mathbb{F}^n)^*$$

has matrix

$$[L_A^*]_{\mathrm{St}^*}^{\mathrm{St}^*} = \left([L_A]_{\mathrm{St}}^{\mathrm{St}}\right)^t.$$

Thus

$$\begin{aligned}
\operatorname{rank}_r(A) &= \operatorname{rank}_C(A^t) = \dim\big(\operatorname{im}(L_{A^t})\big) \\
&= \dim\big(\operatorname{im}(L_A^*)\big) = \dim\big((\ker L_A)^\perp\big) \\
&= n - \dim(\ker L_A) = \dim\big(\operatorname{im}(L_A)\big) \\
&= \operatorname{rank}_C(A).
\end{aligned}$$

□

6 Eigenvalues, Eigenvectors, Diagonalization

Definition

Let V be an $\mathbb{F}$-vector space and $T : V \to V$

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be a linear transformation. An *eigenvector* for T is a

$$v \in V \setminus \{0\}$$

such that

$$T(v) = \lambda v$$

for some $\lambda \in \mathbb{F}$. λ is called the *eigenvalue of* T.

For $A \in M_{n\times n}(\mathbb{F})$, an eigenvalue (vector) for A is an eigenvalue (vector) of

$$L_A : \mathbb{F}^n \longrightarrow \mathbb{F}^n.$$

Example

$$\begin{pmatrix} 1 & 2 & -2 \\ 0 & -1 & 0 \\ 1 & 1 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}.$$

Example

Let

$$T : \mathbb{R}[x] \longrightarrow \mathbb{R}[x]$$

be the differential operator. Then $1 \in \mathbb{R}[x]$ is an eigenvector (sometimes we also call eigenfunction) of T with eigenvalue 0.

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Example

For a general $\mathbb{F}$-vector space V with $T : V \to V$, as long as $\ker(T)$ is not the trivial subspace, $\lambda = 0$ is an eigenvalue of T.

Remark and Discussion

By a previous proposition, we know that if V is a finite-dimensional $\mathbb{F}$-vector space with $\mathcal{B}$ an ordered basis, then $\forall v \in V$, $\lambda \in \mathbb{F}$,

$$Tv = \lambda v \iff [T]_{\mathcal{B}}^{\mathcal{B}}[v]_{\mathcal{B}} = [Tv]_{\mathcal{B}} = \lambda[v]_{\mathcal{B}}.$$

Thus λ is an eigenvalue for T if and only if it is an eigenvalue for $[T]_{\mathcal{B}}^{\mathcal{B}}$.

Moreover, if λ is an eigenvalue for $A \in M_n(\mathbb{F})$, and $Q \in GL_n(\mathbb{F})$, then λ is an eigenvalue for $Q^{-1}AQ$. This

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comes from the change of basis formula

$$[T]_{\mathcal{C}}^{\mathcal{C}} = [\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}}[T]_{\mathcal{B}}^{\mathcal{B}}[\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}.$$

Noticing that $Q = (v_1, \dots, v_n)$ being invertible implies its columns form a basis $\mathcal{B}$, we get

$$\begin{aligned}[L_A]_{\mathcal{B}}^{\mathcal{B}} &= [\mathrm{id}]_{\mathrm{St}}^{\mathcal{B}}[L_A]_{\mathrm{St}}^{\mathrm{St}}[\mathrm{id}]_{\mathcal{B}}^{\mathrm{St}} \\ &= Q^{-1}AQ.\end{aligned}$$

Thus by the last part, the eigenvalue λ of L_A is also an eigenvalue for

$$[L_A]_{\mathcal{B}}^{\mathcal{B}} = Q^{-1}AQ.$$

Furthermore, if v is an eigenvector of A, then $Q^{-1}v$ is an eigenvector of $Q^{-1}AQ$ with the same eigenvalue:

$$Q^{-1}AQ(Q^{-1}v) = Q^{-1}Av = Q^{-1}(\lambda v) = \lambda Q^{-1}v.$$

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6.1 Characteristic Polynomials

Definition

Let V be a finite-dimensional $\mathbb{F}$-vector space, $T : V \to V$ be a linear transformation. The *characteristic polynomial* of T is

$$P_T(t) := \det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right) \in \mathbb{F}[t],$$

where t is a variable, $\mathcal{B}$ is any ordered basis of V. The *characteristic polynomial* of $A \in M_n(\mathbb{F})$ is

$$P_A(t) := \det(tI_n - A) \in \mathbb{F}[t].$$

Example

For

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 0 & -1 & 0 \\ 1 & 1 & 4 \end{pmatrix},$$

$$\begin{aligned}P_A(t) &= \det \begin{pmatrix} t-1 & -2 & 2 \\ 0 & t+1 & 0 \\ -1 & -1 & t-4 \end{pmatrix} \\ &= (t+1)\det \begin{pmatrix} t-1 & 2 \\ -1 & t-4 \end{pmatrix} \\ &= (t+1)((t-1)(t-4)+2).\end{aligned}$$

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Notation

Let $A \in M_{m\times n}(\mathbb{F})$. From now on, we stop distinguishing between A and

$$L_A : \mathbb{F}^n \longrightarrow \mathbb{F}^m, \qquad v \longmapsto Av.$$

Thus

$$\ker(A) := \ker(L_A), \qquad \mathrm{im}(A) := \mathrm{im}(L_A).$$

Proposition

Let V be an $\mathbb{F}$-vector space, $\dim V = n < \infty$. Let $T : V \to V$ be a linear transformation. Then:

1. $$P_T(t) := \det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right)$$
   is independent from the choice of the ordered basis $\mathcal{B}$.
2. Let
   $$A := [T]_{\mathcal{B}}^{\mathcal{B}}$$
   for any ordered basis $\mathcal{B}$ of V. Then
   $$P_T(t) = P_A(t) = t^n - \operatorname{tr}(A)t^{n-1} + \cdots + (-1)^n \det(A).$$

[One can find a concise expression for the middle terms, but that is not so useful.]

Proof. **1.** Let $\mathcal{C}$ be another ordered basis of V. Then, by change of basis matrix,

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$$\begin{aligned}
\det\left(tI_n - [T]_{\mathcal{C}}^{\mathcal{C}}\right) &= \det\left(tI_n - [\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}}[T]_{\mathcal{B}}^{\mathcal{B}}[\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right) \\
&= \det\left([\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}}\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right)[\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right) \\
&= \det\left([\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}}\right)\det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right)\det\left([\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right) \\
&= \det\left(\left([\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right)^{-1}\right)\det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right)\det\left([\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right) \\
&= \det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right),
\end{aligned}$$

since

$$[\mathrm{id}]_{\mathcal{B}}^{\mathcal{C}} = \left([\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}\right)^{-1}.$$

2. $P_T(t) = P_A(t)$ follows from the definition of characteristic polynomials. Recall the definition of determinants:

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$$P_T(t) = P_A(t) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) b_{\sigma(1)1} \cdots b_{\sigma(n)n},$$

where

$$tI_n - A =: (b_{ij})_{i,j=1}^n.$$

If $\sigma \neq e$, then there exist indices $i \neq j$ such that

$$\sigma(i) \neq i, \qquad \sigma(j) \neq j.$$

Thus if $\sigma \neq e$,

$$\deg(b_{\sigma(1)1} \cdots b_{\sigma(n)n}) \leq n - 2$$

(two entries are off the diagonal, yet all the terms with a variable live on the diagonal). Then

$$\begin{aligned}
P_A(t) &= \operatorname{sgn}(e) b_{11} \cdots b_{nn} + \sum_{\sigma \in S_n \setminus \{e\}} \operatorname{sgn}(\sigma) b_{\sigma(1)1} \cdots b_{\sigma(n)n} \\
&= (t - a_{11})(t - a_{22}) \cdots (t - a_{nn}) + \sum_{\sigma \in S_n \setminus \{e\}} \cdots .
\end{aligned}$$

In the first term, notice the degree n term must be t^n. Moreover, the

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degree $(n-1)$ terms must have exactly $(n-1)$ t's from $(n-1)$ factors, and pick up a $-a_{ii}$ from the remaining one factor. Thus

$$\text{degree } n-1 \text{ terms} = \sum_i (-a_{ii}) t^{n-1} = -\operatorname{tr}(A) t^{n-1}.$$

To see that the constant term is $(-1)^n \det(A)$, plug in $t = 0$ to

$$P_A(t) = \det(tI_n - A) = \sum_{i=0}^{n} \text{coefficient}(i) t^i$$

and deduce that

$$\begin{aligned} \text{coefficient}(0) &= \text{constant term} \\ &= \det(-A) \\ &= (-1)^n \det(A). \end{aligned}$$

□

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Example

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{F}),$$

$$P_A(t) = t^2 - (a+d)t + ad - bc.$$

Proposition

Let V be an $\mathbb{F}$-vector space, $\dim(V) = n < \infty$, and $T : V \to V$. The following are equivalent:

1. $\lambda \in \mathbb{F}$ is an eigenvalue of T;
2. $$\ker(T - \lambda \operatorname{id}) \neq \{0\};$$
3. $$P_T(\lambda) = 0$$

   (λ is a root of the characteristic polynomial).

Proof. We will write out the equivalences not at once.

$$\begin{aligned} (1) &\iff \lambda \text{ is an eigenvalue of } T \\ &\iff \exists v \in V \setminus \{0\} \text{ such that } Tv = \lambda v \\ &\iff \exists v \in V \setminus \{0\} \text{ such that } (T - \lambda \operatorname{id})v = 0 \\ &\iff (2). \end{aligned}$$

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Let $\mathcal{B}$ be an ordered basis of V. Then

$$\begin{aligned}(1) &\iff \exists v \in V \setminus \{0\} \text{ such that } [T]_{\mathcal{B}}^{\mathcal{B}}[v]_{\mathcal{B}} = [Tv]_{\mathcal{B}} = \lambda[v]_{\mathcal{B}} \\ &\iff \exists [v]_{\mathcal{B}} \in \mathbb{F}^n \setminus \{0\} \text{ such that } \left([T]_{\mathcal{B}}^{\mathcal{B}} - \lambda I_n\right)[v]_{\mathcal{B}} = 0 \\ &\iff \ker\left([T]_{\mathcal{B}}^{\mathcal{B}} - \lambda I_n\right) = \ker\left(\lambda I_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right) \neq \{0\} \\ &\iff \lambda I_n - [T]_{\mathcal{B}}^{\mathcal{B}} \text{ is not invertible (as a matrix)} \\ &\iff \det\left(\lambda I_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right) = P_T(\lambda) = 0 \\ &\iff (3).\end{aligned}$$

□

Corollary

With T and V as above, T has at most $\dim(V)$ eigenvalues.

Proof. By a result from Algebra 1, we know $P_T(t)$ must have

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at most as many roots as its degree. Then it follows from the previous proposition. □

6.2 Eigenspaces and Multiplicities

Definition

Let $T : V \to V$ be a linear transformation, $\dim(V) < \infty$ ($A \in M_n(\mathbb{F})$). If $\lambda \in \mathbb{F}$ is an eigenvalue of T (of A), we call the set

$$E_\lambda := \{v \in V \mid Tv = \lambda v\} \qquad (Av = \lambda v)$$

the *eigenspace of* λ.

Remark

$$E_\lambda = \ker(\lambda \operatorname{id} - T),$$

thus $E_\lambda \leq V$. This is exactly how we find the eigenspaces!

Example

Let

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 0 & -1 & 0 \\ 1 & 1 & 4 \end{pmatrix}, \qquad P_A(t) = (t+1)(t-2)(t-3).$$

Thus the eigenvalues are $-1, 2, 3$.

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The eigenspaces are

$$\begin{aligned} E_{-1} &= \ker(-I_3 - A) \\ &= \ker \begin{pmatrix} -2 & -2 & 2 \\ 0 & 0 & 0 \\ -1 & -1 & -5 \end{pmatrix} \\ &= \operatorname{Span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}, \end{aligned}$$

$$\begin{aligned} E_2 &= \ker(2I_3 - A) \\ &= \ker \begin{pmatrix} 1 & -2 & 2 \\ 0 & 3 & 0 \\ -1 & -1 & -2 \end{pmatrix} \\ &= \operatorname{SolutionSpace} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 3 & 0 \\ -1 & -1 & -2 \end{pmatrix} \\ &= \operatorname{Span} \left\{ \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \right\}, \end{aligned}$$

and

$$E_3 = \ker(3I_3 - A) = \operatorname{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}.$$

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Example

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R}).$$

$$P_A(t) = t^2 + 1,$$

which has no roots in $\mathbb{R}$. Thus A has no eigenvalues or eigenvectors over $\mathbb{R}$. But if $A \in M_2(\mathbb{C})$, then

$$P_A(t) = (t - i)(t + i),$$

so

$$E_i = \ker(iI_2 - A) = \operatorname{Span} \left\{ \begin{pmatrix} i \\ 1 \end{pmatrix} \right\},$$

$$E_{-i} = \ker(-iI_2 - A) = \operatorname{Span} \left\{ \begin{pmatrix} 1 \\ i \end{pmatrix} \right\}.$$

Definition

Let V be a finite-dimensional $\mathbb{F}$-vector space, $T : V \to V$ be a linear transformation, and $\lambda \in \mathbb{F}$ an eigenvalue of T. The *geometric multiplicity* is

$$m_g(\lambda) := \dim(E_\lambda).$$

The *algebraic multiplicity* $m_a(\lambda)$ is

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given by the unique factorization theorem:

$$P_T(t) = (t - \lambda)^{m_a(\lambda)} g(t),$$

where

$$g(t) \in \mathbb{F}[t], \qquad g(\lambda) \neq 0.$$

Proposition

With the notation in the definition,

$$1 \leq m_g(\lambda) \leq m_a(\lambda) \leq \dim V.$$

Example

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in M_2(\mathbb{F}).$$

$$P_A(t) = (t-1)^2, \qquad m_a(1) = 2.$$

Yet

$$E_1 = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

which means

$$\dim(E_1) = 1.$$

Proof. Since λ is an eigenvalue, $E_\lambda \neq \{0\}$. Thus

$$m_g(\lambda) = \dim(E_\lambda) \geq 1.$$

Since

$$\deg(P_T(t)) = \dim(V),$$

we have

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$$m_a(\lambda) \leq \deg(P_T(t)) = \dim V.$$

Let $\{v_1, \ldots, v_k\}$ be a basis of E_λ. Expand it to a basis

$$\mathcal{B} = \{v_1, \ldots, v_n\}$$

for V. Then

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda I_k & M \\ 0 & N \end{pmatrix}.$$

Therefore

$$\begin{aligned} P_T(t) &= \det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right) \\ &= \det\begin{pmatrix} (t-\lambda)I_k & -M \\ 0 & tI_{n-k} - N \end{pmatrix} \\ &= (t-\lambda)^k \det(tI_{n-k} - N) \qquad \text{(by Question 4 of Homework 4).} \end{aligned}$$

Therefore

$$(t-\lambda)^k \mid P_T(t) \implies m_a(\lambda) \geq k \implies k = m_g(\lambda) \leq m_a(\lambda).$$

□

6.3 Diagonalization

Definition

We say a linear transformation $T : V \to V$ is

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diagonalizable if there exists an ordered basis $\mathcal{B}$ of V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}}$$

is a diagonal matrix. We say $A \in M_{n\times n}(\mathbb{F})$ is *diagonalizable* if there exists an invertible matrix

$$Q \in M_n(\mathbb{F})$$

such that

$$Q^{-1}AQ \text{ is a diagonal matrix.}$$

Remark

For a matrix $A \in M_{n\times n}(\mathbb{F})$, it is diagonalizable if and only if

$$L_A : \mathbb{F}^n \longrightarrow \mathbb{F}^n$$

is diagonalizable. Hint:

$$[L_A]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}]_{\mathrm{St}}^{\mathcal{B}} [L_A]_{\mathrm{St}}^{\mathrm{St}} [\mathrm{id}]_{\mathcal{B}}^{\mathrm{St}}.$$

Lemma

Let $T : V \to V$ be a linear transformation and $\mathcal{B}$ an ordered basis for V. Then

$$[T]_{\mathcal{B}}^{\mathcal{B}} \text{ is diagonal} \iff \text{every } v \in \mathcal{B} \text{ is an eigenvector for } T.$$

Proof. Trivial from

$$[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}}, [Tv_2]_{\mathcal{B}}, \ldots, [Tv_n]_{\mathcal{B}}).$$

□

Corollary

With same setup as last proposition,

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Lemma 6.1. *Let $\overline{T} : V \longrightarrow V$ be a linear transformation and let $\lambda_1, \ldots, \lambda_k$ be distinct eigenvalues for T. For each $1 \leq i \leq k$, let $L_i \subseteq E_{\lambda_i}$ be a linearly independent subset. Then*

$$L_1 \cup L_2 \cup \cdots \cup L_k$$

is linearly independent.

Proof. **Claim.** The lemma holds for L_i singletons. Suppose

$$L_i = \{w_i\}, \qquad i = 1, 2, \ldots, k.$$

(Induction on k.) For $k = 1$: a linearly independent subset is itself independent.
Suppose it holds for $k - 1 \in \mathbb{N}$. Suppose

$$a_1 w_1 + a_2 w_2 + \cdots + a_k w_k = 0. \tag{$*$}$$

Apply T; we have

$$T(a_1 w_1 + a_2 w_2 + \cdots + a_k w_k) = 0,$$

so

$$a_1 \lambda_1 w_1 + a_2 \lambda_2 w_2 + \cdots + a_k \lambda_k w_k = 0. \tag{$\dagger$}$$

Multiply λ_k to $(*)$; we have

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$$a_1 \lambda_k w_1 + a_2 \lambda_k w_2 + \cdots + a_k \lambda_k w_k = 0.$$

Subtract $(\dagger)$ from this; we get

$$a_1(\lambda_1 - \lambda_k) w_1 + a_2(\lambda_2 - \lambda_k) w_2 + \cdots + a_k(\lambda_k - \lambda_k) w_k = 0,$$

hence

$$a_1(\lambda_1 - \lambda_k) w_1 + \cdots + a_{k-1}(\lambda_{k-1} - \lambda_k) w_{k-1} = 0.$$

By the inductive hypothesis,

$$a_i(\lambda_i - \lambda_k) = 0, \qquad i = 1, \ldots, k - 1.$$

Since $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues,

$$\lambda_i - \lambda_k \neq 0 \qquad \text{for } i = 1, \ldots, k - 1.$$

Therefore

$$a_1 = a_2 = \cdots = a_{k-1} = 0.$$

Thus $a_k = 0$.

Suppose

$$\{v_{i1}, v_{i2}, \ldots, v_{in_i}\} = L_i.$$

Suppose

$$\underbrace{a_{11}v_{11} + \cdots + a_{1n_1}v_{1n_1}}_{E_{\lambda_1}} + \cdots + \underbrace{a_{k1}v_{k1} + \cdots + a_{kn_k}v_{kn_k}}_{E_{\lambda_k}} = 0.$$

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Then define

$$\sum_{j=1}^{n_i} a_{ij}v_{ij} = w_i.$$

Clearly $w_i \in E_{\lambda_i}$. Then

$$w_1 + \cdots + w_k = 0.$$

After discarding any zero w_i, the remaining w_i are eigenvectors belonging to distinct eigenvalues. The claim says that no nonempty collection of these vectors can sum to zero. Hence

$$w_1 = \cdots = w_k = 0.$$

Yet the sets L_i are all linearly independent. Therefore it follows that all the coefficients are zero. □

Remark. If $\lambda \neq \mu \in \mathbb{F}$ are distinct eigenvalues of a linear operator $\overline{T} : V \longrightarrow V$, then

$$E_\lambda \cap E_\mu = \{0\}.$$

If there exists $v \in E_\lambda \cap E_\mu$, $v \neq 0$, then

$$\overline{T}v = \lambda v = \mu v \quad \Longrightarrow \quad (\lambda - \mu)v = 0 \quad \Longrightarrow \quad \lambda - \mu = 0 \quad \Longrightarrow \quad \lambda = \mu.$$

Theorem 6.2. *Let V be a finite-dimensional $\mathbb{F}$-vector space, and let $\overline{T}$ be a linear operator on V. Then T is diagonalizable if and only if*

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1) $p_T(t)$ splits over $\mathbb{F}$ (into linear factors);
2) for each root λ of $p_T(t)$, we have

$$m_g(\lambda) = m_a(\lambda).$$

Proof. ($\Rightarrow$) Suppose diagonalizable. Then by lemma, there exists $\mathcal{B}$ a basis of V consisting of eigenvectors of $\overline{T}$. Then there exist $\lambda_1, \ldots, \lambda_k$ eigenvalues of $\overline{T}$, and

$$v_{i1}, \ldots, v_{in_i}$$

eigenvectors of eigenvalue λ_i. Since

$$\mathcal{B} = \{v_{11}, \ldots, v_{1n_1}, v_{21}, \ldots, v_{2n_2}, \ldots, v_{k1}, \ldots, v_{kn_k}\}$$

is a basis, it follows that

$$\{v_{i1}, \ldots, v_{in_i}\} \subseteq E_{\lambda_i}$$

is linearly independent. Hence

$$\dim(E_{\lambda_i}) = m_g(\lambda_i) \geq n_i.$$

Yet

$$\begin{aligned} p_T(t) &= \det\left(tI_n - [T]_{\mathcal{B}}^{\mathcal{B}}\right) \\ &= \prod_{i=1}^{k}\prod_{j=1}^{n_i}(t-\lambda_i) = \prod_{i=1}^{k}(t-\lambda_i)^{n_i}. \end{aligned} \tag{1}$$

Thus

$$m_a(\lambda_i) = n_i, \qquad i = 1, 2, \ldots, k.$$

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By a proposition before,

$$m_g(\lambda_i) \leq m_a(\lambda_i).$$

Thus

$$m_g(\lambda_i) = m_a(\lambda_i) = n_i. \tag{2}$$

($\Leftarrow$) Suppose (1) and (2). Since λ is an eigenvalue if and only if $p_T(\lambda) = 0$, we know

$$\#\text{ eigenvalues} = \#\text{ distinct roots of } p_T(t),$$

and

$$\sum_{\substack{\lambda \text{ eigenvalue} \\ \text{of } T}} m_a(\lambda) = \deg p_T(t) = \dim(V) \qquad \text{by (1)}.$$

Since $m_a(\lambda) = m_g(\lambda)$, we know

$$\sum_{\lambda \text{ eigenvalue}} m_g(\lambda) = \sum_{\lambda \text{ eigenvalue}} \dim(E_\lambda) = \dim(V).$$

By the lemma, we can pick bases of the spaces E_λ, and their union is linearly independent. A set with the same cardinality as $\dim(V)$ forms a basis of V. Then by the other lemma, $\overline{T}$ is diagonalizable. □

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Remark. By the previous remark, a scalar is an eigenvalue of $A \in M_{n\times n}(\mathbb{F})$ if and only if it is an eigenvalue of L_A, and a nonzero vector is an eigenvector of A if and only if it is an eigenvector of L_A as a linear operator on $\mathbb{F}^n$. Thus, by applying the previous theorem to L_A, we also have

$$A \in M_n(\mathbb{F}) \text{ is diagonalizable} \iff (1), (2).$$

~~Corollary~~

Proposition 6.3. *If $\dim(V) = n$ and $T : V \longrightarrow V$ has n distinct eigenvalues, then T is diagonalizable.*

Proof. Pick an eigenvector for each eigenvalue; then clearly T is a diagonal matrices under the basis consisting of these vectors. □

Remark. By the theorems and lemmas established above, we have the Diagonalization algorithm:

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1. Compute $p_T(t)$ and factor completely over $\mathbb{F}$. If it does not split, then T is not diagonalizable.
2. For each eigenvalue (root) λ, compute a basis for

$$E_\lambda = \ker(T - \lambda \operatorname{Id}) = \ker([T]_{\mathcal{B}}^{\mathcal{B}} - \lambda \operatorname{Id}).$$

If $m_g(\lambda) = m_a(\lambda)$ for every λ, take a basis for all E_λ, take the union, and compute $[T]$ under this new basis. Otherwise T is not diagonalizable.

Example

Let

$$T : \mathbb{R}[x]_2 \longrightarrow \mathbb{R}[x]_2$$

be

$$T(f(x)) = f(0) - f''(x).$$

Then for

$$\mathcal{C} = \{1, x, x^2\},$$

we have

$$[T]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

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We compute

$$p_T(t) = \det(tI_3 - [T]_{\mathcal{C}}^{\mathcal{C}}) = t^2(t-1).$$

Thus $\lambda = 0, \lambda = 1$ are eigenvalues of T, and

$$m_a(0) = 2, \qquad m_a(1) = 1.$$

Compute the eigenvectors:

$$\begin{aligned} E_1 &= \ker(I_3 - [T]_{\mathcal{C}}^{\mathcal{C}}) \\ &= \ker \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \cdots = \operatorname{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}, \\ E_0 &= \ker([T]_{\mathcal{C}}^{\mathcal{C}}) \\ &= \ker \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \cdots \\ &= \operatorname{Span} \left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}. \end{aligned}$$

Thus

$$m_g(0) = 2, \qquad m_g(1) = 1.$$

Both equal m_a, thus T is

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diagonalizable. Then notice

$$e\begin{pmatrix}1\\0\\0\end{pmatrix}=1, \qquad e\begin{pmatrix}2\\0\\1\end{pmatrix}=2+x^2, \qquad e\begin{pmatrix}0\\1\\0\end{pmatrix}=x$$

are the eigenvectors of $1,0,0$ respectively of T. Thus for

$$\mathcal{B}=\{1,2+x^2,x\},$$

$$[T]_{\mathcal{B}}^{\mathcal{B}}=\begin{pmatrix}1&0&0\\0&0&0\\0&0&0\end{pmatrix} \qquad \longrightarrow \qquad \text{corresponding eigenvalues.}$$

6.4 Polynomial Evaluation and Cayley–Hamilton

Notation

For $f(t)\in\mathbb{F}[t]$ a polynomial and

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$A\in M_{n\times n}(\mathbb{F})$, if

$$f(t)=\sum_{i=0}^{m}a_it^i,$$

then

$$f(A):=\sum_{i=0}^{m}a_iA^i\in M_{n\times n}(\mathbb{F}).$$

Denote

$$\mathbb{F}[A]:=\left\{\sum_{i=0}^{n}a_iA^i \;\middle|\; a_i\in\mathbb{F},\ n\in\mathbb{N}\right\}.$$

If $B=(b_{ij})\in M_n(\mathbb{F})$, then Bt^m denotes the matrix whose (i,j)-entry is $b_{ij}t^m$.

Observation

1. For all $f(t),g(t)\in\mathbb{F}[t]$, for all $A\in M_n(\mathbb{F})$,

$$f(A)+g(A)=(f+g)(A), \qquad f(A)g(A)=(fg)(A).$$

2. $\mathbb{F}[A]$ is a ring, hence a subring and also a subspace of $M_n(\mathbb{F})$.
3. By (1) and (2), the map

$$\varphi:\mathbb{F}[t]\longrightarrow\mathbb{F}[A], \qquad f(t)\longmapsto f(A)$$

is a ring homomorphism.

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Lemma 6.4. *Let $A \in M_n(\mathbb{F})$ and $f(t) \in \mathbb{F}[t]$ monic. Then there exist $B_0, B_1, \dots, B_a \in M_n(\mathbb{F})$ such that $B_a \neq 0$,*

$$(tI_n - A)(B_a t^a + B_{a-1}t^{a-1} + \cdots + B_0) = f(t) \cdot I_n \tag{$*$}$$

if and only if

$$f(A) = 0.$$

Remark (A matter of notation). False proof: Plug in A to both side of $(*)$; we have $f(A) = 0$. Why? $(*)$ is an equation in $M_n(\mathbb{F})[t]$, with t central and with matrix coefficients. The evaluation rule

$$\sum_i B_i t^i \longmapsto \sum_i B_i A^i$$

is not multiplicative in general, because its matrix coefficients need not commute with A; for example,

$$AB_aA^a \neq B_aA^{a+1}$$

in general (matrix

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multiplication is non commutative). Therefore substituting $t = A$ into the unexpanded product is invalid.

Proof. $(\Rightarrow)$ Compute

$$\begin{aligned}&(tI_n - A)(B_a t^a + \cdots + B_0)\\ &= B_a t^{a+1} + (B_{a-1} - AB_a)t^a + (B_{a-2} - AB_{a-1})t^{a-1}\\ &\quad + \cdots + (B_0 - AB_1)t - AB_0.\end{aligned}$$

Then since $B_a \neq 0$, it follows that

$$\deg[f(t)] = a + 1.$$

Suppose

$$f(t) = t^{a+1} + b_a t^a + \cdots + b_0, \qquad b_i \in \mathbb{F}.$$

Then equating the coefficients, we have

$$\begin{cases} B_a = I, \\ B_{a-1} - AB_a = b_a I. \end{cases}$$

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$$\begin{cases} B_{a-2} - AB_{a-1} = b_{a-1}I, \\ \qquad\qquad \vdots \\ B_0 - AB_1 = b_1 I, \\ -AB_0 = b_0 I. \end{cases}$$

Multiply by A^{a+1} to first equation, A^a to second, …, A^0 to the last; we get

$$\begin{cases} A^{a+1}B_a = A^{a+1}, \\ A^aB_{a-1} - A^{a+1}B_a = b_aA^a, \\ A^{a-1}B_{a-2} - A^aB_{a-1} = b_{a-1}A^{a-1}, \\ \vdots \\ AB_0 - A^2B_1 = b_1A, \\ -AB_0 = b_0I. \end{cases}$$

Summing all the equations, we get

$$0 = A^{a+1} + b_aA^a + \cdots + b_1A + b_0I = f(A).$$

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($\Leftarrow$) Suppose $f(A) = 0$, and

$$f(t) = t^{a+1} + b_at^a + \cdots + b_1t + b_0, \qquad b_i \in \mathbb{F}.$$

Define, using the equations, matrices B_i's:

$$\begin{cases} B_a = I, \\ B_{a-1} = b_aI + AB_a, \\ B_{a-2} = b_{a-1}I + AB_{a-1}, \\ \vdots \\ B_1 = b_2I + AB_2, \\ B_0 = b_1I + AB_1. \end{cases}$$

Then compute

$$\begin{aligned} &(tI - A)(B_at^a + B_{a-1}t^{a-1} + \cdots + B_1t + B_0) \\ &= B_at^{a+1} + B_{a-1}t^a + \cdots + B_0t \\ &\quad - AB_at^a - AB_{a-1}t^{a-1} - \cdots - AB_0 \\ &= B_at^{a+1} + (B_{a-1} - AB_a)t^a + \cdots + (B_0 - AB_1)t - AB_0. \end{aligned}$$

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Continuing the computation,

$$(tI - A)(B_at^a + B_{a-1}t^{a-1} + \cdots + B_1t + B_0) = It^{a+1} + b_aIt^a + \cdots + b_1It - AB_0.$$

This equals $f(t)I_n$ if and only if

$$-AB_0 = b_0I.$$

Yet compute

$$\begin{aligned} -AB_0 &= -A(b_1I + AB_1) \\ &= -b_1A - A^2B_1 \\ &= -b_1A - A^2(b_2I + AB_2) \\ &= -b_1A - b_2A^2 - A^3B_2 \end{aligned}$$

$$\begin{aligned}&\vdots\\ &= -b_1A - b_2A^2 - \cdots - b_aA^a - A^{a+1}B_a\\ &= -b_1A - b_2A^2 - \cdots - b_aA^a - A^{a+1}\\ &= -f(A) + b_0I = 0 + b_0I = b_0I.\end{aligned}$$

Thus this direction is shown. □

Theorem 6.5 (Cayley–Hamilton). *Let $A \in M_n(\mathbb{F})$. Its characteristic polynomial $p_A(t)$ annihilates*

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A, that is,

$$p_A(A) = 0.$$

Proof. Recall that we showed in a previous proposition,

$$(tI - A)\operatorname{Adj}(tI - A) = \det(tI - A)I_n.$$

Each entry of $\operatorname{Adj}(tI - A)$ is a minor of degree at most $n-1$. Thus there exist $B_{n-1}, B_{n-2}, \ldots, B_0 \in M_n(\mathbb{F})$ such that

$$\operatorname{Adj}(tI - A) = B_{n-1}t^{n-1} + B_{n-2}t^{n-2} + \cdots + B_0,$$

where $B_{n-1} = I_n \neq 0$. Thus by lemma,

$$p_A(A) = 0.$$

□

Example

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad p_A(t) = t^2 + 1.$$

Then

$$\begin{aligned}p_A(A) &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\\ &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} + I_2 = 0.\end{aligned}$$

6.5 The Gerstenhaber Problem

Take an $A \in M_{n\times n}(\mathbb{F})$. Cayley–Hamilton

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implies A^n is a linear combination of $I_n, A, \ldots, A^{n-1}$. Hence

$$\operatorname{Span}\{A^k \mid k \geq 0\} \leq M_{n\times n}(\mathbb{F})$$

has dimension $\leq n$.

Theorem 6.6 (Gerstenhaber). *Let $A, B \in M_n(\mathbb{F})$ be commuting. Then*

$$\mathrm{Span}\{A^k B^m \mid k, m \geq 0\} \leq M_{n\times n}(\mathbb{F})$$

has dimension no more than n.

Claim

This fails for 4 or more.

Example

Take $M_{4\times 4}(\mathbb{R})$. Consider

$$E_{ij} := \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & 1 & \vdots \\ 0 & \cdots & 0 \end{pmatrix},$$

where the 1 is the (i, j)-entry. Let

$$A = E_{13}, \qquad B = E_{14}, \qquad C = E_{23}, \qquad D = E_{24}.$$

They all commute – the product

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between any two is 0. Thus

$$\mathrm{Span}\{A^i B^j C^k D^\ell\} = \mathrm{Span}\{A, B, C, D, I_4\},$$

which has dim 5.

Open Question

If $A, B, C \in M_n(\mathbb{F})$ are mutually commuting, then

$$\dim\Big(\mathrm{Span}\{A^i B^j C^k \mid i, j, k \in \mathbb{N}\}\Big) \leq n \ ?$$

6.6 Minimal Polynomials

Proposition 6.7. *Let V be a finite-dimensional vector space over a field $\mathbb{F}$, and let $\overline{T} : V \longrightarrow V$ be linear. Then, by the Cayley–Hamilton theorem,*

$$\{f(t) \in \mathbb{F}[t] \mid f(t) \neq 0,\ f(\overline{T}) = 0\} \neq \varnothing.$$

By the well-ordering principle of $\mathbb{N}$, we can find a monic polynomial $m_T(t)$ such that $m_T(t)$ has minimal degree in the

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set above. Then for every $f(t) \in \mathbb{F}[t]$ with $f(\overline{T}) = 0$,

$$m_T(t) \mid f(t).$$

Moreover, $m_T(t)$ is unique.

Proof. Since $\mathbb{F}[t]$ is a Euclidean domain, write

$$f(t) = g(t)m_T(t) + r(t), \qquad \deg r(t) < \deg m_T(t).$$

Evaluate them at $t = \overline{T}$; we get

$$0 = g(\overline{T}) \cdot 0 + r(\overline{T}) \quad \Longrightarrow \quad r(\overline{T}) = 0.$$

If $r(t) \neq 0$, then it contradicts the fact that $m_T(t)$ has minimal degree as $r(\overline{T}) = 0$. Thus $r(t) = 0$, so

$$m_T(t) \mid f(t).$$

Suppose there exists $\widetilde{m}_T(t)$ with same degree, monic, and annihilates $\overline{T}$. Then

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by first part we proved, we get

$$m_T(t) \mid \widetilde{m}_T(t) \qquad \text{and} \qquad \widetilde{m}_T(t) \mid m_T(t).$$

Thus

$$m_T(t) = a\widetilde{m}_T(t).$$

Yet they are both monic, thus $a = 1$. Hence

$$m_T(t) = \widetilde{m}_T(t).$$

□

Definition 6.8. We call this $m_T(t)$ the minimal polynomial of T.

Theorem 6.9. *The minimal polynomial $m_T(t)$ of $\overline{T}$ and $p_T(t)$, the characteristic polynomial of $\overline{T}$, has the same irreducible factors. More precisely, if $n = \dim V$, then*

$$m_T(t) \mid p_T(t), \qquad p_T(t) \mid m_T(t)^n.$$

Proof. $m_T(t) \mid p_T(t)$ follows from the Cayley–Hamilton theorem and the previous proposition.

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By the last lemma we proved, we know there exist $M_a, \ldots, M_0 \in M_{n\times n}(\mathbb{F})$ such that

$$(tI_n - A)(M_a t^a + \cdots + M_1 t + M_0) = m_T(t)I_n.$$

Take det on both sides; we get

$$\det(tI_n - A) \underbrace{\det(M_a t^a + \cdots + M_1 t + M_0)}_{\in \mathbb{F}[t]} = \det(m_T(t)I_n).$$

Therefore

$$p_A(t) \det(M_a t^a + \cdots + M_1 t + M_0) = m_T(t)^n \cdot 1,$$

so

$$p_A(t) \mid m_T(t)^n.$$

□

Alternative Proof

Lemma 6.10. *Let $\mathbb{E}, \mathbb{F}$ be fields, $\mathbb{E} \subseteq \mathbb{F}$. Then*

$$f(x) \mid g(x) \text{ in } \mathbb{E}[x] \iff f(x) \mid g(x) \text{ in } \mathbb{F}[x],$$

for all $f(x), g(x) \in \mathbb{E}[x]$.

Proof. ($\Rightarrow$) is trivial since $\mathbb{E}[x] \subseteq \mathbb{F}[x]$.
($\Leftarrow$) If $f(x) = 0$, then divisibility on either side is equivalent to $g(x) = 0$. Assume $f(x) \neq 0$. Divide $g(x)$ by $f(x)$ in $\mathbb{E}[x]$:

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$$g(x) = f(x)h(x) + r(x), \qquad h(x), r(x) \in \mathbb{E}[x], \qquad \deg r < \deg f.$$

This is also division with remainder in $\mathbb{F}[x]$.

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Since $f(x) \mid g(x)$ in $\mathbb{F}[x]$, uniqueness of Euclidean division in $\mathbb{F}[x]$ gives $r(x) = 0$. Thus $f(x) \mid g(x)$ in $\mathbb{E}[x]$. □

Lemma 6.11. *Every field has an algebraic closure $\overline{\mathbb{F}}$, unique up to an $\mathbb{F}$-isomorphism.*

Proof (of the theorem). Let λ be a root of $p_T(t)$ in $\overline{\mathbb{F}}$. Then λ is an eigenvalue of $\overline{T}$ in $\overline{\mathbb{F}}$. Suppose

$$m_T(t) = \sum_{i=0}^{m} a_i t^i.$$

Then for an eigenvector v of eigenvalue λ, we have

$$m_T(\overline{T}) = 0,$$
$$\left(\sum_{i=0}^{m} a_i \overline{T}^{\,i}\right)(v) = 0(v) = 0.$$

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Therefore

$$\sum_{i=0}^{m} a_i \lambda^i v = 0 \implies (m_T(\lambda))(v) = 0.$$

Since $v \neq 0$, it follows that

$$m_T(\lambda) = 0.$$

Thus λ is a root of m_T. Therefore $m_T(t)$ has same root as $p_T(t)$. Furthermore, a root has at most multiplicity $\deg p_T(t)$, thus

$$p_T(t) \mid m_T(t)^n \qquad \text{in } \overline{\mathbb{F}}[t].$$

By our lemma, since divisibility is not affected by field extension, it follows that

$$p_T(t) \mid m_T(t)^n \qquad \text{in } \mathbb{F}[t].$$

□

Corollary 6.12. *If $p_T(t)$ is a product of distinct irreducible monic polynomials, then*

$$m_T(t) = p_T(t).$$

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Example

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = A.$$

Then

$$p_A(t) = (t-1)^2(t-2).$$

Then $m_A(t)$ must be either

$$(t-1)(t-2) \qquad \text{or} \qquad (t-1)^2(t-2).$$

By computation, the former does not annihilate A. Thus

$$m_A(t) = (t-1)^2(t-2).$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = A.$$

Again

$$p_A(t) = (t-1)^2(t-2).$$

This time

$$m_A(t) = (t-1)(t-2)$$

as it annihilates A.

Recall

We say V is a finite-dimensional $\mathbb{F}$-vector space.

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6.7 Invariant Subspaces and Primary Decomposition

of $W_1, \ldots, W_k \leq V$, if for every $v \in V$ can be written uniquely as

$$v = w_1 + \cdots + w_k \qquad \text{with } w_i \in W_i.$$

We write

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_k.$$

Definition 6.13. If $T : V \to V$ is linear, a subspace $W \leq V$ is *T-invariant* if

$$T(W) \subseteq W.$$

Example 6.14. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$Tv = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} v.$$

The T-invariant subspaces are $\{0\}$ and $\mathbb{R}^2$.

Example 6.15. $\forall v \in V$,

$$W = \text{Span}\{v, Tv, \ldots\}$$

is T-invariant.

Example 6.16. For $f(t) \in \mathbb{F}[t]$, $\ker f(T)$ is T-invariant.
Exercise.

Theorem 6.17 (Primary Decomposition Theorem). *Let $T : V \to V$ be a linear operator on an*

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$\mathbb{F}$-vector space V. Suppose

$$m_T(t) = f_1(t)^{n_1} \cdots f_k(t)^{n_k}$$

with $f_1(t), \ldots, f_k(t)$ distinct monic irreducibles. Then

$$V = \bigoplus_{i=1}^{k} \ker(f_i(T)^{n_i}),$$

with $f_i(t)^{n_i}$ the minimal polynomials of

$$T|_{\ker(f_i(T)^{n_i})}.$$

Lemma 6.18. *Let $f(t) \in \mathbb{F}[t]$ and*

$$f(t) = g(t)h(t), \qquad g(t), h(t) \in \mathbb{F}[t]$$

such that

$$\gcd(g(t), h(t)) = 1.$$

If $f(T) = 0$, *then*

$$V = W_1 \oplus W_2,$$

with

$$W_1 = \ker g(T) = \operatorname{im} h(T), \qquad W_2 = \ker h(T) = \operatorname{im} g(T).$$

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Proof. From Algebra 1, since $\gcd(g(t), h(t)) = 1$, there exist $a(t), b(t) \in \mathbb{F}[t]$ such that

$$g(t)a(t) + h(t)b(t) = 1. \tag{$*$}$$

Thus

$$g(T)a(T) + h(T)b(T) = \mathrm{Id}_V .$$

Let

$$W_1 = \operatorname{im}\big(h(T)b(T)\big), \qquad W_2 = \operatorname{im}\big(g(T)a(T)\big).$$

Thus $\forall v \in V$,

$$v = \underbrace{g(T)a(T)(v)}_{\in W_2} + \underbrace{h(T)b(T)(v)}_{\in W_1}, \tag{$\dagger$}$$

so

$$V = W_1 + W_2.$$

In A4 we have shown

$$V = W_1 \oplus W_2 \quad \Longleftrightarrow \quad W_1 \cap W_2 = \{0\}.$$

By

$$f(T) = g(T)h(T) = 0,$$

we have

$$W_1 \le \operatorname{im} h(T) \le \ker g(T) \qquad \text{and} \qquad W_2 \le \operatorname{im} g(T) \le \ker h(T).$$

Thus if $v \in W_1 \cap W_2$, then we have

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$$v \in \ker g(T) \cap \ker h(T).$$

Therefore, by ($\dagger$),

$$v = a(T)g(T)v + b(T)h(T)v = 0 + 0 = 0.$$

Thereby

$$W_1 \cap W_2 = \{0\} \quad \Longrightarrow \quad V = W_1 \oplus W_2,$$

and

$$\dim V = \dim W_1 + \dim W_2.$$

Lastly, by rank-nullity theorem applied to $g(T)$, we have

$$\dim V = \dim\big(\ker g(T)\big) + \dim\big(\operatorname{im} g(T)\big).$$

Since

$$W_1 \leq \ker g(T), \qquad W_2 \leq \operatorname{im} g(T),$$

it follows that

$$\dim(W_1) = \dim(\ker g(T)), \qquad \dim(W_2) = \dim(\operatorname{im} g(T)).$$

Hence

$$W_1 = \ker g(T), \qquad W_2 = \operatorname{im} g(T).$$

Moreover,

$$W_1 \subseteq \operatorname{im} h(T) \subseteq \ker g(T) = W_1,$$

so $W_1 = \operatorname{im} h(T)$. Rank–nullity for $h(T)$ now gives

$$\dim \ker h(T) = \dim V - \dim \operatorname{im} h(T) = \dim W_2.$$

Since $W_2 \subseteq \ker h(T)$, it follows that $W_2 = \ker h(T)$. Therefore

$$W_1 = \ker g(T) = \operatorname{im} h(T), \qquad W_2 = \ker h(T) = \operatorname{im} g(T).$$

□

Lemma 6.19. *With the setup of the previous lemma, suppose moreover that $f(t) = m_T(t)$. Then the minimal*

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polynomial of $T|_{W_1}$ is $g(t)$, " of $T|_{W_2}$ is $h(t)$ (g, h monic).

Proof. By a previous proposition, since $g(t)$ annihilates $T|_{W_1}$ and $h(t)$ annihilates $T|_{W_2}$, we have

$$m_1(t) \mid g(t) \qquad \text{and} \qquad m_2(t) \mid h(t).$$

Thus

$$m_1(t)m_2(t) \mid g(t)h(t) = m_T(t).$$

By the previous lemma, for every $v \in V$, we have

$$v = w_1 + w_2, \qquad w_i \in W_i.$$

Thus

$$\begin{aligned}(m_1(T)m_2(T))(v) &= m_1(T)m_2(T)(w_1 + w_2) \\ &= 0 + 0 = 0.\end{aligned}$$

Therefore $m_1(t)m_2(t)$ annihilates T, so

$$m_T(t) \mid m_1(t)m_2(t).$$

Hence

$$m_T(t) = \lambda m_1(t)m_2(t).$$

Since $m_T(t), m_1(t), m_2(t)$ are all

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monic polynomials, it follows that $\lambda = 1$. Thus

$$m_T(t) = m_1(t)m_2(t) = g(t)h(t).$$

Since $m_1(t) \mid g(t)$ and $m_2(t) \mid h(t)$, it follows that

$$m_1(t) = g(t), \qquad m_2(t) = h(t),$$

as they are all monic. □

Proof of the Primary Decomposition Theorem. Induction on k.
Base case $k = 1$ is trivial.
Suppose it holds for $k - 1 \in \mathbb{N}$. Suppose

$$m_T(t) = f_1(t)^{n_1} \cdots f_k(t)^{n_k}$$

with $f_1, \ldots, f_k$ monic distinct irreducibles. Let

$$g(t) = \prod_{i=1}^{k-1} f_i(t)^{n_i}, \qquad h(t) = f_k(t)^{n_k}.$$

Thus by the lemmas,

$$V = \ker g(T) \oplus \ker h(T),$$

and the

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minimal polynomials of $T|_{\ker g(T)}$ and $T|_{\ker h(T)}$ are $g(t)$ and $h(t)$, respectively.
By the inductive hypothesis, we know

$$\ker g(T) = \bigoplus_{i=1}^{k-1} \ker\big(f_i(T)^{n_i}\big),$$

and the minimal polynomial of

$$\Big(T|_{\ker g(T)}\Big)\Big|_{\ker(f_i(T)^{n_i})}$$

is $f_i(t)^{n_i}$, because here $g(t)$ is the minimal polynomial of

$$T|_{\ker g(T)} : \ker g(T) \longrightarrow \ker g(T),$$

but not of $T : V \to V$.
Therefore, we want to show that for $i = 1, \ldots, k-1$,

$$\ker\big(f_i(T)^{n_i}\big) = \ker g(T) \cap \ker\big(f_i(T)^{n_i}\big).$$

Clearly $\supseteq$ is true. On the other hand, if

$$v \in \ker\big(f_i(T)^{n_i}\big),$$

then

$$g(T)(v) = f_1(T)^{n_1} \cdots f_{k-1}(T)^{n_{k-1}}(v) = 0,$$

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so $v \in \ker g(T)$. Therefore $\subseteq$, and thus

$$\ker\big(f_i(T)^{n_i}\big) = \ker g(T) \cap \ker\big(f_i(T)^{n_i}\big).$$

Consequently,

$$\begin{aligned} T|_{\ker(f_i(T)^{n_i})} &= T|_{\ker g(T)\cap\ker(f_i(T)^{n_i})} \\ &= \Big(T|_{\ker g(T)}\Big)\Big|_{\ker(f_i(T)^{n_i})} \end{aligned}$$

has minimal polynomial $f_i(t)^{n_i}$, which proves the missing point. □

6.8 Diagonalizability via the Minimal Polynomial

Corollary 6.20. *Suppose $T : V \to V$ is linear on a finite-dimensional $\mathbb{F}$-vector space V. Then T is diagonalizable if and only if the minimal polynomial $m_T(t)$ splits completely into distinct linear factors, that is,*

$$m_T(t) = (t-\lambda_1)\cdots(t-\lambda_k), \qquad \lambda_i \in \mathbb{F}, \qquad \lambda_i \neq \lambda_j \text{ for } i \neq j.$$

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Proof. ($\Rightarrow$) Suppose T is diagonalizable. Then there exists a basis $\mathcal{B}$ of V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \operatorname{diag}(\underbrace{\lambda_1,\ldots,\lambda_1}_{n_1 \text{ times}},\ldots,\underbrace{\lambda_k,\ldots,\lambda_k}_{n_k \text{ times}}).$$

Define

$$f(t) = \prod_{i=1}^{k}(t-\lambda_i).$$

Then, by the natural isomorphism

$$\operatorname{Hom}(V,V) \longrightarrow M_n(\mathbb{F}), \qquad T \longmapsto [T]_{\mathcal{B}}^{\mathcal{B}}$$

preserving product and addition and scalar multiplication, we have

$$\begin{aligned} [f(T)]_{\mathcal{B}}^{\mathcal{B}} &= f\left([T]_{\mathcal{B}}^{\mathcal{B}}\right) \\ &= \prod_{i=1}^{k}\left([T]_{\mathcal{B}}^{\mathcal{B}} - \lambda_i I\right) = 0, \end{aligned}$$

because in every diagonal position one factor has a zero.

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Thus

$$f(T) = 0 \quad \Longrightarrow \quad m_T(t) \mid f(t),$$

since $f(t)$ annihilates T. Yet $\lambda_1,\ldots,\lambda_k$ are eigenvalues of T, so they must be roots of $m_T(t)$ since $m_T(t)$ and $p_T(t)$ have same roots. Therefore

$$\deg m_T(t) \geq k.$$

Yet $\deg f(t) = k$, so

$$\deg m_T(t) = k \quad \Longrightarrow \quad m_T(t) = f(t).$$

($\Leftarrow$) Suppose $m_T(t)$ splits. By the primary decomposition theorem,

$$V = \bigoplus_{i=1}^{k} \ker(T - \lambda_i \operatorname{Id}),$$

and $t - \lambda_i$ is the minimal polynomial of

$$T|_{\ker(T-\lambda_i \operatorname{Id})}.$$

Thus

$$T|_{\ker(T-\lambda_i \operatorname{Id})} = \lambda_i \operatorname{Id}_{\ker(T-\lambda_i \operatorname{Id})},$$

so vectors in $\ker(T - \lambda_i \operatorname{Id})$ are eigenvectors

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of T with eigenvalue λ_i. Since

$$V = \bigoplus_{i=1}^{k} \ker(T - \lambda_i \operatorname{Id}),$$

take basis $\mathcal{B}_i$ for $\ker(T - \lambda_i \operatorname{Id})$; the union

$$\mathcal{B} = \bigcup_{i=1}^{k} \mathcal{B}_i$$

is a basis of V. Since there exists a basis $\mathcal{B}$ consisting only of eigenvectors of T, T is diagonalizable by a previous proposition. □

Corollary 6.21. *If $T : V \to V$ is diagonalizable, and $W \leq V$ is a T-invariant subspace, then*

$$T|_W : W \longrightarrow W$$

is diagonalizable.

Proof. Notice that $m_T(t)$ annihilates $T|_W$:

$$m_T(T) : V \longrightarrow V$$

is the zero map on V, so

$$m_T\left(T|_W\right) = m_T(T)|_W$$

is the zero map on W. Thus

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by the proposition,

$$m_{T|W}(t) \mid m_T(t).$$

By the previous corollary, $m_T(t)$ splits into linear factors since T is diagonalizable. Thus $m_{T|W}(t)$ splits into distinct linear factors, so $T|_W$ is diagonalizable. □

6.9 Computing Minimal Polynomials

Now we briefly discuss the computational significance of the minimal polynomial.

Discussion 1

Minimal polynomial can be calculated without calculating the characteristic polynomial.

Proposition 6.22. *Let V be an $\mathbb{F}$-vector space, and let $T : V \to V$ be a linear transformation. Suppose*

$$V = \bigoplus_{i=1}^{s} W_i,$$

where W_i are T-invariant subspaces, and $m_i(t)$ minimal polynomials of $T|_{W_i}$. Then

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$$m_T(t) = \operatorname{lcm}\{m_1(t), \ldots, m_s(t)\}.$$

Proof. Let $f(t)$ be an annihilating polynomial of T, that is, $f(T) = 0$. Then

$$f\left(T|_{W_i}\right) = f(T)|_{W_i} = 0,$$

so

$$m_i(t) \mid f(t).$$

On the other hand, let $m(t)$ be a common multiple of $m_1(t), \ldots, m_s(t)$. Let $v \in V$. Then

$$v = \sum_{i=1}^{s} w_i, \qquad w_i \in W_i.$$

Thus

$$\begin{aligned} m(T)(v) &= m(T)\left(\sum_{i=1}^{s} w_i\right) \\ &= \sum_{i=1}^{s} m(T)(w_i) \\ &= \sum_{i=1}^{s} g_i(T) \underbrace{m_i(T)(w_i)}_{0} = \sum_{i=1}^{s} 0 = 0. \end{aligned}$$

Therefore $m(T) = 0$, that is, $m(t)$ annihilates T. Hence

$$\left\{ \text{common multiples of } m_i(t) \right\} \iff \left\{ \text{annihilating polynomial of } T \right.$$

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Thus, by uniqueness of the least common multiple and the minimal polynomial,

$$m_T(t) = \operatorname{lcm}(m_1(t), \ldots, m_s(t)).$$

□

Proposition 6.23 (Assignment 8, Question 1). *Let V be a finite-dimensional $\mathbb{F}$-vector space, let $T : V \to V$, let $0 \neq v \in V$, and*

$$W := \text{Span}\{T^i v \mid i \in \mathbb{N}\}.$$

Let $k \geq 1$ be the smallest integer such that

$$T^k v = a_0 v + a_1 T v + \cdots + a_{k-1} T^{k-1} v$$

for some $a_i \in \mathbb{F}$. Then

① *$\{v, Tv, \ldots, T^{k-1}v\}$ is a basis of W;*

②

$$m_{T|W}(t) = p_{T|W}(t) = t^k - a_{k-1}t^{k-1} - \cdots - a_0.$$

These two propositions give us a way to calculate $m_T(t)$ without factoring. Denote

$$\text{Span}\{T^i v\} \text{ by } W(v).$$

Choose $v_1 \in V \setminus \{0\}$ and compute $W(v_1)$

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and $m_{T|W(v_1)}(t)$ by Proposition 2. If $W(v_1) \neq V$, choose

$$v_2 \in V \setminus W(v_1)$$

and compute $W(v_2)$ and $m_{T|W(v_2)}(t)$. Continue: after choosing $v_1, \ldots, v_s$, if

$$W(v_1) + \cdots + W(v_s) \neq V,$$

choose v_{s+1} outside this sum. Because V is finite-dimensional, this process terminates with

$$V = W(v_1) + \cdots + W(v_s).$$

The sum need not be direct. A polynomial annihilates V if and only if it annihilates every $W(v_i)$, so

$$m_T(t) = \text{lcm}\left(m_{T|W(v_1)}(t), \ldots, m_{T|W(v_s)}(t)\right).$$

Recall we have proven that a matrix

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6.10 Repeated Roots and the Derivative Test

is diagonalizable if and only if its minimal polynomial splits into distinct linear factors. This point, along with the next proposition, makes it extremely fast to determine diagonalizability.

Proposition 6.24. *Let $m(t) \in \mathbb{F}[t]$. Then $m(t)$ has no repeated roots in its algebraically closed field extension if and only if*

$$\gcd(m(t), m'(t)) = 1.$$

Proof. ($\Rightarrow$) Suppose

$$\gcd(m(t), m'(t)) = f(t) \neq 1.$$

Then $\deg f(t) \geq 1$, so

$$\text{there exists } a \in \overline{\mathbb{F}} \text{ such that } (t-a) \mid f(t),$$

where a is in $\overline{\mathbb{F}}$ (the algebraically closed field extension). Thus

$$(t-a) \mid m(t), m'(t).$$

Write

$$m(t) = (t-a)g(t).$$

Then compute

$$m'(t) = (t-a)g'(t) + g(t).$$

Since $m'(t)$ is divisible by $(t-a)$, it follows that

$$(t-a) \mid m'(t) - (t-a)g'(t) = g(t).$$

Therefore

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$$(t-a)^2 \mid m(t),$$

so a is a repeated root of m, a contradiction.
($\Leftarrow$) Suppose $m(t)$ has repeated roots and

$$\gcd(m(t), m'(t)) = 1.$$

Then there exists $a \in \overline{\mathbb{F}}$ such that

$$(t-a)^2 \mid m(t).$$

Then differentiate $m(t)$; we get

$$\begin{aligned}(m(t))' &= \left((t-a)^2 g(t)\right)' \\ &= 2(t-a)g(t) + (t-a)^2 g'(t).\end{aligned}$$

Thus $t-a$ is a common divisor of $m(t), m'(t)$, yet

$$t-a \nmid \gcd(m(t), m'(t)),$$

a contradiction. □

Thus to determine diagonalizability, we compute the $\gcd(m(t), m'(t))$ using the Euclidean algorithm, which is fast. ×

7 Jordan Canonical Form

The existence of the Jordan canonical form is a weaker

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condition than diagonalizability. In this section we require the minimal polynomial to split over the field $\mathbb{F}$.

Proposition 7.1 (Setup). *Let V be a finite-dimensional $\mathbb{F}$-vector space, and let T be a linear operator on V. Suppose*

$$m_T(t) = (t-\lambda_1)^{m_1}\cdots(t-\lambda_r)^{m_r}, \qquad \lambda_i \in \mathbb{F}, \quad \lambda_i \neq \lambda_j \text{ for } i \neq j.$$

Then we first notice the following:

1. $$V = \bigoplus_{i=1}^{r} W_i, \qquad W_i := \ker(T-\lambda_i \operatorname{Id})^{m_i},$$

   and

   $$m_{T|W_i}(t) = (t-\lambda_i)^{m_i}.$$

2. *$p_T(t)$ takes the form*

   $$p_T(t) = (t-\lambda_1)^{n_1}\cdots(t-\lambda_r)^{n_r},$$

   and

   $$p_{T|W_i}(t) = (t-\lambda_i)^{n_i}, \qquad \dim(W_i) = n_i.$$

3. $$\ker(T-\lambda_i \operatorname{Id})^a \subseteq W_i, \qquad \forall a \in \mathbb{N}.$$

   More precisely,

   $$\ker(T-\lambda_i \operatorname{Id})^a = W_i, \qquad a \geq m_i,$$

   and

   $$\ker(T-\lambda_i \operatorname{Id}) \subsetneq \ker(T-\lambda_i \operatorname{Id})^2 \subsetneq \cdots \subsetneq \ker(T-\lambda_i \operatorname{Id})^{m_i} = W_i.$$

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Proof. Denote $T|_{W_i} =: \overline{T}_i$.

1. follows from the primary decomposition theorem.

2. From part 1, we have

$$V = \bigoplus_{i=1}^{r} W_i.$$

Take $\mathcal{B}_i$ basis of W_i, $i = 1, \ldots, r$. Then

$$\mathcal{B} := \bigcup_{i=1}^{r} \mathcal{B}_i$$

is a basis of V. Since the spaces W_i are T-invariant subspaces, it follows that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} A_1 & & & 0 \\ & A_2 & & \\ & & \ddots & \\ 0 & & & A_r \end{pmatrix}.$$

In particular, note that if

$$\mathcal{B}_i = \{v_1, \ldots, v_s\},$$

then

$$\begin{aligned} A_i &= \left([Tv_1]_{\mathcal{B}_i}, \ldots, [Tv_s]_{\mathcal{B}_i}\right) \\ &= [T|_{W_i}]_{\mathcal{B}_i}^{\mathcal{B}_i} \qquad \text{(other components are 0).} \end{aligned}$$

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Furthermore,

$$\begin{aligned} p_T(t) &= \det\left(tI - [T]_{\mathcal{B}}^{\mathcal{B}}\right) \\ &= \det \begin{pmatrix} tI - A_1 & & & 0 \\ & tI - A_2 & & \\ & & \ddots & \\ 0 & & & tI - A_r \end{pmatrix} \\ &= \prod_{i=1}^{r} \det(tI - A_i) \qquad \text{by Assignment 4, Question 4, and an induction argument} \\ &= \prod_{i=1}^{r} \det\left(tI - [T|_{W_i}]_{\mathcal{B}_i}^{\mathcal{B}_i}\right) \\ &= \prod_{i=1}^{r} p_{T|W_i}(t) = \prod_{i=1}^{r} p_{\overline{T}_i}(t). \end{aligned}$$

Since the minimal polynomial and characteristic polynomial have the same irreducible factors, and

$$m_{\overline{T}_i}(t) = (t - \lambda_i)^{m_i}$$

by part 1, it follows that

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$$p_{\overline{T}_i}(t) = (t - \lambda_i)^{n_i}, \qquad n_i = \dim W_i \geq m_i.$$

And by our argument,

$$p_T(t) = \prod_i p_{\overline{T}_i}(t) = (t - \lambda_1)^{n_1} \cdots (t - \lambda_r)^{n_r}.$$

3. The inclusion

$$\ker(T - \lambda_i \operatorname{Id})^a \subseteq \ker(T - \lambda_i \operatorname{Id})^{m_i}, \qquad \forall a \leq m_i,$$

is trivial by definition.
We have

$$\ker(T - \lambda_i \operatorname{Id})^a \neq \ker(T - \lambda_i \operatorname{Id})^{m_i}, \qquad \forall a < m_i.$$

If otherwise

$$\ker(T - \lambda_i \operatorname{Id})^a = \ker(T - \lambda_i \operatorname{Id})^{m_i},$$

then

$$(T - \lambda_i \operatorname{Id})^a(W_i) = \{0\},$$

so

$$(T - \lambda_i \operatorname{Id})^a|_{W_i} = 0.$$

Thus $(t - \lambda_i)^a$ annihilates $T|_{W_i}$. This contradicts the minimality of

$$m_{\overline{T}_i}(t) = (t - \lambda_i)^{m_i}.$$

We also have

$$\ker(T - \lambda_i \operatorname{Id})^a \subsetneq \ker(T - \lambda_i \operatorname{Id})^{a+1}, \qquad \forall a < m_i.$$

Suppose

$$\ker(T - \lambda_i \operatorname{Id})^a = \ker(T - \lambda_i \operatorname{Id})^{a+1}.$$

Then for $v \in V$,

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$$\begin{aligned} v \in \ker(T - \lambda_i \operatorname{Id})^{a+2} &\iff (T - \lambda_i \operatorname{Id})^{a+2}(v) = 0 \\ &\iff (T - \lambda_i \operatorname{Id})^{a+1}(T - \lambda_i \operatorname{Id})(v) = 0 \\ &\iff (T - \lambda_i \operatorname{Id})v \in \ker(T - \lambda_i \operatorname{Id})^{a+1} \\ &\qquad\quad = \ker(T - \lambda_i \operatorname{Id})^a \\ &\iff v \in \ker(T - \lambda_i \operatorname{Id})^{a+1}. \end{aligned}$$

We can inductively show that

$$\ker(T - \lambda_i \operatorname{Id})^a = \ker(T - \lambda_i \operatorname{Id})^{a+1} = \cdots,$$

which contradicts that

$$\ker(T - \lambda_i \operatorname{Id})^a \neq W_i.$$

Finally,

$$\ker(T - \lambda_i \operatorname{Id})^a = \ker(T - \lambda_i \operatorname{Id})^{m_i}, \qquad \forall a \geq m_i.$$

Let $a \geq m_i$. Then notice that

$$m_T(t) = \prod_{j=1}^{r} (t - \lambda_j)^{m_j} \mid (t - \lambda_i)^a \prod_{\substack{1 \leq j \leq r \\ j \neq i}} (t - \lambda_j)^{m_j}.$$

Thus

$$(t - \lambda_i)^a \prod_{\substack{1 \leq j \leq r \\ j \neq i}} (t - \lambda_j)^{m_j}$$

annihilates T. Then by the first lemma before we prove

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the primary decomposition theorem, since $(t-\lambda_i)^a$ is coprime with $(t-\lambda_j)^{m_j}$, we have (by an induction argument)

$$V = ker(T-\lambda_1 \operatorname{Id})^{m_1} \oplus \cdots \oplus \ker(T-\lambda_i \operatorname{Id})^a \oplus \cdots \oplus \ker(T-\lambda_r \operatorname{Id})^{m_r}. \tag{$*$}$$

Now compare the dimension. By part 2,

$$V = \bigoplus_{i=1}^{r} W_i,$$

so

$$\dim V = \sum_{i=1}^{r} \dim(W_i) = \sum_{i=1}^{r} n_i.$$

On the other hand, from $(*)$,

$$\dim V = \sum_{\substack{1\le j\le r \\ j\neq i}} \dim(W_j) + \dim\big(\ker(T-\lambda_i \operatorname{Id})^a\big).$$

Therefore

$$\dim\big(\ker(T-\lambda_i \operatorname{Id})^a\big) = n_i = \dim W_i.$$

Since

$$W_i = \ker(T-\lambda_i \operatorname{Id})^{m_i} \le \ker(T-\lambda_i \operatorname{Id})^a,$$

and their dimensions are equal, it follows that

$$W_i = \ker(T-\lambda_i \operatorname{Id})^a, \qquad \forall a \ge m_i.$$

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Alternative Proof.
First notice that

$$(T-\lambda_i \operatorname{Id})|_{W_j}$$

is invertible for all $j \neq i$. This is true since a linear transformation is invertible

$$\iff 0 \text{ is not an eigenvalue of it} \iff \lambda_i \text{ is not an eigenvalue of } T|_{W_j}.$$

If λ_i is an eigenvalue of $T|_{W_j}$, then there exists $v \in W_j = \ker(T-\lambda_j \operatorname{Id})^{m_j} \setminus \{0\}$ such that

$$Tv = \lambda_i v.$$

Then we can compute

$$\begin{aligned}(T-\lambda_j \operatorname{Id})^{m_j}(v) &= (T-\lambda_j \operatorname{Id})^{m_j-1}(Tv-\lambda_j v) \\ &= (T-\lambda_j \operatorname{Id})^{m_j-1}((\lambda_i-\lambda_j)v) \\ &= (\lambda_i-\lambda_j)(T-\lambda_j \operatorname{Id})^{m_j-1}(v) \\ &= \cdots = (\lambda_i-\lambda_j)^{m_j}(v).\end{aligned}$$

Since $\lambda_i \neq \lambda_j$,

$$(T-\lambda_j \operatorname{Id})^{m_j}(v) \neq 0,$$

so $v \notin W_j$, a contradiction.

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Thus λ_i is not an eigenvalue, and thus

$$(T - \lambda_i \operatorname{Id})|_{W_j}$$

is invertible. As a result,

$$(T - \lambda_i \operatorname{Id})^a|_{W_j}$$

is invertible for all $a \in \mathbb{N}$, $j \neq i$.
Let

$$v \in \ker(T - \lambda_i \operatorname{Id})^a.$$

Then by direct sum decomposition, we have

$$v = w_1 + w_2 + \cdots + w_r, \qquad w_j \in W_j.$$

Therefore

$$(T - \lambda_i \operatorname{Id})^a v = \sum_{j=1}^{r} (T - \lambda_i \operatorname{Id})^a w_j,$$
$$0 = \sum_{j=1}^{r} (T - \lambda_i \operatorname{Id})^a w_j.$$

Since W_j's are T-invariant, we have

$$(T - \lambda_i \operatorname{Id})^a w_j \in W_j.$$

Then by direct sum definition,

$$(T - \lambda_i \operatorname{Id})^a w_j = 0, \qquad j = 1, \ldots, r.$$

By invertibility we proved,

$$w_j = 0, \qquad \forall j \neq i.$$

Thus $v = w_i \in W_i$, so

$$\ker(T - \lambda_i \operatorname{Id})^a \subseteq W_i.$$

□

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$$W_i \subseteq \ker(T - \lambda_i \operatorname{Id})^a, \qquad \forall a \geq m_i,$$

is trivial from the definition. Thus

$$W_i = \ker(T - \lambda_i \operatorname{Id})^a, \qquad \forall a \geq m_i.$$

□

7.1 Nilpotent Operators and Jordan Blocks

Observation

This proposition allows us to focus on nilpotent operators. Consider a linear transformation $T : V \to V$ such that

$$m_T(t) = (t-\lambda)^m, \qquad p_T(t) = (t-\lambda)^n.$$

$$[m_{T_i}(t) = (t-\lambda_i)^{m_i}, \qquad p_{T_i}(t) = (t-\lambda_i)^{n_i}, \qquad T_i : W_i \to W_i]\,.$$

Define

$$U = T - \lambda \mathrm{Id}.$$

The following facts follow trivially:

1. U is a nilpotent operator on V.
2. $m_U(t) = t^m$ and $p_U(t) = t^n$.

We call m the *index of nilpotence* of U. ×

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Let us state an important theorem first and discuss its formal consequences before proving it.

Theorem I

A nilpotent operator $U : V \to V$ is represented, with respect to a suitable basis, by a block-diagonal matrix

$$\mathrm{diag}(N_1, N_2, \ldots, N_d),$$

where every N_i is a standard nilpotent matrix, or Jordan block, of size k_i:

$$N_i = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{pmatrix}.$$

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Proof deferred. □

Consequences

1. Relating this back to T, where $m_T(t) = (t-\lambda)^m$, it follows that in the same basis

$$T = U + \lambda \mathrm{Id} = \mathrm{diag}(\lambda I_{k_1} + N_1, \lambda I_{k_2} + N_2, \ldots, \lambda I_{k_d} + N_d),$$

where each block has the form

$$\lambda I + N = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda & 1 \\ 0 & \cdots & \cdots & 0 & \lambda \end{pmatrix} =: J_a(\lambda),$$

where a is the size of the block.

2. If the basis of one nilpotent block is $\mathcal{B} = \{b_1, \ldots, b_k\}$, then N acts by the chain

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$$b_k \xrightarrow{N} b_{k-1} \longrightarrow \cdots \longrightarrow b_1 \longrightarrow 0.$$

This implies that

$$\mathcal{B} = \{N^{k-1}b_k, N^{k-2}b_k, \ldots, Nb_k, b_k\}.$$

Furthermore, for the standard nilpotent block

$$N = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & 0 & 1 \\ 0 & & & 0 \end{pmatrix},$$

we can compute

$$N^a = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ & & & \ddots & \ddots & & \vdots \\ 0 & & & & 0 & 1 & 0 \\ 0 & \cdots & & & & 0 & 0 \end{pmatrix},$$

where the first row begins with a zeros.

3. With the same settings as Theorem I, we investigate the constituents of the block diagonal matrix. By the computation above, we know

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Hence

$$\operatorname{null}(N^a) = \begin{cases} a, & a \le k, \\ k, & a \ge k, \end{cases} \qquad (3.1)$$

where k is the size of N.

For a block-diagonal nilpotent matrix,

$$\operatorname{diag}(N_1, N_2, \ldots, N_d)^a = \operatorname{diag}(N_1^a, N_2^a, \ldots, N_d^a).$$

Thus, if m is the index of nilpotence of U,

$$m = \max\{k_i \mid i = 1, \ldots, d\} \qquad (3.2),$$

where k_i is the size of each Jordan block. Furthermore,

$$\text{null}(U) = \#\{\text{zero rows in its matrix representation}\} = \sum_{i=1}^{\infty} \#\{\text{size-}i \text{ blocks}\},$$

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since each Jordan block has one zero row. Therefore

$$\text{null}(U) = \sum_{i=1}^{\infty} \#\{i\text{-blocks}\} = \#\{\text{Jordan blocks}\} \qquad (3.3).$$

Likewise,

$$\begin{aligned}\text{null}(U^a) &= \sum_{i=1}^{\infty} \text{null}(i\text{-block})^a \cdot \#\{i\text{-blocks}\} \qquad (3.4)\\ &= \sum_{i=1}^{a} i\,\#\{i\text{-blocks}\} + \sum_{i=a+1}^{\infty} a\,\#\{i\text{-blocks}\}.\end{aligned}$$

Thus we can compute

$$\begin{aligned}\#\{a\text{-blocks}\} &= 2\sum_{i=1}^{a} i\,\#\{i\text{-blocks}\} + 2\sum_{i=a+1}^{\infty} a\,\#\{i\text{-blocks}\}\\ &\quad - \sum_{i=1}^{a-1} i\,\#\{i\text{-blocks}\} - \sum_{i=a}^{\infty}(a-1)\,\#\{i\text{-blocks}\}\\ &\quad - \sum_{i=1}^{a+1} i\,\#\{i\text{-blocks}\} - \sum_{i=a+2}^{\infty}(a+1)\,\#\{i\text{-blocks}\}\\ &= 2\,\text{null}(U^a) - \text{null}(U^{a-1}) - \text{null}(U^{a+1}) \qquad (3.5).\end{aligned}$$

□

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We give the next proposition.

Proposition

With the same setup as in Theorem I:

1. The number of blocks is $\text{null}(U)$.
2. The index of nilpotence of U is the size of the largest block.
3. The number of b-blocks is

$$\begin{aligned}&2\,\text{null}(U^b) - \text{null}(U^{b+1}) - \text{null}(U^{b-1})\\ &= \text{rk}(U^{b+1}) + \text{rk}(U^{b-1}) - 2\,\text{rk}(U^b).\end{aligned}$$

Proof. Part 1 follows from (3.3), and part 2 follows from (3.2). The first equality in part 3 follows from (3.5). For the second equality,

$$\begin{aligned}&2\,\text{null}(U^b) - \text{null}(U^{b+1}) - \text{null}(U^{b-1})\\ &= 2(\dim V - \dim \text{im}\, U^b) - \dim V + \dim \text{im}\, U^{b+1} - \dim V + \dim \text{im}\, U^{b-1}\\ &= \text{rk}(U^{b+1}) + \text{rk}(U^{b-1}) - 2\,\text{rk}(U^b),\end{aligned}$$

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7.2 The Jordan Canonical Form Theorem

since $\dim \operatorname{im} A = \dim \operatorname{Col} A = \operatorname{rk} A$. □

We can now introduce the main theorem.

Theorem II: Jordan Canonical Form

Let $T : V \to V$ be linear, with

$$p_T(t) = \prod_{i=1}^{r} (t - \lambda_i)^{n_i}, \qquad m_T(t) = \prod_{i=1}^{r} (t - \lambda_i)^{m_i}.$$

Then, with respect to a suitable basis, T has a block-diagonal matrix representation called the *Jordan canonical form*, whose blocks are Jordan blocks

$$J_{a(i,j)}(\lambda_i), \qquad a(i,1) \geq a(i,2) \geq \cdots, \qquad i = 1, \ldots, r.$$

Furthermore, the following hold:

1. $$a(i,1) = m_i,$$

   that is, the size

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   of the maximal block $J(\lambda_i)$ is the index of nilpotence of

   $$(T - \lambda_i \mathrm{Id})|_{W_i}.$$

2. For every λ_i,
   $$\sum_{j=1}^{\infty} a(i,j) = n_i.$$

3. For every λ_i,
   $$\#\{\text{blocks } J(\lambda_i)\} = m_g(\lambda_i).$$

4. For every λ_i, set
   $$U_i := T - \lambda_i \mathrm{Id}.$$

   Then the number of blocks of the form $J_b(\lambda_i)$ is

   $$2 \operatorname{null}(U_i^b) - \operatorname{null}(U_i^{b+1}) - \operatorname{null}(U_i^{b-1})$$

   $$= \operatorname{rk}(U_i^{b+1}) + \operatorname{rk}(U_i^{b-1}) - 2 \operatorname{rk}(U_i^b).$$

Proof. Part 1 follows from the previous proposition, part 2, and the setup proposition, part 1. Part 2 follows because the total size of the blocks equals

$$\dim(W_i) = n_i.$$

For part 3,

$$\#\{\text{blocks } J(\lambda_i)\} = \operatorname{null}\left((T - \lambda_i \mathrm{Id})|_{W_i}\right)$$

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$$= \dim \ker(T_i - \lambda_i \mathrm{Id}_{W_i}) = \dim \ker(T - \lambda_i \mathrm{Id}) = m_g(\lambda_i).$$

For part 4, the previous proposition gives

$$\#\{\text{blocks } J_b(\lambda_i)\} = 2\,\mathrm{null}(T_i - \lambda_i \mathrm{Id}_{W_i})^b - \mathrm{null}(T_i - \lambda_i \mathrm{Id}_{W_i})^{b+1} - \mathrm{null}(T_i - \lambda_i \mathrm{Id}_{W_i})^{b-1}$$

$$= 2\,\mathrm{null}(T - \lambda_i \mathrm{Id})^b - \mathrm{null}(T - \lambda_i \mathrm{Id})^{b+1} - \mathrm{null}(T - \lambda_i \mathrm{Id})^{b-1}.$$

[Since

$$\ker(T - \lambda_i \mathrm{Id})^a \subseteq W_i$$

it follows from the definition of the kernel and restriction that

$$\ker(T - \lambda_i \mathrm{Id})^a|_{W_i} = \ker(T - \lambda_i \mathrm{Id})^a, \qquad \forall a.$$

] The rank version follows. □

Corollary

Similar or conjugate matrices $A = QBQ^{-1}$ have identical Jordan normal form up

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to rearrangement of Jordan blocks.

Proof. The minimal polynomial, characteristic polynomial, nullity, and rank are preserved between similar matrices. One way we can see this is that A can be thought of as $[L_A]^{\mathrm{St}}_{\mathrm{St}}$. Taking the columns of Q as a basis $\mathcal{B}$ for $\mathbb{F}^n$, we have

$$B = [L_A]^{\mathcal{B}}_{\mathcal{B}}, \qquad [\mathrm{Id}]^{\mathrm{St}}_{\mathcal{B}} [L_A]^{\mathcal{B}}_{\mathcal{B}} [\mathrm{Id}]^{\mathcal{B}}_{\mathrm{St}} = A.$$

Thus A and B are matrix representations of the same operator.

On the other hand, we can show that A, B have the same annihilating polynomials by computation, and the

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other parts we proved them already. □

7.3 Constructing a Jordan Basis

Remark

This closes our discussion on the Jordan canonical form as soon as we prove Theorem I, which will also give a concrete algorithm for computing bases for each $T_i : W_i \to W_i$. Combining those bases gives a basis of V with respect to which T is represented by its Jordan canonical form. ×

Proof of Theorem I. Recall the setup. Let $U : V \to V$ be nilpotent, with

$$m_U(t) = t^m, \qquad p_U(t) = t^n.$$

By the setup proposition, part 6, choose C_m such that

$$\ker(U^m) = \ker(U^{m-1}) \oplus C_m.$$

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Furthermore:

1. $$U(C_m) \subseteq \ker(U^{m-1}),$$
   since elements of C_m are in $\ker(U^m)$ but not in $\ker(U^{m-1})$.
2. $$U(C_m) \cap \ker(U^{m-2}) = \{0\}.$$
   Indeed, if $Uv \in U(C_m)$ and $Uv \in \ker(U^{m-2})$, then $v \in \ker(U^{m-1})$, so $v = 0$ because $C_m \oplus \ker(U^{m-1})$ is direct.

Choose $C_{m-1} \supseteq U(C_m)$ such that

$$\ker(U^{m-1}) = \ker(U^{m-2}) \oplus C_{m-1}.$$

Inductive step. Suppose we have found

$$V = \ker(U^{i-1}) \oplus C_i \oplus C_{i+1} \oplus \cdots \oplus C_m,$$

with the indicated partial sums equal to successive kernels, and such that

$$U(C_j) \subseteq C_{j-1}.$$

Then notice that

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$$U(C_i) \subseteq \ker(U^{i-1}), \qquad U(C_i) \cap \ker(U^{i-2}) = \{0\},$$

by the same reasoning as on the last page. Hence we may find $C_{i-1} \subseteq \ker(U^{i-1})$ such that

$$C_{i-1} \supseteq U(C_i), \qquad \ker(U^{i-1}) = \ker(U^{i-2}) \oplus C_{i-1}.$$

We can therefore find

$$V = C_1 \oplus C_2 \oplus \cdots \oplus C_m,$$

where the partial sums are

$$C_1 = \ker U, \quad C_1 \oplus C_2 = \ker U^2, \quad \ldots, \quad C_1 \oplus \cdots \oplus C_m = \ker U^m = V.$$

Moreover, $C_{i-1} \supseteq U(C_i)$ for every i, and the map

$$U : C_i \longrightarrow C_{i-1}$$

is injective. If $v \in \ker(U|_{C_i})$, then $v \in \ker U \subseteq \ker U^{i-1}$, so $v = 0$ because $C_i \oplus \ker U^{i-1}$ is direct.

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The derivation gives the following algorithm for computing the basis.

Algorithm

1. Find a basis
$$\{v_1^m, \dots, v_{n(m)}^m\}$$
for C_m. Obtain this by finding a basis of $\ker U^{m-1}$ and expanding it to a basis of $\ker U^m = V$; the added vectors form a basis of C_m.
2. By the boxed statements above,
$$\{Uv_1^m, \dots, Uv_{n(m)}^m\}$$
is linearly independent and can be expanded to a basis of C_{m-1} by adding
$$\{v_1^{m-1}, \dots, v_{n(m-1)}^{m-1}\}.$$
3. The set
$$\{U^2v_1^m, \dots, U^2v_{n(m)}^m, Uv_1^{m-1}, \dots, Uv_{n(m-1)}^{m-1}\}$$
is linearly independent and is a basis of $U(C_{m-1})$. Expand it to a basis of C_{m-2} by adding
$$\{v_1^{m-2}, \dots, v_{n(m-2)}^{m-2}\}.$$

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4. Repeat the procedure in steps 2 or 3.
5. Take all bases of the C_i and combine them, since
$$V = \bigoplus_{i=1}^{m} C_i.$$

_____ × _____

This yields chains of basis vectors of the form

$$U^{m-1}v_j^m,\ U^{m-2}v_j^m, \dots,\ Uv_j^m,\ v_j^m,$$

$$U^{m-2}v_j^{m-1},\ U^{m-3}v_j^{m-1}, \dots,\ Uv_j^{m-1},\ v_j^{m-1},$$

and so on, down to $v_1^1, \dots, v_{n(1)}^1$.

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To conclude that this basis gives the Jordan canonical form of U, notice that for each chain above, the span of the vectors in that chain gives a cyclic subspace of U. It is trivial that:

1. cyclic subspaces are U-invariant;
2. U restricted to each cyclic subspace, under the displayed ordering, is a Jordan block with diagonal entry 0.

This concludes the proof, since the matrix of U is block diagonal with those Jordan blocks. It is the Jordan canonical form of U by uniqueness. □

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7.4 Examples and Computation

Example

The following are equivalent:

$$T \text{ is diagonalizable}$$
$$\iff m_T(t) \text{ splits into distinct linear factors}$$
$$\iff \text{every Jordan block has size 1,}$$

since the maximum size of a Jordan block $J(\lambda)$ is the exponent of $(t-\lambda)$ in the minimal polynomial, by Theorem II. ×

Example

Let

$$T : \mathbb{R}[x]_n \to \mathbb{R}[x]_n, \qquad T(f(x)) = f'(x).$$

Then

$$p_T(t) = t^{n+1}, \qquad m_T(t) = t^k \quad \text{for some } 1 \le k \le n+1.$$

Since

$$T^n(x^n) = \frac{d^n}{dx^n}(x^n) = n! \neq 0,$$

it follows that $m_T(t) = t^{n+1}$. Thus there is a Jordan block with diagonal entry 0 of size $n+1$.

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But the matrix of T has size $n+1 = \dim \mathbb{R}[x]_n$, so its Jordan canonical form has exactly one Jordan block:

$$\begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & 0 & 1 \\ 0 & & & 0 \end{pmatrix}.$$

A basis with respect to which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is the matrix above is

$$\mathcal{B} = \left\{1, x, \frac{1}{2}x^2, \frac{1}{6}x^3, \ldots, \frac{1}{n!}x^n\right\}.$$

×

Example

Since $\mathbb{C}$ is algebraically closed, for every $A \in M_{2\times 2}(\mathbb{C})$,

$$p_A(t) = (t-\lambda_1)(t-\lambda_2), \qquad \lambda_1, \lambda_2 \in \mathbb{C}.$$

Then A is conjugate to one of the following.

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If $\lambda_1 \neq \lambda_2$, then

$$m_A(t) = (t-\lambda_1)(t-\lambda_2), \qquad A \sim \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

If $\lambda_1 = \lambda_2 = \lambda$ and $m_A(t) = t - \lambda$, then the algebraic multiplicity is 2, but the maximum block size is 1, so

$$A \sim \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}.$$

If $\lambda_1 = \lambda_2 = \lambda$ and $m_A(t) = (t-\lambda)^2$, then

$$A \sim \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix},$$

a Jordan block of size 2. ×

Example

Let

$$A = \begin{pmatrix} 2 & -1 & 0 & 1 \\ 0 & 3 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 3 \end{pmatrix} \in M_{4\times 4}(\mathbb{Q}).$$

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Compute

$$p_A(t) = (t-2)^3(t-3).$$

Since the algebraic multiplicity of $\lambda = 2$ is 3, the possibilities for its Jordan blocks are:

- three 1×1 blocks;
- one 2×2 block and one 1×1 block;
- one 3×3 block.

Remark

We may start from either $p_A(t)$ or $m_A(t)$ when calculating the Jordan canonical form, but in general we also need the nullities (or ranks) of powers of $A - \lambda I$, or equivalent information that determines the Jordan-block multiplicities, as we compute next. ×

Compute

$$\ker(A - 2I_4) = \ker \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} = \ker \begin{pmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus

$$\operatorname{null}(A - 2I_4) = 4 - \operatorname{rk}(A - 2I_4) = 2.$$

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Compute

$$(A-2I_4)^2=\begin{pmatrix}0&-2&1&1\\0&0&0&0\\0&0&0&0\\0&-2&1&1\end{pmatrix}.$$

Thus

$$\operatorname{null}(A-2I_4)^2=4-\operatorname{rk}(A-2I_4)^2=4-1=3.$$

By Theorem II, the number of Jordan blocks of size 1 for $\lambda=2$ is

$$2\operatorname{null}(A-2I_4)-\operatorname{null}(A-2I_4)^2-\operatorname{null}(A-2I_4)^0=2\cdot 2-3=1.$$

Thus it is the second case: one Jordan block of size 2 and one of size 1 for $\lambda=2$, together with one block of size 1 for $\lambda=3$.

(This is vacuously true since $m_A(t)$ must have $(t-3)$ as a factor.)

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Therefore

$$J=\begin{pmatrix}2&1&0&0\\0&2&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}.$$

We now find a conjugating basis, looking at $U=(A-2I_4)|_{W_1}$.

Following the basis algorithm, find a basis for C_2 such that

$$\ker(U^2)=\ker(U)\oplus C_2.$$

Here

$$\ker(U^2)=\operatorname{Span}\left\{\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}0\\\frac{1}{2}\\1\\0\end{pmatrix},\begin{pmatrix}0\\\frac{1}{2}\\0\\1\end{pmatrix}\right\},$$

while

$$\ker(U)=\operatorname{Span}\left\{\begin{pmatrix}0\\1\\1\\1\end{pmatrix},\begin{pmatrix}1\\0\\0\\0\end{pmatrix}\right\}.$$

Choose

$$v_1=\begin{pmatrix}0\\\frac{1}{2}\\1\\0\end{pmatrix}$$

as a basis of C_2. Then

$$(A-2I_4)v_1=\begin{pmatrix}-\frac{1}{2}\\-\frac{1}{2}\\-\frac{1}{2}\\-\frac{1}{2}\end{pmatrix}=Uv_1.$$

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This is a linearly independent subset of $\ker U$ and can be expanded to a basis by

$$\{Uv_1, v_2\}, \qquad v_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

Then Uv_1, v_1 is a cyclic subspace for $(A - 2I_4)|_{W_1}$, and v_2 is another cyclic subspace.
Lastly, for $(A - 3I_4)|_{W_2}$, simply choose a basis for its kernel:

$$\ker(A - 3I_4|_{W_2}) = \operatorname{Span}\{v_3\}, \qquad v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Thus

$$(Uv_1, v_1, v_2, v_3)$$

is a basis of

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$\mathbb{Q}^4$ with respect to which the matrix conjugate to A is its Jordan canonical form J.
Indeed, with

$$Q = \begin{pmatrix} -\frac{1}{2} & 0 & 1 & 1 \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ -\frac{1}{2} & 1 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 1 \end{pmatrix},$$

we have

$$Q^{-1}AQ = J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

×

Remark

After computing the Jordan canonical form of an operator or matrix whose minimal polynomial remains unknown, we can recover the minimal polynomial by backward computation.

Example

From the previous example, the maximum size of a Jordan block equals the power of the corresponding factor in the minimal polynomial.

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Thus

$$m_A(t) = (t-2)^2(t-3).$$

Equivalently, from the 'setup' proposition,

$$m_i = \min\{j \geq 1 : \ker((A-\lambda_i I)^j) = \ker((A-\lambda_i I)^{j+1})\}.$$

Thus the kernels stabilize as follows:

$$\ker(A-\lambda_i I) \subsetneq \cdots \subsetneq \ker((A-\lambda_i I)^{m_i}) = \ker((A-\lambda_i I)^{m_i+1}) = \cdots$$

We can determine this while we compute

$$\ker((A-\lambda_i I)^j), \qquad j = 1, 2, \ldots,$$

□

8 Inner Product Spaces

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. For $\alpha = x + iy$,

$$\overline{\alpha} := x - iy$$

is the complex conjugate of α.

Definition

An inner product on an $\mathbb{F}$-vector space V is a function

$$\langle \cdot, \cdot \rangle : V \times V \longrightarrow \mathbb{F}$$

such that $\forall u, v, w \in V$, $\lambda \in \mathbb{F}$,

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1. $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$;
2. $\langle \lambda v, w\rangle = \lambda\langle v, w\rangle$;
3. $\langle v, w\rangle = \overline{\langle w, v\rangle}$;
4. $\langle v, v\rangle \geq 0$, and $\langle v, v\rangle = 0 \iff v = 0 \in V$.

□

Remark

Note that from 3, we have

$$\langle v, v\rangle = \overline{\langle v, v\rangle} \Longrightarrow \langle v, v\rangle \in \mathbb{R}.$$

From 1, 2, 3, we know

$$\begin{aligned} \langle v, u+w\rangle &= \overline{\langle u+w, v\rangle} = \overline{\langle u, v\rangle} + \overline{\langle w, v\rangle} \\ &= \langle v, u\rangle + \langle v, w\rangle, \end{aligned} \tag{$*$}$$

$$\langle v, \lambda u\rangle = \overline{\langle \lambda u, v\rangle} = \overline{\lambda}\,\overline{\langle u, v\rangle} = \overline{\lambda}\langle v, u\rangle.$$

I will refer to $(*)$ along with 1, 2, 3 as ~~conjugate~~/skew linearity,

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and 4 as positive definiteness. □

Definition

An $\mathbb{F}$-vector space equipped with an inner product is called an inner product space. ×

Definition

If V is an inner product space, with inner product $\langle \cdot, \cdot \rangle$, the norm induced by this inner product is

$$\| \cdot \| : V \longrightarrow \mathbb{R}_{\geq 0}$$

defined by

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

×

Example

The standard inner product on $\mathbb{F}^n$ is

$$\langle (x_1, x_2, \ldots, x_n), (y_1, \ldots, y_n) \rangle = \sum_{i=1}^{n} x_i \overline{y_i}.$$

This induces the norm

$$\|(x_1, \ldots, x_n)\| = \sqrt{\sum_{i=1}^{n} x_i \overline{x_i}} = \sqrt{\sum_{i=1}^{n} |x_i|^2}.$$

×

Example

Say $\alpha_1, \ldots, \alpha_n \in \mathbb{R}_{>0}$. Then we can define an inner product on $\mathbb{F}^n$

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by

$$\langle (x_1, x_2, \ldots, x_n), (y_1, \ldots, y_n) \rangle = \sum_{i=1}^{n} \alpha_i x_i \overline{y_i}.$$

More generally, given $A \in M_{n \times n}(\mathbb{F})$, we can further define

$$A^* := (\overline{A})^t,$$

the conjugate transpose (or adjoint) of A. Then we can check if $A = A^*$, in which case such matrices are called Hermitian/self-adjoint, then for

$$\langle \cdot, \cdot \rangle : \mathbb{F}^n \times \mathbb{F}^n \longrightarrow \mathbb{F}$$

defined by

$$\langle x, y \rangle := xAy^* = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i \overline{y_j},$$

we can safely check conjugate linearity holds. However, positive definiteness is harder to characterize, which we'll come back to later and the proof (or I might) supply

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a complete characterization using eigenvalues (and quadratic forms by me). ×

Example

Let $a < b \in \mathbb{R}$, $\mathbb{F} = \mathbb{R}$. Let

$$V := C([a, b]),$$

the set of continuous functions

$$f : [a, b] \longrightarrow \mathbb{R}.$$

Then

$$\langle f(x), g(x) \rangle := \int_a^b f(x)g(x)\, dx$$

defines an inner product on V. Positive definiteness follows from the fact that if $h \in C([a, b])$, $h \geq 0$, and $h(x) > 0$ for some $x \in [a, b]$, then

$$\int_a^b h(x)\, dx > 0.$$

The other axioms are trivial. ×

Sub-example

Let $a = 0$, $b = 1$. Then

$$\langle 2x - 1, x^2 \rangle = \int_0^1 (2x - 1)x^2\, dx = \frac{1}{6}.$$

×

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8.1 Orthogonality and Norm Inequalities

Definition

For an inner product space we say $v, w \in V$ are orthogonal, denoted by $v \perp w$, if

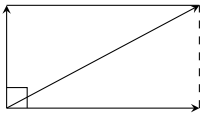
$$\langle v, w \rangle = 0.$$

×

Pythagorean Theorem

For an inner product space V and $v, w \in V$, if $v \perp w$, then

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2.$$



Proof:

$$\begin{aligned}\|v+w\|^2 &= \langle v+w, v+w\rangle \\ &= \langle v,v\rangle + \langle w,v\rangle + \langle v,w\rangle + \langle w,w\rangle \\ &= \langle v,v\rangle + \overline{0} + 0 + \langle w,w\rangle \\ &= \|v\|^2 + \|w\|^2.\end{aligned}$$

□

Definition

Let V be an $\mathbb{F}$-inner product space. Let $v \in V$, $w \in V \setminus \{0\}$. The projection of v onto (the direction of) w is

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$$\operatorname{proj}_w(v) := \frac{\langle v,w\rangle}{\langle w,w\rangle} w.$$

If $\|w\| = 1$, then

$$\operatorname{proj}_w(v) = \langle v,w\rangle w.$$

×

Proposition

For V an $\mathbb{F}$-inner product space, $v, w \in V$, $w \neq 0$,

$$v - \operatorname{proj}_w v \perp w.$$

Proof: Exercise. Same flavour as the last one. □

Corollary

$$\|\operatorname{proj}_w v\| \leq \|v\|.$$

Proof: By the previous proposition,

$$\begin{aligned}&\|v - \operatorname{proj}_w v\|^2 + \|\operatorname{proj}_w v\|^2 = \|v\|^2 \\ \implies \quad &\|\operatorname{proj}_w v\|^2 \leq \|v\|^2.\end{aligned}$$

□

Cauchy–Bunyakovsky–Schwarz

Let V be an $\mathbb{F}$-inner product space. Then

$$\forall v, w \in V, \qquad |\langle v,w\rangle| \leq \|v\|\|w\|.$$

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Proof of part 1: If $w = 0$, then $0 \leq 0$.

Suppose $w \neq 0$. By the previous corollary,

$$\begin{aligned}\|v\| \geq \|\operatorname{proj}_w v\| &= \left\| \frac{\langle v, w\rangle}{\langle w, w\rangle} w \right\| \\ &= |\langle v, w\rangle| \frac{\|w\|}{\|w\|^2}.\end{aligned}$$

Therefore

$$|\langle v, w\rangle| \leq \|v\|\|w\|.$$

Proof 2 (analytic proof). First prove it for $\langle v, w\rangle \in \mathbb{R}$. Let $v, w \in V$. Define, $\forall t \in \mathbb{R}$,

$$\begin{aligned}f(t) &= \|v + tw\|^2 \\ &= \|v\|^2 + t^2\|w\|^2 + 2t\langle v, w\rangle.\end{aligned}$$

Thus $f \geq 0$ and $f \in C^\infty(\mathbb{R})$. Compute

$$f'(t) = 2t\|w\|^2 + 2\langle v, w\rangle.$$

f achieves its minimum when $f'(t) = 0$, hence

$$t = -\frac{\langle v, w\rangle}{\|w\|^2}.$$

Therefore

$$\begin{aligned}f(t) = \|v\|^2 + \frac{\langle v, w\rangle^2}{\|w\|^2} - \frac{2\langle v, w\rangle^2}{\|w\|^2} \geq 0 \\ \implies \quad |\langle v, w\rangle| \leq \|v\|\|w\|.\end{aligned}$$

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To show this for $\langle v, w\rangle \in \mathbb{C} \setminus \mathbb{R}$, note that if

$$\lambda := \overline{\langle v, w\rangle},$$

then

$$\langle \lambda v, w\rangle = \overline{\langle v, w\rangle}\langle v, w\rangle \in \mathbb{R} \setminus \{0\}.$$

Thus

$$\begin{aligned}|\langle \lambda v, w\rangle| \leq \|\lambda v\|\|w\| \\ \implies \quad |\lambda|\,|\langle v, w\rangle| \leq |\lambda|\|v\|\|w\| \\ \implies \quad |\langle v, w\rangle| \leq \|v\|\|w\|.\end{aligned}$$

□

Corollary

The 'norm' induced by the inner product is indeed a norm.

Proof: The homogeneity and positive definiteness trivially follow from conjugate-linearity and positive definiteness of the inner product.

$\triangle$-ineq: Let $v, w \in V$. Then

$$\begin{aligned}\|v+w\|^2 &= \|v\|^2 + \|w\|^2 + 2\,\mathrm{Re}\langle v, w\rangle \\ &\leq \|v\|^2 + \|w\|^2 + 2\|v\|\|w\| = (\|v\| + \|w\|)^2 \qquad \text{by the Cauchy–Schwarz inequality}\end{aligned}$$

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$$\implies \quad \|v+w\| \leq \|v\| + \|w\|.$$

□

8.2 Gram–Schmidt Orthogonalization

Definition

Let V be a finite dimensional $\mathbb{F}$-inner product space. A basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ is called orthogonal if

$$\langle v_i, v_j\rangle = 0, \qquad \forall i \neq j.$$

It is orthonormal if it is orthogonal and

$$\|v_i\| = 1, \qquad \forall i.$$

×

Gram–Schmidt Orthogonalization

Let $\{v_1, \ldots, v_n\}$ be a basis for a finite-dimensional inner product space V. Set

$$w_1 = v_1, \qquad u_1 = \frac{1}{\|w_1\|} w_1,$$

and inductively define, for $2 \leq k \leq n$,

$$w_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i\rangle}{\langle w_i, w_i\rangle} w_i, \qquad u_k = \frac{1}{\|w_k\|} w_k.$$

Then $\{w_1, \ldots, w_n\}$ (respectively $\{u_1, \ldots, u_n\}$) is an orthogonal (respectively, orthonormal) basis

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for V such that $\forall\, 1 \leq k \leq n$,

$$\mathrm{Span}\{w_1, \ldots, w_k\} = \mathrm{Span}\{v_1, \ldots, v_k\} = \mathrm{Span}\{u_1, \ldots, u_k\}.$$

Proof: Clearly

$$\text{Span}\{w_1, \ldots, w_k\} = \text{Span}\{u_1, \ldots, u_k\},$$

and $\{u_1, \ldots, u_n\}$ is orthonormal as soon as $\{w_1, \ldots, w_n\}$ is orthogonal. It's left to show that $\{w_1, \ldots, w_n\}$ is orthogonal and

$$\text{Span}\{w_1, \ldots, w_k\} = \text{Span}\{v_1, \ldots, v_k\}.$$

Base Case: $k = 1$:

$$\text{Span}(w_1) = \text{Span}(v_1)$$

by definition. $\{w_1\}$ is trivially orthogonal.

Inductive Step: Suppose $\{w_1, \ldots, w_{k-1}\}$ is orthogonal and

$$\text{Span}\{w_1, \ldots, w_{k-1}\} = \text{Span}\{v_1, \ldots, v_{k-1}\}, \qquad 1 \leq k \leq n-1.$$

Then notice by definition of w_k, we have

$$w_k \in \text{Span}\{w_1, \ldots, w_{k-1}, v_k\}, \qquad v_k \in \text{Span}\{w_1, \ldots, w_k\}.$$

Hence

$$\text{Span}\{w_1, \ldots, w_k\} = \text{Span}\{w_1, \ldots, w_{k-1}, v_k\}$$

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$$= \text{Span}\{v_1, \ldots, v_{k-1}, v_k\}$$

by the inductive hypothesis. Since $\{v_1, \ldots, v_k\}$ is linearly independent,

$$\dim(\text{Span}\{v_1, \ldots, v_k\}) = k.$$

Thus $\{w_1, \ldots, w_k\}$ is a basis.

It's left to show the orthogonality:

$$\begin{aligned}
\langle w_k, w_j \rangle &= \left\langle v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle} w_i, w_j \right\rangle \\
&= \langle v_k, w_j \rangle - \sum_{i=1}^{k-1} \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle} \langle w_i, w_j \rangle \\
&= \langle v_k, w_j \rangle - \langle v_k, w_j \rangle = 0, \qquad j = 1, \ldots, k-1.
\end{aligned}$$

Thus the theorem is proven by induction. □

Example

Let $\langle \cdot, \cdot \rangle$ be the inner product on $\mathbb{R}[x]_1$ given by

$$\langle f(x), g(x) \rangle = \int_0^1 f(x) g(x)\, dx.$$

Let $\mathcal{B} = \{1, x\}$. Then $w_1 = 1$.

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$$w_2 = x - \frac{\langle x, 1\rangle}{\langle 1, 1\rangle} 1$$
$$= x - \frac{\frac{1}{2}}{1} 1 = x - \frac{1}{2}.$$

Normalizing them, we have

$$u_1 = \frac{w_1}{\|w_1\|} = \frac{1}{1} = 1,$$
$$u_2 = \frac{w_2}{\|w_2\|} = \frac{x - \frac{1}{2}}{\left(\int_0^1 (x - \frac{1}{2})^2 \, dx\right)^{1/2}}$$
$$= \frac{x - \frac{1}{2}}{\left(\frac{1}{12}\right)^{1/2}} = 2\sqrt{3}\, x - \sqrt{3}.$$

Thus

$$\{1,\, 2\sqrt{3}x - \sqrt{3}\}$$

is an orthonormal basis for $(\mathbb{R}[x]_1, \langle \cdot, \cdot \rangle)$. ×

Proposition

Let $\{u_1, \ldots, u_n\}$ be an orthonormal basis of an inner product space V. Then $\forall v \in V$,

$$v = \sum_{i=1}^{n} \langle v, u_i \rangle u_i.$$

Proof: Suppose $v = \sum a_i u_i$. Then for $1 \leq j \leq n$,

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$$\langle v, u_j \rangle = \left\langle \sum_{i=1}^{n} a_i u_i, u_j \right\rangle = \sum_{i=1}^{n} a_i \langle u_i, u_j \rangle = a_j.$$

[Compare this with the dual basis equation:

$$v = \sum f_i(v) v_i,$$

where $\{f_1, \ldots, f_n\}$ is the dual basis of $\{v_1, \ldots, v_n\}$. We show in assignment this can be seen via the Riesz representation theorem.]

□

Definition

Let V be an inner product space and $S \subseteq V$. The orthogonal complement of S is

$$S^{\perp} := \{v \in V \mid \langle u, v \rangle = 0,\ \forall u \in S\}.$$

×

Proposition

Let V be an $\mathbb{F}$-inner product space and $S \subseteq V$. Then

$$S^{\perp} \leq V.$$

[Note the $\perp$ notation now represents orthocomplement instead of the annihilator.]

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Proof: Exercise. □

8.3 Orthogonal Projections

Proposition

Let V be a finite-dimensional inner product space, $W \leq V$. Then

$$V = W \oplus W^{\perp}.$$

($\oplus$ represents internal direct sum here.)

Proof: Recall from an assignment question that

$$V = W \oplus W^{\perp} \iff W \cap W^{\perp} = \{0\}, \quad V = W + W^{\perp}.$$

Let $v \in W \cap W^{\perp}$. Then

$$\langle v, v \rangle = 0 \implies v = 0.$$

Let $v \in V$. Take an orthogonal basis $\{w_1, \ldots, w_s\}$, a basis of W. Then consider

$$\operatorname{proj}_W v = \sum_{i=1}^{s} \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle} w_i;$$

we have

$$v - \operatorname{proj}_W v + \operatorname{proj}_W v = v.$$

Clearly $\operatorname{proj}_W v \in W$. We want to show that

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$$v - \operatorname{proj}_W v \in W^{\perp}.$$

Take

$$w = \sum_{i=1}^{s} a_i w_i \in W.$$

Then

$$\begin{aligned}\left\langle v - \operatorname{proj}_W v, \sum_{i=1}^{s} a_i w_i \right\rangle &= \left\langle v, \sum_{i=1}^{s} a_i w_i \right\rangle - \left\langle \operatorname{proj}_W v, \sum_{i=1}^{s} a_i w_i \right\rangle \\ &= \sum_{i=1}^{s} \overline{a_i}\langle v, w_i\rangle - \left\langle \sum_{i=1}^{s} \frac{\langle v, w_i\rangle}{\langle w_i, w_i\rangle} w_i, \sum_{i=1}^{s} a_i w_i \right\rangle \\ &= \sum_{i=1}^{s} \overline{a_i}\langle v, w_i\rangle - \sum_{i=1}^{s}\sum_{j=1}^{s} \frac{\langle v, w_i\rangle}{\langle w_i, w_i\rangle} \overline{a_j}\langle w_i, w_j\rangle \\ &= \sum_{i=1}^{s} \overline{a_i}\langle v, w_i\rangle - \sum_{i=1}^{s} \frac{\langle v, w_i\rangle}{\langle w_i, w_i\rangle} \overline{a_i}\langle w_i, w_i\rangle \\ &= 0 \qquad \text{by orthogonality}\end{aligned}$$

Hence $v - \operatorname{proj}_W v \in W^\perp$. □

Corollary

For finite-dimensional V and W,

$$(W^\perp)^\perp = W.$$

Proof: Clearly $(W^\perp)^\perp \supseteq W$. But by the previous proposition, we know

$$\dim W = \dim(W^\perp)^\perp$$

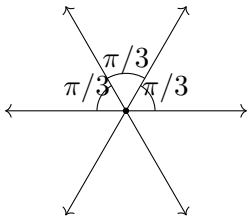
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$$= \dim V - \dim W^\perp.$$

Hence $(W^\perp)^\perp = W$. □

Aside

You can draw 3 equiangular lines in $\mathbb{R}^2$ through the origin. Equiangular means any pair of lines form the same angle.



Recall with the standard inner product on $\mathbb{R}^n$ or $\mathbb{C}^n$, we can define generalized angle as

$$\theta = \arccos\left(\frac{|\langle v, w\rangle|}{\|v\|\|w\|}\right).$$

Then we can generalize the notion of equiangular: let $u_1, \ldots, u_n \in \mathbb{C}^n$ be unit vectors in $\mathbb{C}^n$, $L_i := \operatorname{Span}\{u_i\}$. Then we say $\{L_1, \ldots, L_k\}$ are equiangular if

$$\exists \alpha \in \mathbb{R} \quad \text{such that} \quad |\langle u_i, u_j\rangle| = \alpha \quad \text{for } 1 \le i \ne j \le k.$$

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Example

The following non-unit representatives, all having the common norm $\sqrt{3-\sqrt{3}}$,

$$\{u_1,\dots,u_4\} = \left\{ \left(1, \frac{1+i}{1+\sqrt{3}}\right), \left(1, \frac{-1-i}{1+\sqrt{3}}\right), \left(\frac{1+i}{1+\sqrt{3}}, 1\right), \left(\frac{-1-i}{1+\sqrt{3}}, 1\right) \right\}$$

give 4 equiangular lines in $\mathbb{C}^2$. ×

Zauner's Conjecture

$\forall n \in \mathbb{N}$, $\exists$ a set of n^2 equiangular lines in $\mathbb{C}^n$. □

Remark

- False in $\mathbb{R}^n$ $(n = 2)$.
- Proved if assuming the other conjecture.
- Proved $\leq n^2$.

×

8.4 Adjoint Operators

$$A^* := (\overline{A})^t.$$

$(V, \langle\cdot,\cdot\rangle)$ is an $\mathbb{F}$-inner product space.

Proposition

Let V be finite-dimensional and let $T : V \to V$ be a linear transformation. Then $\exists!$ linear transformation $T^* : V \to V$

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such that

$$\langle Tv, w\rangle = \langle v, T^*w\rangle, \qquad \forall v, w \in V.$$

Proof (uniqueness). Let $S_1, S_2 : V \to V$ be such that

$$\langle Tv, w\rangle = \langle v, S_i w\rangle.$$

Then $\forall v, w \in V$,

$$\begin{aligned}\langle v, (S_1 - S_2)w\rangle &= \langle v, S_1w - S_2w\rangle \\ &= \langle v, S_1w\rangle - \langle v, S_2w\rangle \\ &= \langle Tv, w\rangle - \langle Tv, w\rangle = 0.\end{aligned}$$

Thus $(S_1 - S_2)w = 0$, $\forall w \in V$, hence $S_1 = S_2$.

Existence: Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be an orthonormal basis of V. Let $v, w \in V$. Suppose

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \qquad [w]_{\mathcal{B}} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

We have

$$\begin{aligned} \langle v, w \rangle &= \left\langle \sum_{i=1}^{n} a_i v_i, \sum_{j=1}^{n} b_j v_j \right\rangle \\ &= \sum_{i,j=1}^{n} a_i \overline{b_j} \langle v_i, v_j \rangle = \sum_{i=1}^{n} a_i \overline{b_i} \\ &= [v]_{\mathcal{B}}^{t} \overline{[w]_{\mathcal{B}}}. \end{aligned}$$

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Thus

$$\begin{aligned} \langle Tv, w \rangle &= [Tv]_{\mathcal{B}}^{t} \overline{[w]_{\mathcal{B}}} \\ &= \left([T]_{\mathcal{B}}^{\mathcal{B}} [v]_{\mathcal{B}} \right)^{t} \overline{[w]_{\mathcal{B}}} \\ &= [v]_{\mathcal{B}}^{t} \left([T]_{\mathcal{B}}^{\mathcal{B}} \right)^{t} \overline{[w]_{\mathcal{B}}} \\ &= [v]_{\mathcal{B}}^{t} \overline{\left([T]_{\mathcal{B}}^{\mathcal{B}} \right)^{*} [w]_{\mathcal{B}}}. \end{aligned}$$

Let

$$\left(\left([T]_{\mathcal{B}}^{\mathcal{B}} \right)^{*} \right) = (c_{ij})_{i,j}.$$

Define T^* by

$$T^* w := \sum_{i=1}^{n} \left[\left([T]_{\mathcal{B}}^{\mathcal{B}} \right)^{*} [w]_{\mathcal{B}} \right]_i v_i.$$

Then clearly

$$\langle Tv, w \rangle = \langle v, T^* w \rangle, \qquad [T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}} \right)^{*}.$$

Finally, T^* does not depend on $\mathcal{B}$ by

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the uniqueness. □

Definition

We call such T^* in the previous proposition the adjoint of T. ×

Proposition

Let $S, T : V \to V$ be linear transformations, $a \in \mathbb{F}$. Then

1. $(T + S)^* = T^* + S^*$;
2. $(aT)^* = \overline{a}\, T^*$;
3. $(S \circ T)^* = T^* \circ S^*$;
4. $(T^*)^* = T$.

Proof:

1. Note that both $(T+S)^*$ and T^*+S^* satisfy the adjoint property of $T+S$. Hence
$$(T+S)^* = T^* + S^*$$
by uniqueness.
2. Note that both $(aT)^*$ and $\overline{a}\,T^*$ satisfy the adjoint property by uniqueness.

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3. Same idea:
$$\begin{aligned}\langle S \circ Tv, w\rangle &= \langle Tv, S^*w\rangle \\ &= \langle v, T^* \circ S^* w\rangle.\end{aligned}$$
Hence by uniqueness,
$$(S\circ T)^* = T^* \circ S^*.$$
4. Same idea: $\forall v, w \in V$,
$$\begin{aligned}\langle T^*v, w\rangle = \overline{\langle w, T^*v\rangle} &= \overline{\langle Tw, v\rangle} \\ &= \langle v, Tw\rangle.\end{aligned}$$
Thus $(T^*)^* = T$ by uniqueness.

□

Definition

Let $T: V \to V$ be linear. We say T is self-adjoint if
$$T = T^*,$$
normal if
$$TT^* = T^*T.$$

×

Definition

Let $A \in M_{n\times n}(\mathbb{F})$. A is self-adjoint if $A^* = A$, normal if
$$A^*A = AA^*,$$
unitary if $A \in GL_n(\mathbb{F})$ and $A^{-1} = A^*$.

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In the case of $\mathbb{F} = \mathbb{R}$, we say orthogonal instead of unitary. ×

Remark

$$\left.\begin{matrix}\text{Self-adjoint}\\ \text{Unitary/Orthogonal}\end{matrix}\right\} \Longrightarrow \text{Normal by definition.}$$

- $A \in M_{n\times n}(\mathbb{F})$ is unitary $\iff$ the columns of A form an orthonormal basis of $\mathbb{F}^n$ with respect to the standard inner product on $\mathbb{F}^n$. Both directions follow from the fact that if

$$A = (v_1, \ldots, v_n), \qquad B = (w_1, \ldots, w_n),$$

then

$$(A^*B)_{i,j} = \langle v_i, w_j \rangle.$$

- Product of unitary matrices is unitary. Exercise.

Hence

$$U(n) := \{\text{group of unitary matrices}\} \underset{\text{subgroup}}{\leq} GL(n).$$

□

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Example

- $A \in M_{2\times 2}(\mathbb{C})$ is self-adjoint $\iff$

$$A = \begin{pmatrix} a & z \\ \overline{z} & b \end{pmatrix}, \qquad a, b \in \mathbb{R},\ z \in \mathbb{C}.$$

- $A \in M_{n\times n}(\mathbb{R})$ is self-adjoint $\iff$ A is symmetric; that is, $A = A^t$.
- Diagonal matrices are normal.
- $\forall \theta \in \mathbb{R}$,

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

is orthogonal.

- $\forall \alpha, \beta \in \mathbb{R}$,

$$\begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{i\beta} \end{pmatrix}$$

is unitary.

□

Example

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not normal.

$$A^* = \overline{A}^t = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

$$A^*A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad AA^* = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $A^*A \neq AA^*$, the matrix A is not normal. □

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8.5 Self-Adjoint and Normal Operators

Proposition

Let $T : V \to V$ be self-adjoint. Then $p_T(t)$ splits over $\mathbb{F}$, and eigenvalues of T are $\mathbb{R}$.

Proof: If $\mathbb{F} = \mathbb{C}$: $p_T(t)$ splits by the Fundamental Theorem of Algebra.

Let λ be an eigenvalue of T. Thus for an eigenvector v of λ ($v \neq 0$), we have

$$\langle Tv, v\rangle = \lambda\langle v, v\rangle = \langle v, Tv\rangle = \overline{\lambda}\langle v, v\rangle$$

$$\implies \quad \lambda = \overline{\lambda} \quad \text{since } \langle v, v\rangle \neq 0.$$

Hence $\lambda \in \mathbb{R}$.

If $\mathbb{F} = \mathbb{R}$: Then V is an $\mathbb{R}$-vector space. Notice from the proof of adjoint, we have

$$T = T^* \iff [T]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$$

where $\mathcal{B}$ is an orthonormal basis of V.

Let $\mathcal{B}$ be an orthonormal basis of V. Then $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self-adjoint. Define

$$S : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

by

$$Sv = Av.$$

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The reason for this construction is that $\overline{T}$ is an operator on an $\mathbb{R}$-vector space, but our first argument only applies to $\mathbb{F} = \mathbb{C}$. Then

$$\underbrace{p_T(t) = p_A(t)}_{\text{in an } \mathbb{R}\text{-vector space}}, \qquad \underbrace{p_A(t) = p_S(t)}_{\text{in a } \mathbb{C}\text{-vector space}}.$$

Furthermore $p_S(t)$ splits into real linear polynomials by what we showed already in part 1, $\mathbb{F} = \mathbb{C}$. Therefore $p_T(t)$ splits over $\mathbb{R}$. □

Lemma 1

Let $\overline{T} : V \longrightarrow V$ be a (normal) linear transformation. If $W \leq V$ is $\overline{T}$-invariant, then $W^\perp$ is $\overline{T}^*$-invariant.

Proof: Let $w \in W^\perp$. Hence $\forall v \in W$,

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$$\langle v, \overline{T}^* w\rangle = \langle \overline{T}v, w\rangle = 0,$$

where $\overline{T}v \in W$ and $w \in W^\perp$. Therefore

$$\overline{T}^* w \in W^\perp.$$

□

Lemma 2

Let $\overline{T} : V \longrightarrow V$ be a normal linear transformation. If $v \in V$ is an eigenvector for $\overline{T}$ with eigenvalue λ, then v is an eigenvector for $\overline{T}^*$ with eigenvalue $\overline{\lambda}$.

Proof: By normality,

$$\begin{aligned}(\overline{T}^* - \overline{\lambda}I)(\overline{T} - \lambda I) &= \overline{T}^*\overline{T} - \lambda\overline{T}^* - \overline{\lambda}\,\overline{T} + \lambda\overline{\lambda}I \\ &= \overline{T}\overline{T}^* - \overline{\lambda}\,\overline{T} - \lambda\overline{T}^* + \lambda\overline{\lambda}I \\ &= (\overline{T} - \lambda I)(\overline{T}^* - \overline{\lambda}I).\end{aligned}$$

Hence

$$\begin{aligned}\|\overline{T}^* v - \overline{\lambda}v\|^2 &= \left\langle (\overline{T}^* - \overline{\lambda}\,\mathrm{Id})v, (\overline{T}^* - \overline{\lambda}\,\mathrm{Id})v \right\rangle \\ &= \left\langle v, (\overline{T}^* - \overline{\lambda}\,\mathrm{Id})^*(\overline{T}^* - \overline{\lambda}\,\mathrm{Id})v \right\rangle \\ &= \left\langle v, (\overline{T}^* - \overline{\lambda}\,\mathrm{Id})(\overline{T} - \lambda\,\mathrm{Id})v \right\rangle.\end{aligned}$$

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$$\|\overline{T}^* v - \overline{\lambda}v\|^2 = \left\langle v, (\overline{T}^* - \overline{\lambda}\,\mathrm{Id})(0) \right\rangle = 0.$$

Therefore

$$(\overline{T}^* - \overline{\lambda}\,\mathrm{Id})(v) = 0 \qquad \text{by positive definiteness,}$$

so

$$\overline{T}^* v = \overline{\lambda}v.$$

□

Lemma 3

Let $\overline{T} : V \longrightarrow V$ be a normal linear transformation and let $\lambda \neq \mu$ be eigenvalues of $\overline{T}$. Then

$$E_\lambda \perp E_\mu.$$

Proof: By Lemma 2, for $v \in E_\lambda$, $w \in E_\mu$,

$$\begin{aligned}\lambda\langle v, w\rangle &= \langle \overline{T}v, w\rangle = \langle v, \overline{T}^* w\rangle \\ &= \underbrace{\langle v, \overline{\mu}w\rangle}_{\text{lemma 2}} = \mu\langle v, w\rangle.\end{aligned}$$

Thus

$$(\lambda - \mu)\langle v, w\rangle = 0,$$

and hence

$$\langle v, w\rangle = 0 \qquad \text{since } \lambda \neq \mu.$$

□

Spectral Theorem for Normal Operators

Let $\overline{T} : V \longrightarrow V$ be a normal operator

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whose characteristic polynomial splits over $\mathbb{F}$. Then there exists an orthonormal basis $\mathcal{B}$ of V consisting of eigenvectors of $\overline{T}$.

Proof: By induction on the dimension n.

Base case: $n = 0$ is trivially true.

Inductive Step: Let $n \in \mathbb{N}$. Let λ be an eigenvalue of $\overline{T}$. Then E_λ is $\overline{T}^*$-invariant since E_λ is $\overline{T}^*$'s $\overline{\lambda}$-eigenspace (lemma 2). Hence $E_\lambda^\perp$ is $\overline{T}$-invariant, as $(\overline{T}^*)^* = \overline{T}$, by lemma 1. Then

$$\left(\overline{T}\Big|_{E_\lambda^\perp}\right)^* = \overline{T}^*\Big|_{E_\lambda^\perp},$$

since $\forall v, w \in E_\lambda^\perp$,

$$\begin{aligned}\left\langle \overline{T}\Big|_{E_\lambda^\perp} v, w \right\rangle &= \langle \overline{T}v, w\rangle = \langle v, \overline{T}^* w\rangle \\ &= \left\langle v, \overline{T}^*\Big|_{E_\lambda^\perp} w \right\rangle.\end{aligned}$$

Hence $\overline{T}\Big|_{E_\lambda^\perp}$ is normal and

$$\dim(E_\lambda) > 0 \quad \Longrightarrow \quad \dim(E_\lambda^\perp) < \dim(V) \qquad (V = W \oplus W^\perp).$$

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Furthermore,

$$p_{\overline{T}|_{E_\lambda^\perp}}(t) \mid p_T(t) \qquad \text{(characteristic polynomial)}.$$

Hence $p_{\overline{T}|_{E_\lambda^\perp}}(t)$ splits. By the induction hypothesis, there exists $\mathcal{B}_\perp$, an orthonormal basis of $E_\lambda^\perp$ consisting of eigenvectors of $\overline{T}\Big|_{E_\lambda^\perp}$.

Let $\mathcal{B}_\lambda$ be an orthonormal basis of E_λ. Then

$$E_\lambda \oplus E_\lambda^\perp = V \quad \Longrightarrow \quad \mathcal{B}_\lambda \cup \mathcal{B}_\perp \text{ gives a basis of } V.$$

All the vectors are unit. Lastly $E_\lambda \perp E_\lambda^\perp$ by definition, so $\mathcal{B}_\lambda \cup \mathcal{B}_\perp$ is orthonormal. □

Corollary

Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let $\overline{T}$ be a normal operator. Then $\overline{T}$ is orthogonally diagonalizable; that is, there exists an orthonormal basis of V consisting of eigenvectors of $\overline{T}$.

Proof: By the Fundamental Theorem of Algebra, the characteristic polynomial of $\overline{T}$ must split over $\mathbb{C}$.

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Hence the result follows from the previous theorem. □

Corollary

Let V be a finite-dimensional $\mathbb{R}$-vector space and let $\overline{T}$ be a self-adjoint operator on V. Then $\overline{T}$ is orthogonally diagonalizable.

Proof: By a previous proposition, a self-adjoint operator has a characteristic polynomial that splits over $\mathbb{R}$. Hence the result follows from the previous theorem. □

Corollary

1. For $A \in M_n(\mathbb{C})$, there exists a unitary Q such that

$$Q^*AQ \text{ is diagonal} \iff A \text{ is normal.}$$

2. For $A \in M_n(\mathbb{R})$, there exists an orthogonal Q such that

$$Q^tAQ \text{ is diagonal} \iff A \text{ is symmetric.}$$

Proof: 1. ($\Leftarrow$) By a corollary and a remark, A is normal $\Longrightarrow L_A : \mathbb{C}^n \longrightarrow \mathbb{C}^n$ is normal $\Longrightarrow$ there exists an orthonormal basis of $\mathbb{C}^n$ consisting of eigenvectors

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of L_A. Hence

$$[L_A]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}[L_A]_{\mathrm{St}}^{\mathrm{St}}[\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}}.$$

Therefore $[\mathrm{Id}]_{\mathrm{St}}^{\mathcal{B}}$ is unitary by the remark, and hence $[\mathrm{Id}]_{\mathcal{B}}^{\mathrm{St}}$ is unitary.

($\Rightarrow$) Suppose

$$A = Q^*DQ,$$

where D is diagonal and $Q \in M_n(\mathbb{C})$ is unitary. Then

$$\begin{aligned} A^*A &= (Q^*DQ)^*(Q^*DQ) \\ &= Q^*D^*QQ^*DQ = Q^*D^*DQ \\ &= Q^*DD^*Q = Q^*DQQ^*D^*Q \\ &= Q^*DQ(Q^*DQ)^* = AA^*. \end{aligned}$$

2. If A is symmetric, the real spectral theorem supplies such an orthogonal Q. Conversely, if $Q^tAQ = D$ is real diagonal, then $A = QDQ^t$ and therefore $A^t = QD^tQ^t = A$. □

Algorithm

To perform such a diagonalization in practice,

1. Find the roots of the characteristic polynomial.

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2. For each eigenvalue λ, find an orthonormal basis for E_λ by applying Gram–Schmidt to a basis of E_λ.

The union gives an orthonormal basis of V that orthogonally diagonalizes $\overline{T}$. ☒

Example

Let

$$A = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 4 \end{pmatrix} \in M_{3\times 3}(\mathbb{R}).$$

Hence symmetric $\implies$ the characteristic polynomial splits and the matrix is orthogonally diagonalizable over $\mathbb{R}$; that is, there exists an orthogonal $Q \in M_{3\times 3}(\mathbb{R})$ such that

$$Q^t AQ \text{ is diagonal.}$$

Compute

$$\begin{aligned} p_A(t) &= \det(tI_3 - A) \\ &= \det \begin{pmatrix} t-4 & -2 & 0 \\ -2 & t-2 & -2 \\ 0 & -2 & t-4 \end{pmatrix} = t(t-4)(t-6). \end{aligned}$$

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$$\begin{aligned} E_0 = \ker(A) &= \ker \begin{pmatrix} 4 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 4 \end{pmatrix} \\ &= \cdots = \operatorname{Span}\left\{ \begin{pmatrix} 1 & -2 & 1 \end{pmatrix}^t \right\}, \\ E_4 = \ker(A - 4I) &= \ker \begin{pmatrix} 0 & 2 & 0 \\ 2 & -2 & 2 \\ 0 & 2 & 0 \end{pmatrix} \\ &= \operatorname{Span}\left\{ \begin{pmatrix} -1 & 0 & 1 \end{pmatrix}^t \right\}, \\ E_6 = \ker(A - 6I) &= \ker \begin{pmatrix} -2 & 2 & 0 \\ 2 & -4 & 2 \\ 0 & 2 & -2 \end{pmatrix} \\ &= \operatorname{Span}\left\{ \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^t \right\}. \end{aligned}$$

Normalizing the vectors, we get a basis

$$\left\{ \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

It is orthonormal by our construction. Hence with

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$$Q = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix},$$

we have

$$Q^*AQ = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{pmatrix}.$$

□

Supplementary Material

8.6 Extrema of Multivariable Functions

Let $A \subseteq \mathbb{R}^n$ be open. $f : A \longrightarrow \mathbb{R} \in C^2$.

Recall from 18.101 that the derivative of

$$f : \mathbb{R}^m \longrightarrow \mathbb{R}^n$$

at every point in $\mathbb{R}^m$ is an $n \times m$ matrix

$$\left(\frac{\partial f_i}{\partial x_j}\right)_{\substack{i=1,\dots,n \\ j=1,\dots,m}}.$$

Let $v \in A$ be a critical point; that is,

$$Df[v] = (0, 0, \dots, 0) \qquad \text{or} \qquad \nabla f(v) = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

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Define

$$Hf(v) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(v)\right) \in M_n(\mathbb{R}).$$

H for Hessian matrix.

The second-order Taylor expansion of f at v is, with $h = x - v$,

$$\begin{aligned} f(v+h) &= f(v) + Df(v)h + \frac{1}{2}h^t Hf(v)h + o(\|h\|^2) \\ &= f(v) + \frac{1}{2}h^t Hf(v)h + o(\|h\|^2) \qquad (h \longrightarrow 0). \end{aligned}$$

By Clairaut's theorem,

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

hence $Hf(v)$ is real symmetric. Therefore, by the spectral theorem for self-adjoint real matrices, there exists an orthogonal $P \in M_{n\times n}(\mathbb{R})$ such that

$$P^t Hf(v) P = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \qquad \lambda_i \in \mathbb{R}.$$

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Perform the change of variables

$$y = P^t(x - v).$$

Then

$$Py = (x - v) \qquad \text{since } P^tP = I_n.$$

Then since $y \longrightarrow 0$ as $x \longrightarrow v$,

$$\begin{aligned}(x-v)^t Hf(v)(x-v) &= (Py)^t Hf(v)(Py)\\ &= y^t P^t Hf(v) Py\\ &= y^t \begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{pmatrix} y = \sum_{i=1}^{n} \lambda_i y_i^2.\end{aligned}$$

Thus if

$$\lambda_i > 0, \qquad \forall i,$$

then

$$c := \min_i \lambda_i > 0 \quad \text{and} \quad (x-v)^t Hf(v)(x-v) \geq c\|x-v\|^2.$$

Since the remainder is $o(\|x-v\|^2)$, for all nonzero $x - v$ sufficiently small we have

$$f(x) - f(v) \geq \frac{c}{4}\|x-v\|^2 > 0,$$

hence f has a strict local minimum

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at v.

Likewise, if

$$\lambda_i < 0, \qquad \forall i,$$

then the same remainder estimate, applied to $-f$, shows that f has a strict local maximum at v.

Consider the case with $\boxed{n = 2}$.

Let

$$Hf(v) = \begin{pmatrix} a & b \\ b & d \end{pmatrix},$$

that is,

$$a = f_{xx}(v), \qquad b = f_{xy}(v) = f_{yx}(v), \qquad d = f_{yy}(v).$$

Then

$$\det(Hf(v)) = \lambda_1\lambda_2, \qquad \operatorname{tr}(Hf(v)) = \lambda_1 + \lambda_2.$$

Then

$$\lambda_1 > 0,\ \lambda_2 > 0 \quad \Longleftrightarrow \quad \operatorname{tr}(Hf(v)) > 0,\ \det(Hf(v)) > 0.$$

Now notice that

$$\det(Hf(v)) > 0 \quad \Longrightarrow \quad ad - b^2 > 0 \quad \Longrightarrow \quad ad > 0;$$

Since $ad > 0$, the numbers a, d have the same sign; together with $\operatorname{tr}(Hf(v)) = a + d > 0$, this implies $a > 0$ and $d > 0$. On the other hand, if $a > 0$,

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then

$$ad > b^2 \geq 0 \quad \Longrightarrow \quad d > 0 \quad \Longrightarrow \quad \operatorname{tr}(Hf(v)) = a + d > 0.$$

Hence

$$\operatorname{tr}(Hf(v)) > 0,\ \det(Hf(v)) > 0 \quad \Longleftrightarrow \quad \det(Hf(v)) > 0,\ a > 0$$

and either condition implies that v is a strict local minimum.

Note that I added a section about quadratic form at the end of the notes, which has a deep connection with this part.

Likewise,

$$\lambda_1 < 0,\ \lambda_2 < 0 \quad \Longleftrightarrow \quad \det(Hf(v)) > 0,\ a < 0$$

and either condition implies that v is a strict local maximum. ☒

8.7 Singular Value Decomposition

Theorem

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $A \in M_{m\times n}(\mathbb{F})$, $\operatorname{rk}(A) = r$. Then $\exists$

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1. Unique real numbers
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$
2. Unitary matrices $P \in M_{m\times m}(\mathbb{F})$ and $Q \in M_{n\times n}(\mathbb{F})$ such that, with $\Sigma \in M_{m\times n}(\mathbb{F})$ defined by
$$(\Sigma)_{i,j} := \begin{cases} \sigma_i, & \text{if } i = j \leq r, \\ 0, & \text{otherwise,} \end{cases}$$
we have
$$A = P\Sigma Q^*.$$

This is called a singular value decomposition of A, and $\sigma_1, \ldots, \sigma_r$ are called the singular values of A.

Proof: Let

$$A^* := (\overline{A})^t \in M_{n\times m}(\mathbb{F}).$$

Then

$$A^*A \in M_{n\times n}(\mathbb{F}),$$

and

$$(A^*A)^* = A^*A^{**} = A^*A.$$

Hence A^*A is self-adjoint. By the

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spectral theorem, there exists a unitary matrix $Q \in M_{n\times n}(\mathbb{F})$ such that

$$Q^*A^*AQ = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \qquad \lambda_i \in \mathbb{R}.$$

Let $v_i \in \mathbb{F}^n$ be an eigenvector for A^*A with eigenvalues λ_i. Then

$$\begin{aligned}\lambda_i \|v_i\|^2 &= \lambda_i \langle v_i, v_i \rangle = \langle \lambda_i v_i, v_i \rangle \\ &= \langle A^*Av_i, v_i \rangle = \langle Av_i, Av_i \rangle \\ &= \|Av_i\|^2 \geq 0.\end{aligned}$$

(we proved that adjoint of a matrix is indeed $A^* = \overline{A}^{,t}$ when we study adjoints)

Hence $\lambda_i \geq 0$, and

$$\lambda_i = 0 \quad \iff \quad Av_i = 0.$$

Reorder $\lambda_1, \ldots, \lambda_n$ such that

$$\lambda_1 \geq \cdots \geq \lambda_r > 0, \qquad r = \operatorname{rk}(Q^*A^*AQ) = \operatorname{rk}(A^*A).$$

Notice that

$$\operatorname{rk}(A^*A) = \operatorname{rk}(A) :$$

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$$\ker(A^*A) \supseteq \ker(A) \qquad \text{trivially.}$$

Let $v \in \mathbb{F}^n$ be such that $A^*Av = 0$. Then

$$\langle A^*Av, v \rangle = 0$$

$$\implies \quad \langle Av, Av \rangle = 0 \quad \implies \quad \|Av\|^2 = 0 \quad \implies \quad Av = 0.$$

Hence

$$\ker(A) \supseteq \ker(A^*A) \quad \implies \quad \ker A = \ker(A^*A)$$

$$\implies \quad \operatorname{rk} A = \operatorname{rk}(A^*A) = n - \operatorname{nullity}(A).$$

Hence

$$\operatorname{rk}(A) = \operatorname{rk}(A^*A) = r.$$

For $1 \leq i \leq r$, let

$$\sigma_i = \sqrt{\lambda_i}, \qquad w_i = \frac{1}{\sigma_i} Av_i \in \mathbb{F}^m,$$

v_i being the ith column of Q.

We can easily check $\{w_1, \ldots, w_r\}$ is orthonormal from the fact that Q is unitary and how we defined w_i's.

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Extend it to an orthonormal basis of $\mathbb{F}^m$:

$$\{w_1, \ldots, w_m\}, \qquad \text{set } P = (w_1, \ldots, w_m).$$

Then

$$P\Sigma e_i = \begin{cases} \sigma_i w_i, & 1 \leq i \leq r, \\ 0, & i > r \end{cases}$$

$$\begin{aligned} &= Av_i \\ &= A(\text{column } i \text{ of } Q) \\ &= A(Qe_i). \end{aligned}$$

Therefore

$$P\Sigma = AQ \quad \Longrightarrow \quad A = P\Sigma Q^*.$$

□

This is useful to represent complex matrix easily:

$$A = P\Sigma Q^* = \sum_{i=1}^{r} \sigma_i \underbrace{w_i v_i^*}_{m\times n \text{ matrices}}.$$

Moreover, $w_i v_i^*$ has rank 1 since

$$\operatorname{rk}(w_i v_i^*) \le \operatorname{rk}(v_i^*) = 1 \qquad (\ker(w_i v_i^*) \supseteq \ker(v_i^*)).$$

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Hence we've decomposed A as a sum of rank-1 matrices, and in practice, the geometric property of A is encoded by the terms with larger singular values (σ_i). ×

8.8 Quadratic Forms

I will answer the two questions unsolved from before with the theory of quadratic forms.

1. When does
$$\langle x, y\rangle := x^t A y$$
define an inner product on $\mathbb{R}^n$, for $A \in M_{n\times n}(\mathbb{R})$?
2. What is the other way of viewing the Hessian matrix and second derivatives test?

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Definition

An expression of the form

$$f(x_1, \dots, x_n) = \sum a_{i_1, i_2, \dots, i_n} x_1^{i_1} \cdots x_n^{i_n}$$

is called an n-variable polynomial over $\mathbb{F}$ if

$$i_1, \dots, i_n \in \mathbb{N}, \qquad a_{i_1,\dots,i_n} \in \mathbb{F},$$

and only finitely many coefficients are nonzero. Each term

$$a_{i_1,\dots,i_n} x_1^{i_1} \cdots x_n^{i_n}$$

is called a monomial. We denote

$$\mathbb{F}[x_1, \dots, x_n] = \{\text{all } n\text{-variable polynomials over the field } \mathbb{F}\}.$$

The degree of the monomial

$$a_{i_1,\dots,i_n} x_1^{i_1} \cdots x_n^{i_n}$$

is

$$i_1 + i_2 + \cdots + i_n.$$

The degree of a polynomial $f(x)$, denoted by $\deg(f(x))$, is the maximum degree of its monomials. Define

$$\deg(0) = -\infty.$$

A polynomial is homogeneous if all

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monomials of it have the same degree. □

Remark

$\mathbb{F}[x_1, \dots, x_n]$, equipped with usual addition and multiplication, forms a commutative ring.

□

Definition

A homogeneous polynomial of degree 2 in n variables is called an n-ary quadratic form:

$$f(x_1, \dots, x_n) = \sum_{i,j=1}^{n} a_{ij} x_i x_j.$$

Writing

$$X = (x_1, \dots, x_n)^t, \qquad A = (a_{ij}),$$

we can express a quadratic form as

$$f(x_1, \dots, x_n) = X^t A X.$$

□

Observation

The matrix is not unique. You can verify that if $f = X^t A X$, then

$$f = X^t \left(\frac{1}{2}(A + A^t) \right) X.$$

This motivates the following.

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Proposition

For $\mathbb{F}$ a field, $\operatorname{char}(\mathbb{F}) \neq 2$, every quadratic form is associated to a unique symmetric matrix A by

$$f(X) = X^t A X.$$

Proof: By the definition of $A = (a_{ij})$, we know there exists A such that

$$f(X) = X^t AX.$$

Since $X^t AX$ is a formal 1×1 matrix,

$$(X^t AX)^t = X^t A^t X = X^t AX.$$

Hence

$$\begin{aligned} f(X) &= \frac{2X^t AX}{2} = \frac{X^t(A + A^t)X}{2} \\ &= X^t \frac{A + A^t}{2} X. \end{aligned}$$

The matrix

$$\frac{A + A^t}{2}$$

is symmetric trivially.

Uniqueness: Suppose there exists a symmetric B such that

$$X^t BX = X^t AX.$$

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Then

$$X^t(A - B)X = 0.$$

By substituting $X = e_i$; that is,

$$x_j = \begin{cases} 0, & \text{if } i \neq j, \\ 1, & \text{if } i = j, \end{cases}$$

we get

$$\begin{aligned} & X^t(A - B)X = a_{ii} - b_{ii} = 0 \\ \implies \quad & a_{ii} = b_{ii}, \qquad \forall i = 1, \ldots, n. \end{aligned}$$

Then plugging in $X = e_i + e_j$, we get

$$\begin{aligned} X^t(A - B)X &= a_{ii} - b_{ii} + a_{ji} - b_{ji} + a_{ij} - b_{ij} + a_{jj} - b_{jj} \\ &= a_{ii} - b_{ii} + 2a_{ij} - 2b_{ij} + a_{jj} - b_{jj} \\ &= 0. \end{aligned}$$

Therefore

$$\begin{aligned} & a_{ii} + 2a_{ij} + a_{jj} = b_{ii} + 2b_{ij} + b_{jj} \\ \implies \quad & 2a_{ij} = 2b_{ij}. \end{aligned}$$

Since $\mathrm{char}(\mathbb{F}) \neq 2$, we conclude with

$$a_{ij} = b_{ij}, \qquad \forall i, j.$$

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Since A, B are symmetric, that implies

$$A = B.$$

□

Notation

From now on, the matrix of a quadratic form will refer to this unique symmetric matrix.

×

Definition

Let

$$X = CY, \qquad C \in GL_n(\mathbb{F}).$$

This is called an invertible/nondegenerate linear change of variable:

$$\begin{aligned} f(X) = X^t AX &= (CY)^t ACY \\ &= Y^t (C^t AC) Y =: g(Y). \end{aligned}$$

×

Definition

Two quadratic forms over $\mathbb{F}$ are equivalent if one can be obtained from the other by an invertible linear

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change of variables. If $f(X)$ and $g(X)$ are equivalent, we denote

$$f(X) \sim g(X).$$

Two $n \times n$ matrices A, B are said to be congruent if there exists $C \in GL_n(\mathbb{F})$ such that

$$B = C^t AC.$$

We denote

$$A \simeq B.$$

×

Proposition

Congruency of matrix and equivalence of quadratic forms are equivalence relations.

Proof: Exercise. □

Proposition

For two quadratic forms $X^t AX$, $X^t BX$, where A, B are symmetric, then

$$X^t AX \sim X^t BX \quad \iff \quad A \simeq B.$$

Proof: ($\Leftarrow$) This follows from the definition.

($\Rightarrow$) Suppose

$$X^t AX \sim X^t BX.$$

Then there exists $C \in GL_n(\mathbb{F})$ such that $X = CY$ with

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$$X^t AX = Y^t C^t ACY = Y^t BY.$$

Hence by uniqueness of the symmetric matrix representation, we have

$$C^t AC = B \implies A \cong B.$$

□

Theorem

Let $\mathbb{F}$ be a field with $\text{char}(\mathbb{F}) \neq 2$. Then every quadratic form in n variables over $\mathbb{F}$ is equivalent to a diagonal form.

$$d_1 y_1^2 + \cdots + d_n y_n^2.$$

Remark

There is another cleaner approach proving this theorem comparing to what I am going to present. But that requires a bit introductory theory on bilinear forms, which we did not set up. ×

Proof: (Lagrange's method) Induction on n.

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Base case: $n = 1$ trivial.

Inductive Step: Suppose the claim holds for $(n-1)$ variables. Suppose

$$f(x_1, \ldots, x_n) = \sum_{i,j=1}^{n} a_{ij} x_i x_j, \qquad a_{ji} = a_{ij}.$$

Case 1: $a_{kk} \neq 0$, for some k. Relabel if necessary so that $a_{11} \neq 0$. Then by completing the square,

$$f = a_{11} \left(x_1 + \frac{1}{a_{11}} \sum_{j=2}^{n} a_{1j} x_j \right)^2 + g(x_2, \ldots, x_n),$$

g being a quadratic form in $x_2, \ldots, x_n$. Then setting

$$y_1 = x_1 + \frac{1}{a_{11}} \sum_{j=2}^{n} a_{1j} x_j, \qquad y_j = x_j, \quad j \geq 2$$

(you should verify that this is an invertible linear change of variable),

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we get

$$f \sim a_{11} y_1^2 + g(y_2, \ldots, y_n).$$

By inductive hypothesis, $g(y_2, \ldots, y_n)$ is equivalent to a diagonal form, hence f is.

Case 2: $a_{ii} = 0, \forall i,\ f \neq 0$. Then $\exists a_{ij} \neq 0$. Perform the invertible linear change of variable

$$x_i = y_i + y_j, \qquad x_j = y_i - y_j.$$

Then we get the monomial

$$a_{ij}x_ix_j = a_{ij}y_i^2 - a_{ij}y_j^2, \qquad a_{ij} \neq 0.$$

Furthermore, since $a_{ii} = a_{jj} = 0$,

$$a_{ii}x_i^2 = a_{jj}x_j^2 = 0.$$

Thus the coefficients of y_i^2 and y_j^2 are $2a_{ij}$ and $-2a_{ij}$ (as there are no other factors contributing). Then we reduced the problem to case 1. □

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Corollary

Every symmetric matrix over a field $\mathbb{F}$ of characteristic $\neq 2$ is congruent to a diagonal matrix. □

Remark

In an $\mathbb{R}$-inner product space, the spectral theorem for self-adjoint matrices gives an orthogonal matrix Q with

$$Q^tAQ = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where λ_i's are eigenvalues of A. ☒

Example

Find the diagonal form of

$$f(x_1, x_2) = x_1^2 + 4x_1x_2 + x_2^2.$$

Solution:

$$f = (x_1 + 2x_2)^2 - 3x_2^2 = y_1^2 - 3y_2^2,$$

by substituting

$$y_1 = x_1 + 2x_2, \qquad y_2 = x_2.$$

☒

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Example

Find the diagonal form of

$$f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1.$$

Solution: Substitute first

$$y_1 + y_2 = x_1, \qquad y_1 - y_2 = x_2, \qquad y_3 = x_3.$$

Then

$$\begin{aligned} f(x_1, x_2, x_3) &= y_1^2 - y_2^2 + 2y_1y_3 \\ &= y_1^2 + 2y_1y_3 + y_3^2 - y_2^2 - y_3^2 \\ &= (y_1 + y_3)^2 - y_2^2 - y_3^2. \end{aligned}$$

Substitute

$$z_1 = y_1 + y_3, \qquad z_2 = y_2, \qquad z_3 = y_3;$$

we get

$$f \sim z_1^2 - z_2^2 - z_3^2.$$

×

Definition

If

$$X^tAX \sim d_1x_1^2 + \cdots + d_nx_n^2,$$

then

$$\text{number of nonzero coefficients among the } d_i := \operatorname{rk}(X^tAX),$$

the rank of the quadratic form. ×

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Proposition

$$\operatorname{rk}(X^tAX) = \operatorname{rk}(A).$$

Proof: Congruent matrices have the same rank (exercise). Hence

$$\operatorname{rk}(A) = \operatorname{rk} \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix} = \operatorname{rk}(X^tAX).$$

□

9 Real Quadratic Forms

Let $\mathbb{F} = \mathbb{R}$.

Let X^tAX be a real quadratic form with $\operatorname{rk}(A) = r$. Then it is equivalent to a diagonal form

$$d_1x_1^2 + \cdots + d_rx_r^2, \qquad d_i \neq 0.$$

After rescaling by

$$x_i \longmapsto \frac{x_i}{\sqrt{|d_i|}},$$

the coefficients become ± 1.

Definition

The expression

$$x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2$$

is called the canonical form of the

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real quadratic form X^tAX. Here

$$r = \mathrm{rk}(A), \qquad p = \#\text{ positive squares.}$$

×

Question: Are p and r unique?

Answer: To answer this question, we first present a different way of seeing equivalence between quadratic forms.

Recall the change-of-basis matrix: Given a vector space V with bases $\mathcal{B}, \mathcal{B}'$,

$$[v]_{\mathcal{B}} = [\mathrm{Id}]_{\mathcal{B}'}^{\mathcal{B}}[v]_{\mathcal{B}'}.$$

Compare this to change of variable:

$$X = CY.$$

Hence given a quadratic form X^tAX, a vector space V of dimension n, and a basis $\mathcal{B}$, the quadratic form defines

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a function

$$\widetilde{f} : V \longrightarrow \mathbb{F}$$

via

$$\widetilde{f}(v) = [v]_{\mathcal{B}}^t A [v]_{\mathcal{B}}.$$

Lemma

Two quadratic forms $f \sim g$ if and only if there exist $\mathcal{B}, \mathcal{B}'$ bases of V such that

$$\widetilde{f} = \widetilde{g},$$

defined with respect to $\mathcal{B}, \mathcal{B}'$.

Proof: ($\Rightarrow$) Suppose $X = CY$.

$$f(X) = X^tAX = Y^tC^tACY = g(Y).$$

Let $\mathcal{B}$ be a basis of V. Then

$$\widetilde{f}(v) = [v]_{\mathcal{B}}^t A [v]_{\mathcal{B}}.$$

Suppose

$$\mathcal{B} = \{v_1, \ldots, v_n\}.$$

Then define

$$\begin{aligned}\mathcal{B}' = \{&c_{11}v_1 + c_{21}v_2 + \cdots + c_{n1}v_n,\\ &c_{12}v_1 + c_{22}v_2 + \cdots + c_{n2}v_n,\\ &\vdots\\ &c_{1n}v_1 + c_{2n}v_2 + \cdots + c_{nn}v_n\}.\end{aligned}$$

Then clearly

$$[\mathrm{Id}]^{\mathcal{B}}_{\mathcal{B}'} = C.$$

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Hence, with respect to this $\mathcal{B}'$,

$$\begin{aligned}\widetilde{g}(v) &= [v]^t_{\mathcal{B}'} C^t A C [v]_{\mathcal{B}'}\\ &= \left([\mathrm{Id}]^{\mathcal{B}}_{\mathcal{B}'}[v]_{\mathcal{B}'}\right)^t A \left([\mathrm{Id}]^{\mathcal{B}}_{\mathcal{B}'}[v]_{\mathcal{B}'}\right)\\ &= [v]^t_{\mathcal{B}} A [v]_{\mathcal{B}} = \widetilde{f}(v).\end{aligned}$$

($\Leftarrow$) Suppose

$$f(X) = X^t A X, \qquad g(X) = X^t M X,$$

A, M symmetric matrices. Suppose $\widetilde{f} = \widetilde{g}$ defined with respect to bases $\mathcal{B}, \mathcal{B}'$ of V. Then $\forall v \in V$,

$$\widetilde{f}(v) = [v]^t_{\mathcal{B}} A [v]_{\mathcal{B}}, \qquad \widetilde{g}(v) = [v]^t_{\mathcal{B}'} M [v]_{\mathcal{B}'}.$$

Hence

$$\begin{aligned}[v]^t_{\mathcal{B}} A [v]_{\mathcal{B}} &= [v]^t_{\mathcal{B}'} M [v]_{\mathcal{B}'}\\ &= \left([\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}}[v]_{\mathcal{B}}\right)^t M \left([\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}}[v]_{\mathcal{B}}\right)\\ &= [v]^t_{\mathcal{B}} \left([\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}}\right)^t M [\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}} [v]_{\mathcal{B}}.\end{aligned}$$

By the same proof for uniqueness of

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symmetric matrices for quadratic forms, we have

$$A = \left([\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}}\right)^t M [\mathrm{Id}]^{\mathcal{B}'}_{\mathcal{B}}.$$

Thus

$$A \cong M \implies f \sim g.$$

□

9.1 Sylvester's Law of Inertia

Now we can give the answer to the question:

Sylvester's Law of Inertia

If a real quadratic form has two canonical forms:

$$\begin{aligned} X^tAX &\sim x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2 = f(X), \\ &\sim y_1^2 + \cdots + y_q^2 - y_{q+1}^2 - \cdots - y_r^2 = g(Y), \end{aligned}$$

then

$$p = q.$$

Proof: Pick any n-dimensional vector space V.

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Then there exist bases $\mathcal{B}, \mathcal{B}'$ of V such that

$$\widetilde{f} = \widetilde{g}$$

defined with respect to $\mathcal{B}, \mathcal{B}'$, respectively. Define

$$\begin{aligned} V_1 &= \{v \in V \mid v_{p+1} = \cdots = v_n = 0 \text{ in } \mathcal{B}\text{-coordinates}\}, \\ V_2 &= \{v \in V \mid v_1 = \cdots = v_q = 0 \text{ in } \mathcal{B}'\text{-coordinates}\}. \end{aligned}$$

Then $\forall v \in V_1$,

$$\begin{aligned} \widetilde{f}(v) &= [v]_{\mathcal{B}}^t \operatorname{diag}(\underbrace{1, \ldots, 1}_{p}, \underbrace{-1, \ldots, -1}_{r-p}, 0, \ldots, 0)[v]_{\mathcal{B}} \\ &= v_1^2 + \cdots + v_p^2 - 0 \geq 0. \end{aligned}$$

For $v \in V_2$,

$$\begin{aligned} \widetilde{g}(v) &= [v]_{\mathcal{B}'}^t \operatorname{diag}(\underbrace{1, \ldots, 1}_{q}, \underbrace{-1, \ldots, -1}_{r-q}, 0, \ldots, 0)[v]_{\mathcal{B}'} \\ &= 0 - v_{q+1}^2 - \cdots - v_r^2 \leq 0. \end{aligned}$$

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Furthermore, by how we defined V_1, V_2,

$$\dim(V_1) = p, \qquad \dim(V_2) = n - q.$$

Furthermore,

$$V_1 \cap V_2 = \{0\}.$$

Let $v \in V_1 \cap V_2$. Suppose $v \neq 0$. Then, by the definition of V_1, at least one of $v_1, \ldots, v_p$ is $\neq 0$. Hence

$$\widetilde{f}(v) > 0.$$

Likewise, by the inequality already established for V_2,

$$\widetilde{g}(v) \leq 0.$$

Yet $\widetilde{f} = \widetilde{g}$, $\rightarrow\leftarrow$. Hence $v = 0$, so $V_1 \cap V_2 = \{0\}$.

Thereby

$$\begin{aligned}\dim(V_1 + V_2) &= \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2) \\ &= p + n - q - 0.\end{aligned}$$

Since $V_1 + V_2 \leq V$,

$$p + n - q \leq n \implies p \leq q.$$

By a symmetric argument defining

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$$\begin{aligned}V_1 &= \{v \in V \mid v_1 = \cdots = v_p = 0 \text{ in } \mathcal{B}\text{-coordinates}\}, \\ V_2 &= \{v \in V \mid v_{q+1} = \cdots = v_n = 0 \text{ in } \mathcal{B}'\text{-coordinates}\},\end{aligned}$$

we get

$$\begin{aligned}\dim(V_1) + \dim(V_2) &= n - p + q \leq n \\ &\implies q \leq p.\end{aligned}$$

Henceforth

$$q \leq p, \quad p \leq q \implies p = q.$$

□

Definition

For a real quadratic form with canonical form

$$x_1^2 + \cdots + x_p^2 - x_{p+1}^2 - \cdots - x_r^2,$$

the integer p is called positive inertia index and $r - p$ the negative inertia index. The pair

$$(p, r - p)$$

is called the signature of the form. ×

Remark

That said, the signature is an invariant under the equivalence relation of quadratic

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forms. ×

Example

With

$$f(X) = x_1x_2 + x_2x_3 + x_3x_1,$$

its canonical form is

$$z_1^2 - z_2^2 - z_3^2.$$

Hence

$$r = 3, \qquad p = 1, \qquad r - p = 2$$

as we have computed from the Lagrange's method. ×

10 Positive-Definite Quadratic Forms

Definition

Let X^tAX be a real quadratic form. We say it is positive definite if

$$x^tAx > 0, \qquad \forall\, 0 \neq x \in \mathbb{R}^n;$$

negative definite if

$$x^tAx < 0, \qquad \forall\, 0 \neq x \in \mathbb{R}^n;$$

positive semidefinite if

$$x^tAx \geq 0, \qquad \forall x \in \mathbb{R}^n;$$

negative semidefinite if

$$x^tAx \leq 0, \qquad \forall x \in \mathbb{R}^n;$$

we say it is indefinite if it is none of these.

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These definition works as is for symmetric matrices since every quadratic form corresponds to a unique symmetric matrix. ×

We now investigate the equivalent conditions for positive definiteness.

Proposition

For a real quadratic form, the following are equivalent:

(1) X^tAX is positive definite;
(2) the canonical form of X^tAX is
$$x_1^2 + \cdots + x_n^2,$$
in other words, the positive inertia index $= n$;
(3) $A \cong I_n$.

Proof: (2) $\Rightarrow$ (1) Trivial.

(1) $\Rightarrow$ (2) Suppose the positive index is $< n$. Then in canonical coordinates there is some $i \in \{1, \ldots, n\}$ such that the

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coefficient of x_i^2 is either -1 or 0. Choosing $x = e_i$, we get

$$e_i^t A e_i \leq 0.$$

Hence $X^t AX$ is not positive definite. $*$

Hence

$$r = p = n.$$

(3) $\Longleftrightarrow$ (2): Immediately from the fact that congruence $\Longleftrightarrow$ equivalence of quadratic forms. $\square$

Proposition

Let $X^t AX$ be a real quadratic form. The following are equivalent:

(1) $X^t AX$ is positive definite;
(2) A is positive definite;
(3) its eigenvalues are strictly positive.

Proof: (1) $\Longleftrightarrow$ (2) follows from the definition.

(2) $\Longleftrightarrow$ (3): By the spectral theorem of real symmetric matrices, there exists an orthogonal C,

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such that

$$C^t AC = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

λ_i being eigenvalues of A. By the orthogonal substitution $X = CY$, we have

$$\begin{aligned} f(X) = X^t AX &= Y^t C^t ACY \\ &= \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 \\ &\sim \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2. \end{aligned}$$

This quadratic form has positive inertia index $= n$ if and only if

$$\lambda_1, \lambda_2, \ldots, \lambda_n > 0.$$

$\square$

Definition

Let $A = (a_{ij})_{n \times n}$. Denote for $k = 1, \ldots, n$,

$$A_k := (a_{ij})_{i,j=1}^k,$$

the kth leading principal submatrix. Denote

$$\Delta_k := \det(A_k), \qquad k = 1, \ldots, n,$$

the kth leading principal minor of A. ×

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10.1 Sylvester's Criterion

Sylvester's Criterion

A real symmetric A is positive definite if and only if

$$\Delta_i > 0, \qquad i = 1, 2, \ldots, n.$$

Proof: We will use block matrix multiplication.

($\Rightarrow$) Suppose A is positive definite. $\forall k \in \{1, \ldots, n\}$, $u \in \mathbb{R}^k \setminus \{0\}$,

$$u^t A_k u = \begin{pmatrix} u \\ 0 \end{pmatrix}^t A \begin{pmatrix} u \\ 0 \end{pmatrix} > 0$$

by positive definiteness of A. Hence the A_k are positive definite. By the previous proposition, the eigenvalues of A_k are > 0. Thus

$$\forall k, \qquad \Delta_k = \det(A_k) = \text{product of eigenvalues} > 0.$$

($\Leftarrow$) By induction on n.

$n = 1$: Clear.

Inductive Step: Suppose the statement holds for $n - 1$. Thus $\exists C \in GL_{n-1}(\mathbb{R})$,

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such that

$$C^t A_{n-1} C = I_{n-1}$$

by the second last proposition. Write

$$A = \begin{pmatrix} A_{n-1} & \beta \\ \beta^t & a_{nn} \end{pmatrix}$$

and congruent by the invertible matrix

$$\begin{pmatrix} C & 0 \\ 0 & 1 \end{pmatrix}$$

with inverse

$$\begin{pmatrix} C^{-1} & 0 \\ 0 & 1 \end{pmatrix}:$$

$$\begin{aligned}&\begin{pmatrix} C & 0 \\ 0 & 1 \end{pmatrix}^t \begin{pmatrix} A_{n-1} & \beta \\ \beta^t & a_{nn} \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} C^t & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} A_{n-1} & \beta \\ \beta^t & a_{nn} \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} C^t A_{n-1} C & C^t \beta \\ \beta^t C & a_{nn} \end{pmatrix} = \begin{pmatrix} I_{n-1} & \gamma \\ \gamma^t & a_{nn} \end{pmatrix},\end{aligned}$$

where

$$\gamma = C^t \beta.$$

Further congruent it by

$$\begin{pmatrix} I_{n-1} & -\gamma \\ 0 & 1 \end{pmatrix},$$

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whose inverse is given by

$$\begin{pmatrix} I_{n-1} & \gamma \\ 0 & 1 \end{pmatrix},$$

we obtain

$$\begin{pmatrix} I_{n-1} & -\gamma \\ 0 & 1 \end{pmatrix}^t \begin{pmatrix} I_{n-1} & \gamma \\ \gamma^t & a_{nn} \end{pmatrix} \begin{pmatrix} I_{n-1} & -\gamma \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} I_{n-1} & 0 \\ 0 & a_{nn} - \gamma^t \gamma \end{pmatrix}.$$

Notice that congruence preserves sign:

$$\det(C^t A C) = (\det C)^2 \det A > 0 \iff \det A > 0$$

since $\det C \neq 0$ by invertibility of C. Thereby

$$\det \begin{pmatrix} I_{n-1} & 0 \\ 0 & a_{nn} - \gamma^t \gamma \end{pmatrix} = a_{nn} - \gamma^t \gamma$$

has same sign as $\det(A)$, which is > 0. Hence

$$A \cong \begin{pmatrix} I_{n-1} & 0 \\ 0 & a_{nn} - \gamma^t \gamma \end{pmatrix} \implies X^t A X$$

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has positive inertia index

$$= \text{positive inertia index of } x_1^2 + \cdots + x_{n-1}^2 + \underbrace{(a_{nn} - \gamma^t \gamma)}_{>0} x_n^2 = n.$$

Thus A (and $X^t A X$) is positive definite by the second-to-last proposition. □

Example

$$f(X) = x_1 x_2 + x_2 x_3 + x_3 x_1.$$

Corresponding symmetric matrix is

$$A := \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

We can compute its eigenvalues (counted with multiplicity):

$$1,\ -\frac{1}{2},\ -\frac{1}{2}.$$

This matches with the signature $(1, 2)$ we computed via the Lagrange's method.

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We can also see it is not positive definite since

$$\Delta_1 = 0 \not> 0.$$

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Sylvester's Criterion for Negative Definiteness

A real symmetric matrix A is negative definite if and only if the leading principal minors have alternating signs:

$$\Delta_1 < 0, \qquad \Delta_2 > 0, \qquad \Delta_3 < 0, \qquad \dots$$

Proof: Notice that A is negative definite if and only if $-A$ is positive definite, by definition. Then apply the previous theorem to $-A$. □

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Now I will answer the first question, leaving the second question as an open

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exercise.

$$\langle x, y \rangle = x^t A y$$

is an inner product on $\mathbb{R}^n$ if A is symmetric and positive definite, equivalently, if A has all positive eigenvalues, equivalently, if

$$\Delta_k(A) > 0, \qquad \forall k.$$

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